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FLUID END WITH THREADED DYNAMIC SECTION

Abstract

A high-pressure hydraulic fracturing pump. The pump is in-line in orientation, with a static section having a high-pressure discharge chamber and a dynamic section having a dynamic internal bore threaded to the static section. A fluid routing plug is disposed within a bore which includes the dynamic internal bore. The pump is designed for ease of access and for reducing wear. Large, external threads on the dynamic section pair with internal threads on the static section to reduce wear associated with repeated pressurization of hydraulic fracturing fluid found within the pump. Seals and wear rings are found at a nose of the dynamic section to further prevent wear and damage.

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Background/Summary

SUMMARY

[0001] The present invention is directed to a high-pressure pump. The high-pressure pump comprises a fluid end body. The fluid end body comprises a static section and a dynamic section. The static section comprises at least one internal flow bore having internally disposed threads. The dynamic section comprises a dynamic internal flow bore, the dynamic section having external threads disposed about an external surface of the dynamic section. The dynamic section is configured for threaded attachment to the static section such that the at least one internal flow bore and the dynamic internal flow bore are aligned.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is an isometric view of a high-pressure pump using one embodiment of a multi-piece fluid end.

[0003] FIG. 2 is an isometric view of the multi-piece fluid end shown in FIG. 1.

[0004] FIG. 3 is a back, top, left isometric view of the multi-piece fluid end shown in FIG. 2.

[0005] FIG. 4 is a cross-sectional view of the multi-piece fluid end shown in FIG. 3, taken along line A-A.

[0006] FIG. 5 is a cross-sectional view of the multi-piece fluid end shown in FIG. 3, taken along line B-B. Some components are cropped or not shown for clarity.

[0007] FIG. 6 is an enlarged view of area C shown in FIG. 5.

[0008] FIG. 7 is an exploded, back, top, left isometric view of the multi-piece fluid end shown in FIG. 2.

[0009] FIG. 8 is an isometric view of a dynamic body used in the multi-piece fluid end shown in FIG. 2.

[0010] FIG. 9 is a top plan view of the dynamic body shown in FIG. 8.

[0011] FIG. 10 is a cross-sectional view of the dynamic body shown in FIG. 9, taken along line D-D.

[0012] FIG. 11 is an isometric view of a retainer used in the multi-piece fluid end shown in FIG. 2.

[0013] FIG. 12 is a back, top, left isometric view of the retainer shown in FIG. 11.

[0014] FIG. 13 is a top plan view of the retainer shown in FIG. 11.

[0015] FIG. 14 is a back elevation view of the retainer shown in FIG. 11.

[0016] FIG. 15 is a cross-sectional view of the retainer shown in FIG. 13, taken along line E-E.

[0017] FIG. 16 is a cross-sectional view of the retainer shown in FIG. 14, taken along line F-F.

[0018] FIG. **17** is an isometric view of another embodiment of a multi-piece fluid end that may be used in the high-pressure pump shown in FIG. **1**.

[0019] FIG. **18** is a back, top, left isometric view of the multi-piece fluid end shown in FIG. **17**.

[0020] FIG. **19** is a cross-sectional view of the multi-piece fluid end shown in FIG. **17** taken along line G-G.

[0021] FIG. **20** is an enlarged view of area H shown in FIG. **19**.

[0022] FIG. **21** is an exploded, back, top, left isometric view of the multi-piece fluid end shown in FIG. **17**.

[0023] FIG. **22** is an isometric view of a retainer used in the multi-piece fluid end shown in FIG. **17**.

[0024] FIG. **23** is a back, top, left isometric view of the retainer shown in FIG. **22**.

[0025] FIG. **24** is a top plan view of the retainer shown in FIG. **22**.

[0026] FIG. **25** is a cross-sectional view of the retainer shown in FIG. **24**, taken along line I-I.

[0027] FIG. **26** is an isometric view of a packing nut used in the multi-piece fluid end shown in FIG. **17**.

[0028] FIG. **27** is a back, top, left isometric view of the packing nut shown in FIG. **26**.

[0029] FIG. **28** is a back elevation view of the packing nut shown in FIG. **26**.

[0030] FIG. **29** is a top plan view of the packing nut shown in **26**.

[0031] FIG. **30** is a cross-sectional view of the packing nut shown in FIG. **28**, taken along line J-J.

[0032] FIG. **31** is a cross-sectional view of the packing nut shown in FIG. **28**, taken along line K-K.

[0033] FIG. **32** is a back, top, left isometric view of another embodiment of a multi-piece fluid end.

[0034] FIG. **33** is a cross-sectional view of the multi-piece fluid end shown in FIG. **32**, taken along line L-L.

[0035] FIG. **34** is a cross-sectional view of the multi-piece fluid end shown in FIG. **32**, taken along line M-M. Some components are cropped or not shown for clarity.

[0036] FIG. **35** is an enlarged view of area N shown in FIG. **33**.

[0037] FIG. **36** is an enlarged view of area O shown in FIG. **34**.

[0038] FIG. **37** is an exploded back, top, left, isometric view of a dynamic section of the multi-piece fluid end shown in FIG. **32**.

[0039] FIG. **38** is an exploded, back, top, left isometric view of the multi-piece fluid end shown in FIG. **32**.

[0040] FIG. **39** is an isometric view of a dynamic body used in the dynamic section shown in FIG. **37**.

[0041] FIG. **40** is a back, top, left isometric view of the dynamic body shown in FIG. **39**.

[0042] FIG. **41** is a top plan view of the dynamic body shown in FIG. **39**.

[0043] FIG. **42** is a cross-sectional view of the dynamic body shown in FIG. **41**, taken along line P-P.

[0044] FIG. **43** is an isometric view of a retainer used in the multi-piece fluid end shown in FIG. **32**.

[0045] FIG. **44** is a back, top, left isometric view of the retainer shown in FIG. **43**.

[0046] FIG. **45** is a top plan view of the retainer shown in FIG. **43**.

[0047] FIG. **46** is a back elevation view of the retainer shown in FIG. **43**.

[0048] FIG. **47** is a cross-sectional view of the retainer shown in FIG. **45**, taken along line Q-Q.

[0049] FIG. **48** is a cross-sectional view of the retainer shown in FIG. **46**, taken along line R-R.

[0050] FIG. **49** is an enlarged view of area S shown in FIG. **47**.

[0051] FIG. **50** is a top plan view of a centering ring used in the multi-piece fluid end shown in FIG. **32**.

[0052] FIG. **51** is an isometric view of the centering ring shown in FIG. **50**.

[0053] FIG. **52** is a front view of the centering ring shown in FIG. **50**.

[0054] FIG. **53** is a cross-sectional view of the centering ring shown in FIG. **52**, taken along line T-T.

[0055] FIG. **54** is an isometric view of another embodiment of a high-pressure pump.

[0056] FIG. **55** is an isometric view of a fluid end used on the high-pressure pump shown in FIG. **54**.

[0057] FIG. **56** is an isometric view of a fluid end section used on the fluid end shown in FIG. **55**.

[0058] FIG. **57** is a back, top, left isometric view of the fluid end section shown in FIG. **56**.

[0059] FIG. **58** is a cross-sectional view of the fluid end section shown in FIG. **56**, taken along line U-U.

[0060] FIG. **59** is a cross-sectional view of the fluid end section shown in FIG. **57**, taken along line V-V.

[0061] FIG. **60** is an exploded, back, top, left isometric view of the fluid end section shown in FIG. **56**.

[0062] FIG. **61** is an isometric view of the dynamic body used in the dynamic section shown in FIG. **56**.

[0063] FIG. **62** is a top plan view of the dynamic body shown in FIG. **61**.

[0064] FIG. **63** is a cross-sectional view of the dynamic body shown in FIG. **62**, taken along line W-W.

[0065] FIG. **64** is an isometric view of the retainer used in the plunger system shown in FIG. **56**.

[0066] FIG. **65** is a back, top, left isometric view of the retainer shown in FIG. **64**.

[0067] FIG. **66** is a top plan view of the retainer shown in FIG. **64**.

[0068] FIG. **67** is a back elevation view of the retainer shown in FIG. **64**.

[0069] FIG. **68** is a cross-sectional view of the retainer shown in FIG. **66**, taken along line X-X.

[0070] FIG. **69** is a cross-sectional view of the retainer shown in FIG. **67**, taken along line Y-Y.

[0071] FIG. **70** is an enlarged view of area Z shown in FIG. **68**.

[0072] FIG. **71** is a back, top, left isometric view of another embodiment of a multi-piece fluid end.

[0073] FIG. **72** is a cross-sectional view of the multi-piece fluid end shown in FIG. **71**, taken along line AA-AA.

[0074] FIG. **73** is an enlarged view of area AB shown in FIG. **72**.

[0075] FIG. **74** is an exploded, back, top, left isometric view of the multi-piece fluid end shown in FIG. **71**.

[0076] FIG. **75** is an isometric view of a dynamic body used in the multi-piece fluid end shown in FIG. **71**.

[0077] FIG. **76** is a top plan view of the dynamic body shown in FIG. **75**.

[0078] FIG. **77** is a cross-sectional view of the dynamic body shown in FIG. **76**, taken along line AC-AC.

[0079] FIG. **78** is an isometric view of a retainer assembly used in the multi-piece fluid end shown in FIG. **71**.

[0080] FIG. **79** is a back, top, left isometric view of the retainer assembly shown in FIG. **78**.

[0081] FIG. **80** is a cross-sectional view of the retainer assembly shown in FIG. **78**, taken along line AD-AD.

[0082] FIG. **81** is an exploded, back, top, left isometric view of the retainer assembly shown in FIG. **78**.

[0083] FIG. **82** is an isometric view of a coupling used in the retainer assembly shown in FIG. **78**.

[0084] FIG. **83** is a back, top, left isometric view of the coupling shown in FIG. **82**.

[0085] FIG. **84** is a cross-sectional view of the coupling shown in FIG. **82**, taken along line AE-AE.

[0086] FIG. **85** is an isometric view of a packing nut retainer used in the retainer assembly shown in FIG. **78**.

[0087] FIG. **86** is a back, top, left isometric view of the packing nut retainer shown in FIG. **85**.

[0088] FIG. **87** is a cross-sectional view of the packing nut retainer shown in FIG. **85**, taken along

line AF-AF.

[0089] FIG. **88** is an isometric view of another embodiment of a fluid end section that may be used in the fluid end shown in FIG. **55**.

[0090] FIG. **89** is a back, top, left isometric view of the fluid end section shown in FIG. **88**.

[0091] FIG. **90** is a cross-sectional view of the fluid end section shown in FIG. **89**, taken along line AG-AG.

[0092] FIG. **91** is a cross-sectional view of the fluid end section shown in FIG. **89**, taken along line AH-AH.

[0093] FIG. **92** is a cross-sectional view of the fluid end section shown in FIG. **89**, taken along line AI-AI.

[0094] FIG. **93** is an exploded, back, top, left isometric view of the fluid end section shown in FIG. **88**.

[0095] FIG. **94** is an exploded isometric view of the dynamic section shown in FIG. **93**.

[0096] FIG. **95** is an isometric view of the dynamic body used in the dynamic section shown in FIG. **88**.

[0097] FIG. **96** is a top plan view of the dynamic body shown in FIG. **95**.

[0098] FIG. **97** is a cross-sectional view of the dynamic body shown in FIG. **95**, taken along line AJ-AJ.

[0099] FIG. **98** is an isometric view of the spacer sleeve shown in FIG. **89**.

[0100] FIG. **99** is a back, top, left isometric view of the spacer sleeve shown in FIG. **98**.

[0101] FIG. **100** is a front elevation view of the spacer sleeve shown in FIG. **98**.

[0102] FIG. **101** is a cross-sectional view of the spacer sleeve shown in FIG. **100**, taken along line AK-AK.

[0103] FIG. **102** is a cross-sectional view of the spacer sleeve shown in FIG. **100**, taken along line AL-AL.

[0104] FIG. **103** is a cross-sectional view of the spacer sleeve shown in FIG. **100**, taken along line AM-AM.

[0105] FIG. **104** is a cross-sectional view of the spacer sleeve shown in FIG. **100**, taken along line AN-AN.

[0106] FIG. **105** is an isometric view of the retainer used in the plunger system shown in FIG. **88**.

[0107] FIG. **106** is a back, top, left isometric view of the retainer shown in FIG. **105**.

[0108] FIG. **107** is a top plan view of the retainer shown in FIG. **105**.

[0109] FIG. **108** is a back elevation view of the retainer shown in FIG. **105**.

[0110] FIG. **109** is a cross-sectional view of the retainer shown in FIG. **107**, taken along line AO-AO.

[0111] FIG. **110** is a cross-sectional view of the retainer shown in FIG. **108**, taken along line AP-AP.

[0112] FIG. **111** is an isometric view of another embodiment of a fluid end section that may be used in the fluid end shown in FIG. **55**.

[0113] FIG. **112** is a back, top, left isometric view of the fluid end section shown in FIG. in.

[0114] FIG. **113** is a cross-sectional view of the fluid end section shown in FIG. **112**, taken along line AQ-AQ.

[0115] FIG. **114** is a cross-sectional view of the fluid end section shown in FIG. **112**, taken along line AR-AR.

[0116] FIG. **115** is an exploded, back, top, left isometric view of the fluid end section shown in FIG. **111**.

[0117] FIG. **116** is an isometric view of the dynamic body used in the dynamic section shown in FIG. **111**.

[0118] FIG. **117** is a top plan view of the dynamic body shown in FIG. **116**.

[0119] FIG. **118** is a cross-sectional view of the dynamic body shown in FIG. **117**, taken along line

AS-AS.

[0120] FIG. **119** is an isometric view of the retainer used in the plunger system shown in FIG. **111**.

[0121] FIG. **120** is the back, top, left isometric view of the retainer shown in FIG. **119**.

[0122] FIG. **121** is a top plan view of the retainer shown in FIG. **119**.

[0123] FIG. **122** is a back elevation view of the retainer shown in FIG. **119**.

[0124] FIG. **123** is a cross-sectional view of the retainer shown in FIG. **121**, taken along line AT-AT.

[0125] FIG. **124** is a cross-sectional view of the retainer shown in FIG. **122**, taken along line AU-AU.

[0126] FIG. **125** is an enlarged view of area AV shown in FIG. **123**.

[0127] FIG. **126** is a back, top, left isometric view of another embodiment of a multi-piece fluid end.

[0128] FIG. **127** is a cross-sectional view of the multi-piece fluid end shown in FIG. **126**, taken along line AW-AW.

[0129] FIG. **128** is an enlarged view of area AX shown in FIG. **127**.

[0130] FIG. **129** is a cross-sectional view of the multi-piece fluid end shown in FIG. **126**, taken along line AY-AY.

[0131] FIG. **130** is an enlarged view of area AZ shown in FIG. **129**.

[0132] FIG. **131** is an exploded, back, top, left isometric view of the multi-piece fluid end shown in FIG. **126**.

[0133] FIG. **132** is a back, top, left isometric view of a dynamic body used in the multi-piece fluid end shown in FIG. **126**.

[0134] FIG. **133** is a top plan view of the dynamic body shown in FIG. **132**.

[0135] FIG. **134** is a cross-sectional view of the dynamic body shown in FIG. **133**, taken along line BA-BA.

[0136] FIG. **135** is an enlarged view of area BB shown in FIG. **134**.

[0137] FIG. **136** is an enlarged view of area BC shown in FIG. **134**.

[0138] FIG. **137** is an isometric view of a retainer used in the multi-piece fluid end shown in FIG. **126**.

[0139] FIG. **138** is a back, top, left isometric view of the retainer shown in FIG. **137**.

[0140] FIG. **139** is a top plan view of the retainer shown in FIG. **137**.

[0141] FIG. **140** is a back elevation view of the retainer shown in FIG. **137**.

[0142] FIG. **141** is a cross-sectional view of the retainer shown in FIG. **139**, taken along line BD-BD.

[0143] FIG. **142** is a cross-section view of the retainer shown in FIG. **140**, taken along line BE-BE.

[0144] FIG. **143** is a back, top, left isometric view of another embodiment of a multi-piece fluid end.

[0145] FIG. **144** is a cross-sectional view of the multi-piece fluid end shown in FIG. **143**, taken along line BF-BF.

[0146] FIG. **145** is an enlarged view of area BG shown in FIG. **144**.

[0147] FIG. **146** is a cross-sectional view of the multi-piece fluid end shown in FIG. **143**, taken along line BH-BH.

[0148] FIG. **147** is a cross-sectional view of the multi-piece fluid end shown in FIG. **143**, taken along line BI-BI.

[0149] FIG. **148** is an exploded back, top, left, isometric view of a dynamic section of the multi-piece fluid end shown in FIG. **143**.

[0150] FIG. **149** is an exploded, back, top, left isometric view of the multi-piece fluid end shown in FIG. **143**.

[0151] FIG. **150** is an isometric view of a dynamic body used in the multi-piece fluid end shown in FIG. **143**.

[0152] FIG. **151** is a back, top, left isometric view of a dynamic body shown in FIG. **150**.

[0153] FIG. **152** is a cross-sectional view of the dynamic body shown in FIG. **151**, taken along line BJ-BJ.

[0154] FIG. **153** is a cross-sectional view of the dynamic body shown in FIG. **151**, taken along line BK-BK.

[0155] FIG. **154** is an isometric view of a retainer used in the multi-piece fluid end shown in FIG. **143**.

[0156] FIG. **155** is a back, top, left isometric view of the retainer shown in FIG. **154**.

[0157] FIG. **156** is a front elevation view of the retainer shown in FIG. **154**.

[0158] FIG. **157** is a cross-sectional view of the retainer shown in FIG. **156**, taken along line BL-BL.

[0159] FIG. **158** is a cross-sectional view of the retainer shown in FIG. **156**, taken along line BM-BM.

[0160] FIG. **159** is a cross-sectional view of the retainer shown in FIG. **156**, taken along line BN-BN.

[0161] FIG. **160** is an isometric view of a front plunger system wear ring used in the multi-piece fluid end shown in FIG. **143**.

[0162] FIG. **161** is a back, top, left isometric view of the front plunger system wear ring shown in FIG. **160**.

[0163] FIG. **162** is a front elevation view of the front plunger system wear ring shown in FIG. **160**.

[0164] FIG. **163** is a cross-sectional view of the front plunger system wear ring shown in FIG. **162**, taken along line BO-BO.

[0165] FIG. **164** is an isometric view of a back plunger system wear ring used in the multi-piece fluid end shown in FIG. **143**.

[0166] FIG. **165** is a back, top, left isometric view of the back plunger system wear ring shown in FIG. **164**.

[0167] FIG. **166** is a front elevation view of the back plunger system wear ring shown in FIG. **164**.

[0168] FIG. **167** is a cross-sectional view of the back plunger system wear ring shown in FIG. **166**, taken along line BP-BP.

[0169] FIG. **168** is an isometric view of a front lantern ring used in the multi-piece fluid end shown in FIG. **143**.

[0170] FIG. **169** is a back, top, left isometric view of the front lantern ring shown in FIG. **168**.

[0171] FIG. **170** is a front elevation view of the front lantern ring shown in FIG. **168**.

[0172] FIG. **171** is a cross-sectional view of the front lantern ring shown in FIG. **170**, taken along line BQ-BQ.

[0173] FIG. **172** is an isometric view of a back lantern ring used in the multi-piece fluid end shown in FIG. **143**.

[0174] FIG. **173** is a back, top, left isometric view of the back lantern ring shown in FIG. **172**.

[0175] FIG. **174** is a front elevation view of the back lantern ring shown in FIG. **172**.

[0176] FIG. **175** is a cross-sectional view of the back lantern ring shown in FIG. **174**, taken along line BR-BR.

[0177] FIG. **176** is an isometric view of another embodiment of a multi-piece fluid end.

[0178] FIG. **177** is a rear isometric view of the multi-piece fluid end shown in FIG. **176**.

[0179] FIG. **178** is a cross-sectional view of the multi-piece fluid end shown in FIG. **176**, taken along line BS-BS.

[0180] FIG. **179** is a cross-sectional view of the multi-piece fluid end shown in FIG. **176**, taken along line BT-BT.

[0181] FIG. **180** is an enlarged view of area BU shown in FIG. **178**.

[0182] FIG. **181** is an enlarged view of area BV shown in FIG. **178**.

[0183] FIG. **182** is an enlarged view of area BW shown in FIG. **181**.

[0184] FIG. **183** is an isometric view of a discharge valve used in the multi-piece fluid end shown in FIG. **176**.

[0185] FIG. **184** is a rear isometric view of the discharge valve shown in FIG. **183**.

[0186] FIG. **185** is a cross-sectional view of the discharge valve shown in FIG. **183**, taken along line BX-BX.

[0187] FIG. **186** is an isometric view of a fluid routing plug used in the multi-piece fluid end shown in FIG. **176**.

[0188] FIG. **187** is a front elevation view of the fluid routing plug shown in FIG. **186**.

[0189] FIG. **188** is a rear isometric view of the fluid routing plug shown in FIG. **186**.

[0190] FIG. **189** is a rear elevation view of the fluid routing plug shown in FIG. **186**.

[0191] FIG. **190** is a cross-sectional view of the fluid routing plug shown in FIG. **187**, taken along line BY-BY.

[0192] FIG. **191** is an isometric view of a suction valve used in the multi-piece fluid end shown in FIG. **176**.

[0193] FIG. **192** is a rear isometric view of the suction valve shown in FIG. **191**.

[0194] FIG. **193** is a cross-sectional view of the suction valve shown in FIG. **191**, taken along line BZ-BZ.

[0195] FIG. **194** is an isometric view of another embodiment of a multi-piece fluid end.

[0196] FIG. **195** is a rear isometric view of the multi-piece fluid end shown in FIG. **194**.

[0197] FIG. **196** is a cross-sectional view of the multi-piece fluid end shown in FIG. **194**, taken along line CA-CA.

[0198] FIG. **197** is a cross-sectional view of the multi-piece fluid end shown in FIG. **194**, taken along line CB-CB.

[0199] FIG. **198** is an enlarged view of area CC shown in FIG. **196**.

[0200] FIG. **199** is an enlarged view of area CD shown in FIG. **196**.

[0201] FIG. **200** is an isometric view of a discharge valve used in the multi-piece fluid end shown in FIG. **194**.

[0202] FIG. **201** is a rear isometric view of the discharge valve shown in FIG. **200**.

[0203] FIG. **202** is a cross-sectional view of the discharge valve shown in FIG. **200**, taken along line CE-CE.

[0204] FIG. **203** is an isometric view of another embodiment of a multi-piece fluid end.

[0205] FIG. **204** is a rear isometric view of the multi-piece fluid end shown in FIG. **203**.

[0206] FIG. **205** is a cross-sectional view of the multi-piece fluid end shown in FIG. **203**, taken along line CF-CF.

[0207] FIG. **206** is an offset cross-sectional view of the multi-piece fluid end shown in FIG. **203**, taken along line CG-CG.

[0208] FIG. **207** is an enlarged view of area CH shown in FIG. **205**.

[0209] FIG. **208** is an enlarged view of area CI shown in FIG. **205**.

[0210] FIG. **209** is an enlarged view of area CJ shown in FIG. **206**.

[0211] FIG. **210** is an isometric view of a discharge valve used in the multi-piece fluid end shown in FIG. **203**.

[0212] FIG. **211** is a rear isometric view of the discharge valve shown in FIG. **210**.

[0213] FIG. **212** is a cross-sectional view of the discharge valve shown in FIG. **210**, taken along line CK-CK.

[0214] FIG. **213** is an isometric view of a fluid routing plug used in the multi-piece fluid end shown in FIG. **203**.

[0215] FIG. **214** is a front elevation view of the fluid routing plug shown in FIG. **213**.

[0216] FIG. **215** is a rear isometric view of the fluid routing plug shown in FIG. **213**.

[0217] FIG. **216** is a rear elevation view of the fluid routing plug shown in FIG. **213**.

[0218] FIG. **217** is a cross-sectional view of the fluid routing plug shown in FIG. **214**, taken along

line CL-CL.

[0219] FIG. **218** is a cross-sectional view of the fluid routing plug shown in FIG. **214**, taken along line CL-CL. This view illustrates the relationship between discharge fluid passage circle and the counterbore of the discharge surface.

[0220] FIG. **219** is a cross-sectional view of the multi-piece fluid end shown in FIG. **203**, taken along line CF-CF. The multi-piece fluid end is shown with the installation tool being utilized to insert the fluid routing plug.

[0221] FIG. **220** is an enlarged view of area CM shown in FIG. **219**

[0222] FIG. **221** is an isometric view of the installation tool used in the assembly of the multi-piece fluid end shown in FIG. **203**.

[0223] FIG. **222** is a rear isometric view of the installation tool shown in FIG. **221**.

[0224] FIG. **223** is a cross-sectional view of the installation tool shown in FIG. **221**, taken along line CN-CN.

[0225] FIG. **224** is an isometric view of another embodiment of a multi-piece fluid end.

[0226] FIG. **225** is a rear isometric view of the multi-piece fluid end shown in FIG. **224**.

[0227] FIG. **226** is a cross-sectional view of the multi-piece fluid end shown in FIG. **224**, taken along line CO-CO.

[0228] FIG. **227** is an offset cross-sectional view of the multi-piece fluid end shown in FIG. **224**, taken along line CP-CP.

[0229] FIG. **228** is a cross-sectional view of the multi-piece fluid end shown in FIG. **224**, taken along line CQ-CQ.

[0230] FIG. **229** is an enlarged view of area CR shown in FIG. **226**.

[0231] FIG. **230** is an enlarged view of area CS shown in FIG. **226**.

[0232] FIG. **231** is an enlarged view of area CT shown in FIG. **227**.

[0233] FIG. **232** is an enlarged view of area CU shown in FIG. **228**.

[0234] FIG. **233** is an isometric view of a discharge plug used in the multi-piece fluid end shown in FIG. **224**.

[0235] FIG. **234** is a rear isometric view of the discharge plug shown in FIG. **233**.

[0236] FIG. **235** is a cross-sectional view of the discharge plug shown in FIG. **233** taken along line CV-CV.

[0237] FIG. **236** is an enlarged view of area CW shown in FIG. **235**.

[0238] FIG. **237** is an isometric view of a discharge valve used in the multi-piece fluid end shown in FIG. **224**.

[0239] FIG. **238** is a cross-sectional view of the discharge valve shown in FIG. **237**, taken along line CX-CX.

[0240] FIG. **239** is an isometric view of a fluid routing plug used in the multi-piece fluid end shown in FIG. **224**.

[0241] FIG. **240** is a front elevation view of the fluid routing plug shown in FIG. **239**.

[0242] FIG. **241** is a rear isometric view of the fluid routing plug shown in FIG. **239**.

[0243] FIG. **242** is a rear elevation view of the fluid routing plug shown in FIG. **239**.

[0244] FIG. **243** is a cross-sectional view of the fluid routing plug shown in FIG. **240**, taken along line CY-CY.

[0245] FIG. **244** is a cross-sectional view of the fluid routing plug shown in FIG. **240**, taken along line CY-CY. This view illustrates the relationship between discharge fluid passage circle and the counterbore of the discharge surface.

DETAILED DESCRIPTION

[0246] This patent application describes an apparatus that simplifies the assembly, disassembly, and maintenance of a high-pressure pump shown in the figures. The design of the pump reduces wear, and transfers forces away from hard-to-replace parts and weak points of the pump. The application also describes a method for using the apparatus. The application further describes additional

embodiments of such a high-pressure pump and components which may aid in its assembly, maintenance, and repair, as shown in the following figures and paragraphs. Features of the pump, shown, reduce and transfer wear in advantageous ways, resulting in better life and performance. [0247] In general, the figures show improvements of a patented pump design shown in U.S. Pat. No. 11,346,339, issued to Nowell, et. al., and U.S. Pat. No. 12,018,662, issued to Keith, et. al., the contents of each of the foregoing applications are incorporated by reference herein.

[0248] However, while the pumps of the incorporated references are effective, the “in-line” nature of the fluid end of the pump has led to inventive features and designs which continue to improve operation. In particular, the pump of the following figures includes what is referred to herein as a “dynamic section.” This phrase, generally, refers to the portion of the fluid-handling system of the pump's fluid end in which the plunger operates. This section is “dynamic” because fluid enters this section at a lower pressure, from a suction manifold, and is pressurized while forced across a fluid routing plug by an extension of a plunger, past a discharge valve. Once past the discharge valve, the fluid is now in a high pressure environment, and may exit at a fluid manifold.

[0249] Thus, in the “static section”, as defined below, pressures are always generally the same—high pressure exists once fluid passes the discharge valve and enters the discharge conduits to exit into the discharge manifold. Low pressure exists around the intermediate section of a fluid routing plug where fluid enters. These areas are well-sealed and pressure fluctuations are minor. However, in the “dynamic section”, the retraction and extension of the plunger causes repeated, dramatic pressure changes.

[0250] As a result, the connection between the “static section” and “dynamic section” is critical, as it may be a point at which wear can become evident and failures to valves and seals may occur. In addition, the manner in which components within the dynamic section are accessible is of the utmost importance, as the proper maintenance and replacement of worn parts will prevent the wear or failure of larger components.

[0251] To that end, this disclosure introduces a threaded dynamic section. Each threaded dynamic section is joined to the static sections by external threads about the periphery of the dynamic section, rather than being bolted to the static section. Various preloading methods and thread designs may be utilized to reduce stress concentrations and preserve thread life on these threads.

[0252] In addition, the dynamic section has a shoulder or nose at which a fluid routing plug, and the static section, seats. At this point, hardened inserts of various geometries are disclosed that may bear repeated compression force and thus improve the life of components, and various sealing methods may be utilized to address the potential for high pressure, abrasive fluid to create a failure point. These hardened inserts may be made of carbide, but the use of tool steel or another hardened metal alloy may also be utilized.

[0253] Various retaining mechanisms are described to limit movement and wear such that the life of the pump is extended and wear is transferred to easily-replaced components. Specifically, a discharge plug on the static section may allow access to many components for inspection, removal, and replacement.

[0254] High pressure fluid leaving the static section may benefit from symmetrical paths, and a discharge manifold having robust design and both a top and bottom portion is provided to eliminate wear to the static section and outlets caused by a single route.

[0255] Tools and procedures to install components, such as the fluid routing plug, should be used which prevent stress to critical points within the system. Such tools and procedures are inventive and described herein.

[0256] Dual guide valves and passages are used to clear material trapped within the pump. Such improvements reduce misalignment of valves due to gravity and the resulting uneven wear. Likewise, the components are utilized to promote laminar flow and eliminate or reduce direct impact (or its effects) of fluid flow on key components, such as the fluid routing plug, discharge valve face, flow bore, discharge plug, and suction valve guide.

[0257] High-pressure pump **100** is shown in FIG. 1. The high-pressure pump **100** comprises a power end **101** connected to a multi-piece fluid end **102**. The multi-piece fluid end **102** shown in FIGS. 1-7, comprises a fluid end body **103** and a plurality of plunger systems **104**. The fluid end body **103** comprises a static section **105** and a plurality of dynamic sections **106**. Each dynamic section **106** comprises a dynamic body **107**, a plunger system wear ring **108**, a plunger system wear ring seal **109**, a flow control system wear ring **110**, and a flow control system wear ring seal **111**. Each plunger system **104** comprises a retainer seal **112**, a retainer **113**, a plurality of tension bolts **114**, packing **115**, a packing nut **116**, a plunger seal **117**, and a plunger **118**.

[0258] The dynamic body **107**, shown in FIGS. 8-10, has a cylindrical shape. The dynamic body **107** comprises opposing front **119** and back **120** surfaces connected by an outer surface **121** and a flow bore **122**. The outer surface **121** and flow bore **122** are concentric and their cylindrical axis is the longitudinal axis of the dynamic body **107**.

[0259] The outer surface **121** comprises multiple sections, all the sections are concentric. Beginning at the front surface **119** of the dynamic body **107** and continuing along the longitudinal axis to the back surface **120** the outer surface **121** comprises a static seal section **123**, static threads **124**, intermediate section **125**, and retainer threads **126**. The intermediate section **125** comprises a plurality of spanner wrench holes **127**. The spanner wrench holes **127** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **107**. Each spanner wrench hole **127** originates from the intermediate section **125** of the outer surface **121** but does not intersect the flow bore **122**. In this embodiment the spanner wrench holes **127** are proximate the retainer threads **126**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the intermediate section **125** as long as access for the spanner wrench (not shown) is available.

[0260] The flow bore **122** also comprises multiple sections and is configured to receive the plunger system wear ring seal **109**, plunger system wear ring **108**, flow control system wear ring seal in, and flow control system wear ring **110**. The flow bore **122** is not the focus of this improvement so no further details about the flow bore **122** are necessary to understand this disclosure.

[0261] The retainer **113**, shown in FIGS. 11-16, has a generally cylindrical shape. The retainer **113** comprises a front surface **128** and back surface **129** connected by an outer surface **130** and plunger bore **131**. The outer surface **130** and plunger bore **131** are concentric and their cylindrical axis is the longitudinal axis of the retainer **113**.

[0262] The outer surface **130** comprises a straight section **132** and a tapered section **133**. The straight section **132** begins at the front surface **128** and continues along the longitudinal axis until it connects with the tapered section **133**. The straight section **132** occupies approximately 80% of the outer surface **130** but may be more or less. The straight section **132** comprises a front chamfer **134**, a lubrication port **135**, and a plurality of spanner wrench holes **136**. The lubrication port **135** is a through bore connecting the straight section **132** of the outer surface **130** to the intermediate section **140** of the plunger bore **131**. The lubrication port **135** comprises a threaded section **137** adjacent the outer surface **130** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0263] The spanner wrench holes **136** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the retainer **113**. Each spanner wrench hole **136** originates from the straight section **132** of the outer surface **130** but does not intersect the plunger bore **131**. In this embodiment the spanner wrench holes **136** are proximate the tapered section **133**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the straight section **132** as long as access for the spanner wrench (not shown) is available.

[0264] The plunger bore **131** comprises multiple sections. Beginning at the front surface **128** and continuing along the longitudinal axis to the back surface **129** the plunger bore **131** comprises dynamic threads **138**, a dynamic shoulder **139**, an intermediate section **140**, a packing nut shoulder **141**, and packing nut threads **142**. The dynamic shoulder **139** comprises a seal groove **143**. The seal groove **143** is circular and concentric with the plunger bore **131**. The seal groove **143** is configured

to receive the retainer seal **112**.

[0265] The retainer **113** further comprises a plurality of tension bolt holes **144**. Each tension bolt hole **144** is a partially threaded through hole originating on the back surface **129** and terminating at the dynamic shoulder **139**. The threaded portion of the tension bolt hole **144** extends from the dynamic shoulder **139** approximately half the tension bolt hole **144** length to the back surface **129** and is configured to receive a tension bolt **114**. The tension bolt holes **144** are distributed evenly on a bolt circle that is concentric with the retainer **113**.

[0266] Referring now to FIGS. 4-7, the assembly of the multi-piece fluid end **102** begins installation of a flow control system wear ring seal **111**, a flow control system wear ring **110**, a plunger system wear ring seal **109**, and a plunger system wear ring **108** into each dynamic body **107** completing the assembly of the dynamic sections **106**. Each dynamic section **106** is then attached to the static section **105** by threading the static threads **124** of the outer surface **121** of the dynamic body **107** into the matching threads of the static section **105**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **127** in the intermediate section **125** of the outer surface **121** of the dynamic body **107**. Finally the flow control components are installed from the other side of the static section.

[0267] Next, for each dynamic section **106**, a retainer seal **112** is installed in the seal groove **143** of the dynamic shoulder **139** of the plunger bore **131** of a retainer **113**. Then, the retainer **113** is attached to a dynamic section **106** by threading the dynamic threads **138** of the plunger bore **131** of the retainer **113** onto the retainer threads **126** of the outer surface **121** of the dynamic body **107**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **136** in the straight section **132** of the outer surface **130** of the retainer **113**. The plurality of tension bolts **114** are then inserted, on a one-to-one basis, into the tension bolt holes **144** from the back surface **129** of the retainer **113** and threaded into the threaded section of the tension bolt holes **144** until the front surface of each tension bolt **114** contacts the back surface **120** of the dynamic body **107**. Once contact is made by all tension bolts **114** the tension bolts **114** may be torqued to specification in a piecewise manner.

[0268] One example of a piecewise manner is applying half the specified torque to a pair of diametrically opposed tension bolts **114** then applying half the specified torque to a second pair of diametrically opposed tension bolts **114** spaced 90 degrees from the first pair. Then applying half the specified torque to the remaining two pairs in a similar manner. After applying half the specified torque to all the tension bolts **114**, the full specified torque may be applied in the same manner.

[0269] Next, the packing **115** is inserted into the dynamic section **106** and the intermediate section **140** of the plunger bore **131** of the retainer **113**. Next, the plunger seal **117** is installed in the packing nut **116** and the packing nut **116** is threaded into the packing nut thread **142** of the plunger bore **131** of the retainer **113**. Next, the plunger **118** is installed in the packing nut **116**, packing **115**, and dynamic section **106**. Lastly, the packing nut **116** is torqued to specification.

[0270] In operation the tension bolts **114** place the threaded joint formed by the retainer threads **126** of the dynamic body **107** and the dynamic threads **138** of the retainer **113** in tension providing an additional tensile load above that produced by the torquing of the threads together. This additional tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **106**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. These benefits result in an easier and quicker removal of the retainer **113** for maintenance or replacement when necessary.

[0271] Referring now to FIGS. 17-31, another embodiment of a multi-piece fluid end **202** is shown. The multi-piece fluid end **202**, shown in FIGS. 17-21, comprises a fluid end body **203** and a plurality of plunger systems **204**. The fluid end body **203** comprises a static section **205**, a plurality of dynamic sections **206**. Each dynamic section **206** comprises a dynamic body **207**, a plunger

system wear ring **208**, a plunger system wear ring seal **209**, a flow control system wear ring **210**, and a flow control system wear ring seal **211**. Each plunger system **204** comprises a retainer seal **212**, a retainer **213**, a plurality of tension bolts **214**, packing **215**, a packing nut **216**, a plunger seal **217**, and a plunger **218**.

[0272] The fluid end body **203** may be identical to the fluid end body **103**. The retainer **213**, shown in FIGS. **22-25**, has a generally cylindrical shape. The retainer **213** comprises a front surface **228** and back surface **229** connected by an outer surface **230** and plunger bore **231**. The outer surface **230** and plunger bore **231** are concentric and their cylindrical axis is the longitudinal axis of the retainer **213**.

[0273] The outer surface **230** comprises a straight section **232** and a tapered section **233**. The straight section **232** begins at the front surface **228** and continues along the longitudinal axis until it connects with the tapered section **233**. The straight section **232** occupies approximately 80% of the outer surface **230** but may be more or less. The straight section **232** comprises a front chamfer **234** and a lubrication port **235**. The lubrication port **235** is a through bore connecting the straight section **232** of the outer surface **230** to the intermediate section **240** of the plunger bore **231**. The lubrication port **235** comprises a threaded section **237** adjacent to the outer surface **230** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0274] The plunger bore **231** comprises multiple sections. Beginning at the front surface **228** and continuing along the longitudinal axis to the back surface **229** the plunger bore **231** comprises dynamic threads **238**, a dynamic shoulder **239**, an intermediate section **240**, a packing nut shoulder **241**, and packing nut threads **242**. The dynamic shoulder **239** comprises a seal groove **243**. The seal groove **243** is circular and concentric with the plunger bore **231**. The seal groove **243** is configured to receive the retainer seal **212**.

[0275] The packing nut **216**, shown in FIGS. **26-31**, is generally cylindrical in shape and comprises a front surface **244** and back surface **245** connected by an outer surface **246**, and plunger bore **247**. The outer surface **246** and plunger bore **247** are concentric and their cylindrical axis is the longitudinal axis of the packing nut **216**. The plunger bore **247** comprises a plunger seal groove **257** configured to receive the plunger seal **217**.

[0276] The outer surface **246** comprises multiple sections, all the sections are concentric. Beginning at the front surface **244** of the packing nut **216** and continuing along the longitudinal axis to the back surface **245** the outer surface **246** comprises a packing nose **248**, a front shoulder **249**, a threaded section **250**, a back shoulder **251**, straight spanner wrench section **252**, and tapered spanner wrench section **253**.

[0277] The straight spanner wrench section **252** comprises a plurality of spanner wrench holes **254**. The spanner wrench holes **254** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the packing nut **216**. Each spanner wrench hole **254** originates from the straight spanner wrench section **252** of the outer surface **246** but does not intersect the plunger bore **247**. In this embodiment the spanner wrench holes **254** are approximately centered longitudinally in the straight spanner wrench section **252** and evenly spaced circumferentially but may be spaced in any manner on the straight spanner wrench section **252** as long as access for the spanner wrench (not shown) is available.

[0278] The tapered spanner wrench section **253** comprises a plurality of spanner wrench holes **255**. The spanner wrench holes **255** are radial through bores with a bore axis that is perpendicular to the tapered face of the tapered spanner wrench section **253**. Each spanner wrench hole **255** originates from the tapered spanner wrench section **253** of the outer surface **246** and intersects the plunger bore **247**. In this embodiment the spanner wrench holes **255** are approximately centered longitudinally in the tapered spanner wrench section **253**. The spanner wrench holes **255** are evenly spaced circumferentially but offset circumferentially from the spanner wrench holes **254** of the straight spanner wrench section **252** to allow greater access by a spanner wrench (not shown). The spanner wrench holes **255**, however, may be spaced in any manner on the tapered spanner wrench

section **253** as long as access for the spanner wrench (not shown) is available.

[0279] The packing nut **216** further comprises a plurality of tension bolt holes **256**. Each tension bolt hole **256** is a partially threaded through hole connecting the front shoulder **249** of the outer surface **246** and the back shoulder **251** of the outer surface **246**. The threaded portion of the tension bolt hole **256** extends from the front shoulder **249** approximately half the tension bolt hole **256** length to the back shoulder **251** and is configured to receive a tension bolt **214**. The tension bolt holes **256** are distributed evenly on a bolt circle that is concentric with the packing nut **216**.

[0280] Referring now to FIGS. **19-21**, the assembly of the multi-piece fluid end **202** begins with the assembling the dynamic sections **206** which are then attached to the static section **205** as described for dynamic sections **106** and static section **105** above. Then the flow control components are installed from the other side of the static section.

[0281] Next, for each dynamic section **206**, the retainer seal **212** is installed in the seal groove **243** of the dynamic shoulder **239** of the plunger bore **231** of the retainer **213**. Then, the retainer **213** is attached to the dynamic section **206** by threading the dynamic threads **238** of the plunger bore **231** of the retainer **213** onto the dynamic body **207**.

[0282] Next, the packing **215** is inserted into the dynamic section **206** and the intermediate section **240** of the plunger bore **231** of the retainer **213**. Next, the plunger seal **217** is installed in plunger seal groove **257** of the packing nut **216** and the packing nut **216** is threaded into the packing nut thread **242** of the plunger bore **231** of the retainer **213**. Next, the plunger **218** is installed in the packing nut **216**, packing **215**, and dynamic section **206**. Next, the packing nut **216** is torqued to specification. Lastly the plurality of tension bolts **214** are then inserted, on a one-to-one basis, into the tension bolt holes **256** from the back shoulder **251** of the packing nut **216** and threaded into the threaded section of the tension bolt holes **256** until the front surface of each tension bolt **214** contacts the packing nut shoulder **241** of the plunger bore **231** of the retainer **213**. Once contact is made by all tension bolts **214**, the tension bolts **214** may be torqued to specification in a piecewise manner.

[0283] In operation the tension bolts **214** place the threaded joint formed by the packing nut threads **242** of the plunger bore **231** of the retainer **213** and threaded section **250** of the packing nut **216** in tension providing an additional tensile load above that produced by the torquing of the threads together. This additional tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **206**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. These benefits result in an easier and quicker removal of the packing nut **216** for maintenance or replacement when necessary.

[0284] Referring now to FIGS. **32-53**, another embodiment of a multi-piece fluid end **302** is shown. The multi-piece fluid end **302** comprises a fluid end body **303** and a plurality of plunger systems **304**. The fluid end body **303** comprises a static section **305** and a plurality of dynamic sections **306**. Each dynamic section **306** comprises a dynamic body **307**, a plunger system wear ring **308**, a plunger system wear ring seal **309**, a flow control system wear ring **310**, a flow control system wear ring seal **311**, and a centering ring **358** as shown in FIG. **37**. Each plunger system **304** comprises a retainer seal **312**, a retainer **313**, a plurality of tension bolts **314**, packing **315**, a packing nut **316**, a plunger seal **317**, and a plunger **318**, as shown in FIG. **36**.

[0285] The dynamic body **307**, shown in FIGS. **39-42**, has a cylindrical shape. The dynamic body **307** comprises opposing front **319** and back **320** surfaces connected by an outer surface **321** and a flow bore **322**. The outer surface **321** and flow bore **322** are concentric and their cylindrical axis is the longitudinal axis of the dynamic body **307**.

[0286] The outer surface **321**, as shown in FIG. **42**, comprises multiple concentric sections. Beginning at the front surface **319** of the dynamic body **307** and continuing along the longitudinal axis to the back surface **320** the outer surface **321** comprises a static seal section **323**, static threads **324**, locating shoulder **359**, spanner section **360**, transition taper **361**, retainer section **362**, retainer

threads **326**, centering ring shoulder **363**, and centering ring section **364**. The spanner section **360** comprises a plurality of spanner wrench holes **327**. The spanner wrench holes **327** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **307**. Each spanner wrench hole **327** originates from the spanner section **360** of the outer surface **321** but does not intersect the flow bore **322**. In this embodiment the spanner wrench holes **327** are approximately centered and aligned longitudinally in the spanner section **360**, the spanner wrench holes **327** are also evenly spaced circumferentially but may be spaced in any manner on the spanner section **360** as long as access for the spanner wrench (not shown) is available.

[0287] The flow bore **322** also comprises multiple sections and is configured to receive the plunger system wear ring seal **309**, plunger system wear ring **308**, flow control system wear ring seal **311**, and flow control system wear ring **310**. The flow bore **322** is not the focus of this improvement so no further details about the flow bore **322** are necessary to understand this disclosure.

[0288] The retainer **313**, shown in FIGS. **43-49**, has a generally cylindrical shape. The retainer **313** comprises a front surface **328** and back surface **329** connected by an outer surface **330** and plunger bore **331**. The outer surface **330** and plunger bore **331** are concentric and their cylindrical axis is the longitudinal axis of the retainer **313**.

[0289] The outer surface **330** comprises a straight section **332** and a tapered section **333**. The straight section **332** begins at the front surface **328** and continues along the longitudinal axis until it connects with the tapered section **333**. The straight section **332** occupies approximately 60% of the outer surface **330** but may be more or less. The straight section **332** comprises a front chamfer **334** and a lubrication port **335**. The lubrication port **335** is a through bore connecting the straight section **332** of the outer surface **330** to the intermediate section **340** of the plunger bore **331**. The lubrication port **335** comprises a threaded section **337** adjacent the outer surface **330** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0290] The tapered section **333** comprises a plurality of spanner wrench holes **336**. The spanner wrench holes **336** are radial bores with a bore axis that is perpendicular to the tapered face of the tapered section **333**. Each spanner wrench hole **336** originates from the tapered section **333** of the outer surface **330** and does not intersect the plunger bore **331**. In this embodiment the spanner wrench holes **336** are proximate the straight section **332** longitudinally. The spanner wrench holes **336** are evenly spaced circumferentially, however, they may be spaced in any manner on the tapered section **333** as long as access for the spanner wrench (not shown) is available.

[0291] The plunger bore **331** comprises multiple sections, as shown in FIG. **48**. Beginning at the front surface **328** and continuing along the longitudinal axis to the back surface **329** the plunger bore **331** comprises dynamic threads **338**, a centering ring section **365**, a dynamic shoulder **339**, an intermediate section **340**, a packing nut shoulder **341**, and packing nut threads **342**. The dynamic shoulder **339** comprises a seal groove **343**. The seal groove **343** is circular and concentric with the plunger bore **331**. The seal groove **343** is configured to receive the retainer seal **312**.

[0292] The dynamic threads **338** comprise a thread relief **366** proximate the centering ring section **365**. Referring now to FIG. **49**, the thread relief **366** comprises a plurality of straight sections, straight section one **367**, straight section two **368**, and straight section three **369**. The thread relief **366** further comprises a plurality of curved sections, curved section one **370**, curved section two **371**, and curved section three **372**. Straight section one **367** extends radially from the last thread. Since the threads are angled, no thread face is perpendicular to the longitudinal axis of the retainer **313**. Straight section one **367** makes a planar surface at the end of the threads. Curved section one **370** is concave and provides a transition from straight section one **367** to straight section two **368**. Straight section two **368** is parallel with the longitudinal axis of the retainer **313** and provides a transition from curved section one **370** to curved section two **371** and the ability to extend the thread relief **366** if desired. Curved section two **371** is concave and provides a transition from straight section two **368** to straight section three **369**. Straight section three **369** is angled relative to the longitudinal axis and provides a transition from curved section two **371** to curved section three

372. Curved section three **372** is convex and provides a transition from straight section three **369** to the centering ring section **365** of the plunger bore **331**.

[0293] In this embodiment curved section two **371** has a larger radius than curved section one **370** but they may be the same or the radius of curved section one **370** may be larger than that of curved section two **371**. It is also possible that straight section two **368** may be eliminated and curved section one **370** may transition directly to curved section two **371**. In this embodiment straight section three **369** has an angle of approximately 45 degrees with the longitudinal axis however it may be any angle between 0 and 90 degrees. It is also contemplated that the curved sections **370**, **371**, and **372** may be spline curves with infinitely varying radii. If both curved section one **370** and curved section two **371** are spline curves and straight section two **368** is removed, then curved section one **370** and curved section two **371** may be considered a single curve.

[0294] The retainer **313** further comprises a plurality of tension bolt holes **344**. Each tension bolt hole **344** is a partially threaded through hole originating on the back surface **329** and terminating at the dynamic shoulder **339**. The threaded portion of the tension bolt hole **344** extends from the dynamic shoulder **339** approximately half the tension bolt hole **344** length to the back surface **329** and is configured to receive a tension bolt **314**. The tension bolt holes **344** are distributed evenly on a bolt circle that is concentric with the retainer **313**.

[0295] The centering ring **358**, shown in FIGS. **50-53**, is ring shaped and comprises a front surface **373** and back surface **374** connected by an outer surface **375** and an inner surface **376**. The outer surface **375** comprises a chamfer **377** proximate the back surface **374**. The front and back surfaces **373**, **374** are planar, parallel to each other, and perpendicular to the longitudinal axis. The inner surface **376** is parallel to the longitudinal axis and configured to receive the centering ring section **364** of the outer surface **321** of the dynamic body **307**. The outer surface **375**, with the exception of the chamfer **377**, is parallel to the longitudinal axis and configured to be received by the centering ring section **365** of the plunger bore **331** of the retainer **313**. The centering ring **358** is formed of a harder material than dynamic body **307**. The material may be carbide, tool steel, carburized **8620** steel or any of the many other harder materials well known in the art.

[0296] Referring now to FIGS. **33-38**, the assembly of the multi-piece fluid end **302** begins the installation of a flow control system wear ring seal **311**, a flow control system wear ring **310**, a plunger system wear ring seal **309**, and a plunger system wear ring **308** into each dynamic body **307**, as shown in FIG. **37**. Next a centering ring **358** is pressed over the centering ring section **364** of the outer surface **321** of each dynamic body **307** until the front surface **373** of the centering ring **358** abuts the centering ring shoulder **363** of the dynamic body **307**. This completes the assembly of the dynamic sections **306**, as shown in FIG. **36**.

[0297] Each dynamic section **306** is then attached to the static section **305** by threading the static threads **324** of the outer surface **321** of the dynamic body **307** into the matching threads of the static section **305** until the locating shoulder **359** abuts the static section **305**, as shown in FIG. **35**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **327** in the spanner section **360** of the outer surface **321** of the dynamic body **307**. The flow control components may now be installed front eh opposite side of the static section **305**.

[0298] Next, for each dynamic section **306**, a retainer seal **312** is installed in the seal groove **343** of the dynamic shoulder **339** of the plunger bore **331** of a retainer **313**. Then, the retainer **313** is attached to a dynamic section **306** by threading the dynamic threads **338** of the plunger bore **331** of the retainer **313** onto the retainer threads **326** of the outer surface **321** of the dynamic body **307**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **336** in the tapered section **333** of the outer surface **330** of the retainer **313**. As the retainer **313** is being threaded on the dynamic body **307** the chamfer **377** of the centering ring **358** will facilitate the insertion of the centering ring **358** into the centering ring section **365** of the plunger bore **331** of the retainer **313**, as shown in FIG. **36**. The outer surface **375** and centering ring section **365** may have a line-to-line, or a light interference fit, to maximize the longitudinal alignment between the

retainer **313** and the dynamic body **307**.

[0299] The plurality of tension bolts **314** are then inserted, on a one-to-one basis, into the tension bolt holes **344** from the back surface **329** of the retainer **313** and threaded into the threaded section of the tension bolt holes **344** until the front surface of each tension bolt **314** contacts the back surface **374** of the centering ring **358**. Once contact is made by all tension bolts **314** the tension bolts **314** may be torqued to specification in a piecewise manner. Since the tension bolts **314** engage the harder centering ring **358** there is no damage done to the dynamic body **307** allowing many assembly and disassembly cycles of the retainer **313** before having to consider the damage inflicted by the tension bolts **314** on the centering ring **358**. Once the damage to the centering ring **358** is considered intolerable the centering ring **358** may be replaced without replacing the much more expensive dynamic body **307**.

[0300] Next, the packing **315** is inserted into the dynamic section **306** and the intermediate section **340** of the plunger bore **331** of the retainer **313**. Next, the plunger seal **317** is installed in the packing nut **316** and the packing nut **316** is threaded into the packing nut thread **342** of the plunger bore **331** of the retainer **313**. Next, the plunger **318** is installed in the packing nut **316**, packing **315**, and dynamic section **306**. Lastly, the packing nut **316** is torqued to specification.

[0301] In operation the tension bolts **314** place the threaded joint formed by the retainer threads **326** of the dynamic body **307** and the dynamic threads **338** of the retainer **313** in tension providing an additional tensile load above that produced by the torquing of the threads together. This additional tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **306**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. Also, the retainer seal **312** protects the retainer threads **326** and dynamic threads **338** from contamination when the packing **315** fails. These benefits result in easier and quicker removal of the retainer **313** for maintenance or replacement when necessary. Also during operation, the thread relief **366** reduces stress in the retainer **313** at the transition from the dynamic threads **338** to the dynamic shoulder **339**.

[0302] Another embodiment of a high-pressure pump **400** is shown in FIG. **54**. The high-pressure pump **400** comprises a power end **401** connected to a fluid end **402**. The fluid end **402** shown in FIG. **55**, comprises a plurality of fluid end sections **478**. Each fluid end section **478** comprises a fluid end body **403** and a plunger system **404**. The fluid end body **403** comprises a static section **405** and a dynamic section **406**.

[0303] Each static section **405**, shown in FIGS. **58** and **59**, comprises a flow bore **479** and a back surface **480**. The flow bore **479** is configured to receive flow control components. The flow bore **479** comprises a section of dynamic threads **481** proximate the back surface **480** of the static section **405**. The dynamic threads **481** are configured to receive the static threads **424** of the dynamic section **406**.

[0304] Each dynamic section **406** comprises a dynamic body **407**, a plunger system wear ring **408**, a plunger system wear ring seal **409**, a flow control system wear ring **410**, and a flow control system wear ring seal **411**. Each plunger system **404** comprises a retainer seal **412**, a retainer **413**, a plurality of tension bolts **414**, packing **415**, a packing nut **416**, a plunger seal **417**, and a plunger **418**.

[0305] The dynamic body **407**, shown in FIGS. **61-63**, has a cylindrical shape. The dynamic body **407** comprises opposing front **419** and back **420** surfaces connected by an outer surface **421** and a flow bore **422**. The outer surface **421** and flow bore **422** are concentric and their cylindrical axis is the longitudinal axis of the dynamic body **407**.

[0306] The outer surface **421** comprises multiple sections, all the sections are concentric. Beginning at the front surface **419** of the dynamic body **407** and continuing along the longitudinal axis to the back surface **420** the outer surface **421** comprises a static seal section **423**, static threads **424**, locating shoulder **459**, intermediate section **425**, back shoulder **461**, and retainer threads **426**.

The intermediate section **425** comprises a plurality of spanner wrench holes **427**. The spanner wrench holes **427** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **407**. Each spanner wrench hole **427** originates from the intermediate section **425** of the outer surface **421** but does not intersect the flow bore **422**. In this embodiment the spanner wrench holes **427** are proximate the locating shoulder **459**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the intermediate section **425** as long as access for the spanner wrench (not shown) is available.

[0307] The flow bore **422** also comprises multiple sections and is configured to receive the plunger system wear ring seal **409**, plunger system wear ring **408**, flow control system wear ring seal **411**, and flow control system wear ring **410**. The flow bore **422** is not the focus of this improvement so no further details about the flow bore **422** are necessary to understand this disclosure.

[0308] The retainer **413**, shown in FIGS. **64-70**, has a generally cylindrical shape. The retainer **413** comprises a front surface **428** and back surface **429** connected by an outer surface **430** and plunger bore **431**. The outer surface **430** and plunger bore **431** are concentric and their cylindrical axis is the longitudinal axis of the retainer **413**.

[0309] The outer surface **430** comprises a straight section **432** and a tapered section **433**. The straight section **432** begins at the front surface **428** and continues along the longitudinal axis until it connects with the tapered section **433**. The straight section **432** occupies approximately 60% of the outer surface **430** but may be more or less. The straight section **432** comprises a front chamfer **434** and plurality of spanner wrench holes **436**.

[0310] The spanner wrench holes **436** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the retainer **413**. Each spanner wrench hole **436** originates from the straight section **432** of the outer surface **430** but does not intersect the plunger bore **431**. In this embodiment the spanner wrench holes **436** are proximate the tapered section **433**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the straight section **432** as long as access for the spanner wrench (not shown) is available.

[0311] The tapered section **433** comprises a lubrication port **435**. The lubrication port **435** is a radial through bore connecting the tapered section **433** of the outer surface **430** to the intermediate section **440** of the plunger bore **431**. The bore axis of the lubrication port **435** is perpendicular to the tapered face of the tapered section **433**. The lubrication port **435** comprises a threaded section **437** adjacent the outer surface **430** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0312] The plunger bore **431** comprises multiple sections, as shown in FIG. **69**. Beginning at the front surface **428** and continuing along the longitudinal axis to the back surface **429** the plunger bore **431** comprises dynamic threads **438**, a dynamic shoulder **439**, an intermediate section **440**, a packing nut shoulder **441**, and packing nut threads **442**. The dynamic shoulder **439** comprises a seal groove **443**. The seal groove **443** is circular and concentric with the plunger bore **431**. The seal groove **443** is configured to receive the retainer seal **412**.

[0313] The dynamic threads **438** comprise a thread relief **466** proximate the dynamic shoulder **439**. Referring now to FIG. **70**, the thread relief **466** comprises a plurality of straight sections, straight section one **468** and straight section two **469**. The thread relief **466** further comprises a plurality of curved sections, curved section one **470**, curved section two **471**, and curved section three **472**. Curved section one **470** is concave and provides a transition from the last thread of the dynamic threads **438** to straight section one **468**. Straight section one **468** is parallel with the longitudinal axis of the retainer **413** and provides a transition from curved section one **470** to curved section two **471** and the ability to extend the thread relief **466** if desired. Curved section two **471** is concave and provides a transition from straight section one **468** to straight section two **469**. Straight section two **469** is angled relative to the longitudinal axis and provides a transition from curved section two **471** to curved section three **472**. Curved section three **472** is concave and provides a transition from straight section two **469** to the dynamic shoulder **439** of the plunger bore **431**.

[0314] In this embodiment curved sections two **471** and three **472** have equal radii that are larger than the radius of curved section one **470** but the radii of the curved sections **470**, **471**, and **472** may be any value that still allows the geometric construction of the thread relief **466**. It is also possible that straight section one **468** may be eliminated and curved section one **470** may transition directly to curved section two **471**. In this embodiment straight section two **469** has an angle of approximately 35 degrees with the longitudinal axis however it may be any angle between 0 and 90 degrees. It is also contemplated that the curved sections **470**, **471**, and **472** may be spline curves with infinitely varying radii. If both curved section one **470** and curved section two **471** are spline curves and straight section one **468** is removed, then curved section one **470** and curved section two **471** may be considered a single curve. Also, if both curved section two **471** and curved section three **472** are spline curves and straight section two **469** is removed then curved section two **471** and curved section three **472** may be considered a single curve.

[0315] The retainer **413** further comprises a plurality of tension bolt holes **444**. Each tension bolt hole **444** is a partially threaded through hole originating on the back surface **429** and terminating at the dynamic shoulder **439**. The threaded portion of the tension bolt hole **444** extends from the dynamic shoulder **439** approximately half the tension bolt hole **444** length to the back surface **429** and is configured to receive a tension bolt **414**. The tension bolt holes **444** are distributed evenly on a bolt circle that is concentric with the retainer **413**.

[0316] Referring now to FIGS. **56-60**, the assembly of a fluid end section **478** begins with the installation of a flow control system wear ring seal **411**, a flow control system wear ring **410**, a plunger system wear ring seal **409**, and a plunger system wear ring **408** into the dynamic body **407**, as shown in FIG. **58**. This completes the assembly of the dynamic section **406**.

[0317] The dynamic section **406** is then attached to the static section **405** by threading the static threads **424** of the outer surface **421** of the dynamic body **407** into the dynamic threads **481** of the static section **405** until the locating shoulder **459** abuts the back surface **480** of the static section **405**, as shown in FIG. **58**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **427** in the intermediate section **425** of the outer surface **421** of the dynamic body **407**. The flow control components may now be installed in the flow bore **479** from the opposite side of the static section **405**.

[0318] Next, a retainer seal **412** is installed in the seal groove **443** of the dynamic shoulder **439** of the plunger bore **431** of a retainer **413**. Then, the retainer **413** is attached to a dynamic section **406** by threading the dynamic threads **438** of the plunger bore **431** of the retainer **413** onto the retainer threads **426** of the outer surface **421** of the dynamic body **407**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **436** in the straight section **432** of the outer surface **430** of the retainer **413**.

[0319] The plurality of tension bolts **414** are then inserted, on a one-to-one basis, into the tension bolt holes **444** from the back surface **429** of the retainer **413** and threaded into the threaded section of the tension bolt holes **444** until the front surface of each tension bolt **414** contacts the back surface **420** of the dynamic body **407**, as shown in FIG. **59**. Once contact is made by all tension bolts **414** the tension bolts **414** may be torqued to specification in a piecewise manner.

[0320] Next, the packing **415** is inserted into the dynamic section **406** and the intermediate section **440** of the plunger bore **431** of the retainer **413**. Next, the plunger seal **417** is installed in the packing nut **416** and the packing nut **416** is threaded into the packing nut threads **442** of the plunger bore **431** of the retainer **413**. Next, the plunger **418** is installed in the packing nut **416**, packing **415**, and dynamic section **406**. Lastly, the packing nut **416** is torqued to specification.

[0321] In operation the tension bolts **414** place the threaded joint formed by the retainer threads **426** of the dynamic body **407** and the dynamic threads **438** of the retainer **413** in tension providing an additional tensile load above that produced by the torquing of the threads together. This additional tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **406**. This reduction in the magnitude of cyclic

loading reduces stress to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. Also, the retainer seal **412** protects the retainer threads **426** and dynamic threads **438** from contamination when the packing **415** fails. These benefits result in easier and quicker removal of the retainer **413** for maintenance or replacement when necessary. Also during operation, the thread relief **466** reduces stress in the retainer **413** at the transition from the dynamic threads **438** to the dynamic shoulder **439**.

[0322] Referring now to FIGS. **71-87**, another embodiment of a multi-piece fluid end **502** is shown. The multi-piece fluid end **502** comprises a fluid end body **503** and a plurality of plunger systems **504**. The fluid end body **503** comprises a static section **505** and a plurality of dynamic sections **506**. Each dynamic section **506** comprises a dynamic body **507**, a plunger system wear ring **508**, a plunger system wear ring seal **509**, a flow control system wear ring **510**, and a flow control system wear ring seal **511** as shown in FIG. **72**. Each plunger system **504** comprises a retainer seal **512**, a retainer assembly **513**, a plurality of tension bolts **514**, packing **515**, a packing nut **516**, a plunger seal **517**, and a plunger **518**, as shown in FIG. **73**.

[0323] The dynamic body **507**, shown in FIGS. **75-77**, has a cylindrical shape. The dynamic body **507** comprises opposing front **519** and back **520** surfaces connected by an outer surface **521** and a flow bore **522**. The outer surface **521** and flow bore **522** are concentric and their cylindrical axis is the longitudinal axis of the dynamic body **507**.

[0324] The outer surface **521**, as shown in FIG. **77**, comprises multiple concentric sections. Beginning at the front surface **519** of the dynamic body **507** and continuing along the longitudinal axis to the back surface **520** the outer surface **521** comprises a static seal section **523**, static threads **524**, intermediate section **525**, retainer threads **526**, and a pilot section **582**. The intermediate section **525** comprises a plurality of spanner wrench holes **527**. The spanner wrench holes **527** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **507**. Each spanner wrench hole **527** originates from the intermediate section **525** of the outer surface **521** but does not intersect the flow bore **522**. In this embodiment the spanner wrench holes **527** are proximate the retainer threads **526**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the intermediate section **525** as long as access for the spanner wrench (not shown) is available.

[0325] The flow bore **522** also comprises multiple sections and is configured to receive the plunger system wear ring seal **509**, plunger system wear ring **508**, flow control system wear ring seal **511**, and flow control system wear ring **510**. The flow bore **522** is not the focus of this improvement so no further details about the flow bore **522** are necessary to understand this disclosure.

[0326] The retainer assembly **513**, shown in FIGS. **78-81** comprises a coupling **583** and a packing nut retainer **584**. The coupling **583**, shown in FIGS. **82-84**, is cylindrical and comprises a front surface **585** and back surface **586** connected by an outer surface **587** and central bore **588**. The outer surface **587** and central bore **588** are concentric and their cylindrical axes are the longitudinal axis of the coupling **583**.

[0327] The central bore **588**, shown in FIG. **84**, comprises multiple sections. Beginning at the front surface **585** and continuing along the longitudinal axis to the back surface **586** the central bore **588** comprises a pilot section **589**, dynamic threads **538**, a center relief section **590**, and packing nut retainer threads **591**. The center relief section **590** shown is straight but may be similar to thread relief **366**.

[0328] The packing nut retainer **584**, shown in FIGS. **85-87**, is cylindrical and comprises a front surface **592** and back surface **593** connected by a threaded outer surface **594** and plunger bore **531**. The threaded outer surface **594** and plunger bore **531** are concentric and their cylindrical axes are the longitudinal axis of the packing nut retainer **584**.

[0329] The front surface **592** comprises a seal groove **543**. The seal groove **543** is circular and concentric with the plunger bore **531**. The seal groove **543** is configured to receive the retainer seal **512**.

[0330] The plunger bore **531** comprises multiple sections. Beginning at the front surface **592** and continuing along the longitudinal axis to the back surface **593** the plunger bore **531** comprises a packing section **540**, a packing nut shoulder **541** and packing nut threads **542**.

[0331] The packing nut retainer **584** further comprises a plurality of tension bolt holes **544**. Each tension bolt hole **544** is a partially threaded through hole connecting the front **592** and back **593** surfaces. The threaded portion of the tension bolt hole **544** extends from the front surface **592** approximately half the tension bolt hole **544** length to the back surface **593** and is configured to receive a tension bolt **514**. The tension bolt holes **544** are distributed evenly on a bolt circle that is concentric with the retainer assembly **513**.

[0332] Referring now to FIGS. **72-74** and **81**, the assembly of the multi-piece fluid end **502** begins with the installation of a flow control system wear ring seal **511**, a flow control system wear ring **510**, a plunger system wear ring seal **509**, and a plunger system wear ring **508** into each dynamic body **507** completing the assembly of the dynamic sections **506**.

[0333] Each dynamic section **506** is then attached to the static section **505** by threading the static threads **524** of the outer surface **521** of the dynamic body **507** into the matching threads of the static section **505**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **527** in the intermediate section **525** of the outer surface **521** of the dynamic body **507**. The flow control components are then inserted in the static section **505** from the opposite side.

[0334] Referring now to FIG. **81**, the retainer assembly **513** is assembled by first orienting the front surface **592** of the packing nut retainer **584** to face the back surface **586** of the coupling **583**. Second, the coupling **583** and packing nut retainer **584** should be aligned concentrically. Third, the packing nut retainer **584** is threaded into the coupling **583** until the back surface **593** of the packing nut retainer **584** is flush with the back surface **586** of the coupling **583**.

[0335] Referring again to FIG. **73**, for each dynamic section **506**, a retainer seal **512** is installed in the seal groove **543** of the front surface **592** of the packing nut retainer **584**. Then, the retainer assembly **513** is attached to a dynamic section **506** by threading the dynamic threads **538** of the coupling **583** onto the retainer threads **526** of the outer surface **521** of the dynamic body **507** until the front surface **592** of the packing nut retainer **584** contacts the back surface **520** of the dynamic body **507**. No additional torque is applied to the retainer assembly **513** after contact. The plurality of tension bolts **514** are then inserted, on a one-to-one basis, into the tension bolt holes **544** from the back surface **593** of the packing nut retainer **584** and threaded into the threaded section of the tension bolt holes **544** until the front surface of each tension bolt **514** contacts the back surface **520** of the dynamic body **507**. Once contact is made by all tension bolts **514** the tension bolts **514** may be torqued to specification in a piecewise manner.

[0336] Next, the packing **515** is inserted into the dynamic section **506** and the packing section **540** of the plunger bore **531** of the packing nut retainer **584**. Next, the plunger seal **517** is installed in the packing nut **516** and the packing nut **516** is threaded into the packing nut thread **542** of the plunger bore **531** of the packing nut retainer **584**. Next, the plunger **518** is installed in the packing nut **516**, packing **515**, and dynamic section **506**. Lastly, the packing nut **516** is torqued to specification.

[0337] In operation the tension bolts **514** place the threaded joint formed by the retainer threads **526** of the dynamic body **507** and the dynamic threads **538** of the coupling **583** in tension providing a tensile load to the threads above that achievable by only torquing the threads to specification. Although as noted above, no additional torque is applied to the retainer assembly **513** after contact is made with the dynamic section **506**. This tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **506**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. These benefits result in an easier and quicker removal of the retainer assembly **513** for maintenance or

replacement when necessary. The lack of additional assembly torque after contact between the retainer assembly **513** and dynamic body **507** results in an even easier disassembly when desired. Once the tension bolts **514** are removed only a small amount of torque is needed to unthread the coupling **583** from the dynamic body **507**.

[0338] Another embodiment of a fluid end section **678** is shown in FIGS. **88-110**. The fluid end section **678** may be used in the fluid end **402** shown in FIGS. **54-55**. The fluid end section **678** comprises a fluid end body **603** and a plunger system **604**. The fluid end body **603** comprises a static section **605**, a dynamic section **606**, a dynamic seal **695**, a plurality of studs **696**, a plurality of nuts **697**, and a plurality of dowel pins **698**.

[0339] Each static section **605**, shown in FIGS. **90-93**, comprises front **699** and back **680** surfaces connected by a flow bore **679**. The flow bore **679** is configured to receive flow control components.

[0340] The back surface **680** comprises seal groove **6100**, a plurality of threaded holes **6101**, and a plurality of dowel pin holes **6102**. The seal groove **6100** is concentric with the flow bore **679** and configured to receive the dynamic seal **695**. The plurality of threaded holes **6101** are blind holes configured to receive the studs **696** arranged in a circular pattern, and equally spaced around the flow bore **679**. The diameter of the circular pattern is larger than the diameter of the seal groove **6100**. This embodiment has twelve threaded holes **6101** but may have more or less, may not be arranged in a circular pattern, and may not be equally spaced around the flow bore **679**. The dowel pin holes **6102** are blind holes configured to receive dowel pins **698**. In this embodiment there are two dowel pin holes **6102** diametrically opposed on the transverse plane of the static section **605** at approximately the same diameter as the circular pattern of the threaded holes **6101** but spaced between them. There may be more or less dowel pin holes **6102** spaced in any manner as long as they do not intersect the threaded holes **6101**.

[0341] Each dynamic section **606**, shown in FIG. **94**, comprises a dynamic body **607**, a plunger system wear ring **608**, a plunger system wear ring seal **609**, a flow control system wear ring **610**, a flow control system wear ring seal **611**, a spacer sleeve **6103**, a plurality of dowel pins **698**, and a plurality of screws **6104**. Each plunger system **604**, shown in FIG. **93**, comprises a retainer seal **612**, a retainer **613**, a plurality of tension bolts **614**, packing **615**, a packing nut **616**, a plunger seal **617**, and a plunger **618**.

[0342] The dynamic body **607**, shown in FIGS. **95-97**, has a generally cylindrical shape. The dynamic body **607** comprises opposing front **619** and back **620** surfaces connected by an outer surface **621** and a flow bore **622**. The outer surface **621** and flow bore **622** are concentric and their cylindrical axes are the longitudinal axis of the dynamic body **607**.

[0343] The outer surface **621** comprises multiple sections, all the sections are concentric. Referring now to FIG. **97**, beginning at the front surface **619** of the dynamic body **607** and continuing along the longitudinal axis to the back surface **620** the outer surface **621** comprises a spacer sleeve section **6105**, locating shoulder **659**, mounting flange **6106**, mounting flange back surface **6107**, transition taper **6108**, wrench clearance section **6109**, wrench clearance shoulder **6110**, intermediate section **6111**, and retainer threads **626**.

[0344] The mounting flange **6106** comprises a plurality of clearance notches **6112** on the outer surface **621** and a plurality of stud holes **6113**. There are four clearance notches **6112** configured to avoid interference with the stay rods and spacers when mounting the fluid end section **678** to a power end. The stud holes **6113** are through bores that connect the locating shoulder **659** to the mounting flange back surface **6107**. The stud holes **6113** are configured to allow the passage of studs **696**. There are the same number of stud holes **6113**, with the same spacing, as the threaded holes **6101** in the back surface **680** of the static section **605**.

[0345] The locating shoulder **659** comprises a plurality of dowel pin holes **6114** and a plurality of threaded holes **6115**. The dowel pin holes **6114** are blind holes configured to receive dowel pins **698**. There are the same number of dowel pin holes **6114**, with the same spacing, as dowel pin holes **6102** in the back surface **680** of the static section **605**. The threaded holes **6115** are blind

holes configured to receive screws **6104**. In this embodiment there are two threaded holes **6115** diametrically opposed on the same bolt circle as the stud holes **6113** and dowel pin holes **6114**. The threaded holes **6115** are circumferentially spaced so that they do not intersect either the stud holes **6113** or the dowel pin holes **6114**.

[0346] The flow bore **622** also comprises multiple sections and is configured to receive the plunger system wear ring seal **609**, plunger system wear ring **608**, flow control system wear ring seal **611**, and flow control system wear ring **610**. The flow bore **622** is not the focus of this improvement so no further details about the flow bore **622** are necessary to understand this disclosure.

[0347] The spacer sleeve **6103**, shown in FIGS. **98-104**, is ring shaped and comprises front **6116** and back **6117** surfaces connected by an outer surface **6118** and central bore **6119**, a plurality of stud holes **6120**, a plurality of dowel pin holes **6121**, and a plurality of mounting screw holes **6122**. [0348] The stud holes **6120** are through holes connecting the front **6116** and back **6117** surfaces and are configured to allow the passage of the studs **696**. There are the same number of stud holes **6120**, with the same spacing, as the threaded holes **6101** in the back surface **680** of the static section **605**.

[0349] The dowel pin holes **6121** are through holes connecting the front **6116** and back **6117** surfaces. Each dowel pin hole **6121** is configured to receive two dowel pins **698**. One dowel pin **698** is inserted from the front surface **6116** and one from the back surface **6117**. There are the same number of dowel pin holes **6121**, with the same spacing, as dowel pin holes **6102** in the back surface **680** of the static section **605**.

[0350] The mounting screw holes **6122** are through holes connecting the front **6116** and back **6117** surfaces and are configured to receive screws **6104**. Each mounting screw hole **6122** comprises a counterbore **6123** and shoulder **6124**. The counterbore **6123** originates from the front surface **6116** and is deep enough to allow the head of the screw **6104** to be completely below the front surface **6116** when assembled. There are the same number of mounting screw holes **6122**, with the same spacing, as threaded holes **6115** in the mounting flange **6106** of the dynamic body **607**.

[0351] The outer surface **6118** comprises a plurality of clearance notches **6125** and a lift hole **6126**. The clearance notches **6125** extend from the front surface **6116** to the back surface **6117** and are shaped and positioned the same as the clearance notches **6112** of the dynamic body **607**. The lift hole **6126** is a threaded blind radial bore originating from the outer surface **6118** positioned so that when fluid end section **678** is installed the threaded lift hole **6126** will be facing up.

[0352] The retainer **613**, shown in FIGS. **105-110**, has a generally cylindrical shape. The retainer **613** comprises a front surface **628** and back surface **629** connected by an outer surface **630** and plunger bore **631**. The outer surface **630** and plunger bore **631** are concentric and their cylindrical axis is the longitudinal axis of the retainer **613**.

[0353] The outer surface **630** comprises a straight section **632** and a tapered section **633**. As shown in FIG. **109**, the straight section **632** begins at the front surface **628** and continues along the longitudinal axis until it connects with the tapered section **633**. The straight section **632** occupies approximately 80% of the outer surface **630** but may be more or less. The straight section **632** comprises a front chamfer **634**, a lubrication port **635**, and a spanner wrench hole **636**. The lubrication port **635** is a through bore connecting the straight section **632** of the outer surface **630** to the intermediate section **640** of the plunger bore **631**. The lubrication port **635** comprises a threaded section **637** adjacent to the outer surface **630** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0354] The spanner wrench hole **636** is a radial blind bore with a bore axis that is perpendicular to the longitudinal axis of the retainer **613**. The spanner wrench hole **636** originates from the straight section **632** of the outer surface **630** but does not intersect the plunger bore **631**. In this embodiment the spanner wrench hole **636** is proximate the tapered section **633** and aligned longitudinally with the lubrication port **635** but may be spaced in any manner on the straight section **632** as long as access for the spanner wrench (not shown) is available.

[0355] The plunger bore **631** comprises multiple sections, as shown in FIG. **110**. Beginning at the front surface **628** and continuing along the longitudinal axis to the back surface **629** the plunger bore **631** comprises dynamic threads **638**, a dynamic shoulder **639**, an intermediate section **640**, a packing nut shoulder **641**, and packing nut threads **642**. The dynamic shoulder **639** comprises a seal groove **643**. The seal groove **643** is circular and concentric with the plunger bore **631**. The seal groove **643** is configured to receive the retainer seal **612**.

[0356] The dynamic threads **638** comprise a thread relief **666** proximate the dynamic shoulder **639**. In this embodiment the thread relief **666** is straight walled but may be similar to the thread relief **466** shown in FIG. **70**.

[0357] The retainer **613** further comprises a plurality of tension bolt holes **644**. Each tension bolt hole **644** is a partially threaded through hole originating on the back surface **629** and terminating at the dynamic shoulder **639**. The threaded portion of the tension bolt hole **644** extends from the dynamic shoulder **639** approximately half the tension bolt hole **644** length to the back surface **629** and is configured to receive a tension bolt **614**. The tension bolt holes **644** are distributed evenly on a bolt circle that is concentric with the retainer **613**.

[0358] Referring now to FIGS. **88-94**, the assembly of a fluid end section **678** begins with the dynamic section **606**. Referring now to FIG. **94**, the dynamic section **606** is assembled by first installing a flow control system wear ring seal **611**, a flow control system wear ring **610**, a plunger system wear ring seal **609**, and a plunger system wear ring **608** into the dynamic body **607**. The depth of the dowel pin holes **6114** is such that, once inserted, a portion of each dowel pin **698** will be protruding from the locating shoulder **659**. Third, the spacer sleeve **6103** is oriented to be concentric with the dynamic body **607**, align the dowel pin holes **6121** with the protruding dowel pins **698**, and to have the back surface **6117** facing the locating shoulder **659**. Fourth, the spacer sleeve section **6105** of the outer surface **621** of the dynamic body **607** is then inserted in the central bore **6119** of the spacer sleeve **6103**, while simultaneously inserting the protruding dowel pins **698** in the dowel pin holes **6121**, until the back surface **6117** contacts the locating shoulder **659**. Fifth, the screws **6104** are inserted in the mounting screw holes **6122**, threaded into the threaded holes **6115** of the locating shoulder **659** and torqued to specification, thus attaching the spacer sleeve **6103** to the dynamic body **607** and completing the assembly of the dynamic section **606**.

[0359] Referring now to FIGS. **89, 91, and 93**, the dynamic section **606** is attached to the static section **605** by first, threading the studs **696** into the threaded holes **6101** in the back surface **680** of the static section **605** and torquing them to specification. Second, a dowel pin **698** is inserted in each of the dowel pin holes **6102** in the back surface **680** of the static section **605**. The depth of the dowel pin holes **6102** is such that, once inserted, a portion of each dowel pin **698** will protrude from the back surface **680**. Third, dynamic seal **695** is installed in seal groove **6100** of the back surface **680**. Fourth, the dynamic section **606** is oriented to be concentric with the flow bore **679** of the static section **605**, align the dowel pin holes **6121** with the protruding dowel pins **698**, face the lift hole **6126** up, and to have the front surface **6116** facing the back surface **680** of the static section **605**. Fifth, the dynamic section **606** is moved toward the static section **605** until the front surface **6116** of the spacer sleeve **6103** contacts the back surface **680** of the static section **605**, simultaneously, the studs **696** are inserted in the stud holes **6120** and the dowel pins **698** are inserted in the dowel pin holes **6121**. Sixth, the nuts **697** are threaded on the studs **696** and torqued to specification completing the attachment of the dynamic section **606** to the static section **605**. Now the flow control components may be installed in the flow bore **679**.

[0360] Referring now to FIGS. **89, 92, and 93**, the plunger system **604** is assembled to the dynamic section **606** by first, installing a retainer seal **612** in the seal groove **643** of the dynamic shoulder **639** of the plunger bore **631** of a retainer **613**. Second, the retainer **613** is attached to a dynamic section **606** by threading the dynamic threads **638** of the plunger bore **631** of the retainer **613** onto the retainer threads **626** of the outer surface **621** of the dynamic body **607**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench hole **636** in the straight

section **632** of the outer surface **630** of the retainer **613**.

[0361] Third, the plurality of tension bolts **614** are then inserted, on a one-to-one basis, into the tension bolt holes **644** from the back surface **629** of the retainer **613** and threaded into the threaded section of the tension bolt holes **644** until the front surface of each tension bolt **614** contacts the back surface **620** of the dynamic body **607**, as shown in FIG. **92**. Once contact is made by all tension bolts **614** the tension bolts **614** may be torqued to specification in a piecewise manner.

[0362] Fourth, the packing **615** is inserted into the dynamic section **606** and the intermediate section **640** of the plunger bore **631** of the retainer **613**. Fifth, the plunger seal **617** is installed in the packing nut **616** and the packing nut **616** is threaded into the packing nut threads **642** of the plunger bore **631** of the retainer **613**. Sixth, the plunger **618** is installed in the packing nut **616**, packing **615**, and dynamic section **606**. Lastly, the packing nut **616** is torqued to specification.

[0363] In operation the tension bolts **614** place the threaded joint formed by the retainer threads **626** of the dynamic body **607** and the dynamic threads **638** of the retainer **613** in tension providing an additional tensile load above that produced by the torquing of the threads together. This additional tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **606**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. Also, the retainer seal **612** protects the retainer threads **626** and dynamic threads **638** from contamination when the packing **615** fails. These benefits result in easier and quicker removal of the retainer **613** for maintenance or replacement when necessary. Also the use of the studs **696** and nuts **697** to attach the dynamic section **606** to the static section **605** may require less torque than a single large thread simplifying disassembly when needed.

[0364] Another embodiment of a fluid end section **778** is shown in FIGS. **111-125**. The fluid end section **778** may be used in the fluid end **402** shown in FIGS. **54-55**. The fluid end section **778** comprises a fluid end body **703** and a plunger system **704**. The fluid end body **703** comprises a static section **705**, a dynamic section **706**, and a plurality of tension bolts **7127**.

[0365] Each static section **705**, shown in FIGS. **113** and **114**, comprises a flow bore **779** and a back surface **780**. The flow bore **779** is configured to receive flow control components. The flow bore **779** comprises a section of dynamic threads **781** proximate the back surface **780** of the static section **705**. The dynamic threads **781** are configured to receive the static threads **724** of the dynamic section **706**.

[0366] Referring now to FIG. **114**, each dynamic section **706** comprises a dynamic body **707**, a plunger system wear ring **708**, a plunger system wear ring seal **709**, a flow control system wear ring **710**, a flow control system wear ring seal **711**, and a spacer ring **7129**. Each plunger system **704** comprises a retainer seal **712**, a retainer **713**, a plurality of tension bolts **714**, packing **715**, a packing nut **716**, a plunger seal **717**, and a plunger **718**.

[0367] The dynamic body **707**, shown in FIGS. **116-118**, has a cylindrical shape. The dynamic body **707** comprises opposing front **719** and back **720** surfaces connected by an outer surface **721** and a flow bore **722**. The outer surface **721** and flow bore **722** are concentric and their cylindrical axes are the longitudinal axis of the dynamic body **707**.

[0368] Referring now to FIG. **118**, the outer surface **721** comprises multiple sections, all the sections are concentric. Beginning at the front surface **719** of the dynamic body **707** and continuing along the longitudinal axis to the back surface **720** the outer surface **721** comprises a static seal section **723**, static threads **724**, locating shoulder **759**, mounting flange **7106**, mounting flange back surface **7107**, intermediate section **725**, back shoulder **761**, and retainer threads **726**. The intermediate section **725** comprises a plurality of spanner wrench holes **727**. The spanner wrench holes **727** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **707**. Each spanner wrench hole **727** originates from the intermediate section **725** of the outer surface **721** but does not intersect the flow bore **722**. In this embodiment the spanner

wrench holes **727** are proximate the back shoulder **761**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the intermediate section **725** as long as access for the spanner wrench (not shown) is available.

[0369] The mounting flange **7106** comprises a plurality of tension bolt holes **7128**. The tension bolt holes **7128** are threaded through holes that connect the locating shoulder **759** to the mounting flange back surface **7107**. The tension bolt holes **7128** are configured to receive the tension bolts **7127**. The tension bolt holes **7128** are formed in a circular pattern concentric with the outer surface **721**. In this embodiment there are thirty-six tension bolt holes **7128** spaced evenly in the circular pattern but there may be more or less spaced in any manner as desired.

[0370] The flow bore **722** also comprises multiple sections and is configured to receive the plunger system wear ring seal **709**, plunger system wear ring **708**, flow control system wear ring seal **711**, flow control system wear ring **710**, and spacer ring **7129**. The flow bore **722** is not the focus of this improvement so no further details about the flow bore **722** are necessary to understand this disclosure.

[0371] The retainer **713**, shown in FIGS. **119-124**, has a generally cylindrical shape. The retainer **713** comprises a front surface **728** and back surface **729** connected by an outer surface **730** and plunger bore **731**. The outer surface **730** and plunger bore **731** are concentric and their cylindrical axes are the longitudinal axis of the retainer **713**.

[0372] The outer surface **730** comprises a straight section **732** and a tapered section **733**. The straight section **732** begins at the front surface **728** and continues along the longitudinal axis until it connects with the tapered section **733**. The straight section **732** occupies approximately 60% of the outer surface **730** but may be more or less. The straight section **732** comprises a front chamfer **734** and plurality of spanner wrench holes **736**.

[0373] The spanner wrench holes **736** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the retainer **713**. Each spanner wrench hole **736** originates from the straight section **732** of the outer surface **730** but does not intersect the plunger bore **731**. In this embodiment the spanner wrench holes **736** are proximate the tapered section **733**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the straight section **732** as long as access for the spanner wrench (not shown) is available.

[0374] The tapered section **733** comprises a lubrication port **735**. The lubrication port **735** is a radial through bore connecting the tapered section **733** of the outer surface **730** to the intermediate section **740** of the plunger bore **731**. The bore axis of the lubrication port **735** is perpendicular to the tapered face of the tapered section **733**. The lubrication port **735** comprises a threaded section **737** adjacent to the outer surface **730** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0375] The plunger bore **731** comprises multiple sections, as shown in FIG. **124**. Beginning at the front surface **728** and continuing along the longitudinal axis to the back surface **729** the plunger bore **731** comprises dynamic threads **738**, a dynamic shoulder **739**, an intermediate section **740**, a packing nut shoulder **741**, and packing nut threads **742**. The dynamic shoulder **739** comprises a seal groove **743**. The seal groove **743** is circular and concentric with the plunger bore **731**. The seal groove **743** is configured to receive the retainer seal **712**.

[0376] The dynamic threads **738** comprise a thread relief **766** proximate the dynamic shoulder **739**. Referring now to FIG. **125**, the thread relief **766** comprises a straight section **768** and a plurality of curved sections, curved section one **770**, curved section two **771**, and curved section three **772**. Curved section one **770** is concave and provides a transition from the last thread of the dynamic threads **738** to curved section two **771**. Curved section two **771** is concave and provides a transition from curved section one **770** to the straight section **768**. Straight section **768** is angled relative to the longitudinal axis and provides a transition from curved section two **771** to curved section three **772**. Curved section three **772** is concave and provides a transition from straight section **768** to the dynamic shoulder **739** of the plunger bore **731**.

[0377] In this embodiment the radius of curved section one **770** is less than the radius of curved section two **771** and the radius of curved section two **771** is less than the radius of curved section three **772** but the radii of the curved sections **770**, **771**, and **772** may be any value that still allows the geometric construction of the thread relief **766**. In this embodiment straight section **768** has an angle of approximately 35 degrees with the longitudinal axis however it may be any angle between 0 and 90 degrees. It is also contemplated that the curved sections **770**, **771**, and **772** may be spline curves with infinitely varying radii. If both curved section one **770** and curved section two **771** are spline curves, then curved section one **770** and curved section two **771** may be considered a single curve. Also, if both curved section two **771** and curved section three **772** are spline curves and straight section **768** is removed then curved section two **771** and curved section three **772** may be considered a single curve.

[0378] Referring now to FIG. **124**, the retainer **713** further comprises a plurality of tension bolt holes **744**. Each tension bolt hole **744** is a partially threaded through hole originating on the back surface **729** and terminating at the dynamic shoulder **739**. The threaded portion of the tension bolt hole **744** extends from the dynamic shoulder **739** approximately half the tension bolt hole **744** length to the back surface **729** and is configured to receive a tension bolt **714**. The tension bolt holes **744** are distributed evenly on a bolt circle that is concentric with the retainer **713**.

[0379] Referring now to FIGS. **111-115**, the assembly of a fluid end section **778** begins with the dynamic section **706**. Referring now to FIG. **113**, the dynamic section is assembled by installing a flow control system wear ring seal **711**, a flow control system wear ring **710**, a plunger system wear ring seal **709**, plunger system wear ring **708**, and spacer ring **7129** into the dynamic body **707**.

[0380] The dynamic section **706** is then attached to the static section **705** by threading the static threads **724** of the outer surface **721** of the dynamic body **707** into the dynamic threads **781** of the static section **705** until the locating shoulder **759** abuts the back surface **780** of the static section **705**, as shown in FIG. **114**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **727** in the intermediate section **725** of the outer surface **721** of the dynamic body **707**.

[0381] Next, the plurality of tension bolts **7127** are then threaded, on a one-to-one basis, into the tension bolt holes **7128** from the mounting flange back surface **7107** of the dynamic body **707** until the front surface of each tension bolt **7127** contacts the back surface **780** of the static section **705**, as shown in FIG. **113**. Once contact is made by all tension bolts **7127** the tension bolts **7127** may be torqued to specification in a piecewise manner.

[0382] Next, a retainer seal **712** is installed in the seal groove **743** of the dynamic shoulder **739** of the plunger bore **731** of a retainer **713**. Then, the retainer **713** is attached to a dynamic section **706** by threading the dynamic threads **738** of the plunger bore **731** of the retainer **713** onto the retainer threads **726** of the outer surface **721** of the dynamic body **707**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **736** in the straight section **732** of the outer surface **730** of the retainer **713**.

[0383] The plurality of tension bolts **714** are then inserted, on a one-to-one basis, into the tension bolt holes **744** from the back surface **729** of the retainer **713** and threaded into the threaded section of the tension bolt holes **744** until the front surface of each tension bolt **714** contacts the back surface **720** of the dynamic body **707**, as shown in FIG. **114**. Once contact is made by all tension bolts **714** the tension bolts **714** may be torqued to specification in a piecewise manner. The flow control components may now be installed in the flow bore **779** of static section **705**.

[0384] Next, the packing **715** is inserted into the dynamic section **706** and the intermediate section **740** of the plunger bore **731** of the retainer **713**. Next, the plunger seal **717** is installed in the packing nut **716** and the packing nut **716** is threaded into the packing nut threads **742** of the plunger bore **731** of the retainer **713**. Next, the plunger **718** is installed in the packing nut **716**, packing **715**, and dynamic section **706**. Lastly, the packing nut **716** is torqued to specification.

[0385] In operation the tension bolts **714** place the threaded joint formed by the retainer threads

726 of the dynamic body **707** and the dynamic threads **738** of the retainer **713** in tension providing an additional tensile load above that produced by the torquing of the threads together. This additional tensile load reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **706**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. Also, the retainer seal **712** protects the retainer threads **726** and dynamic threads **738** from contamination when the packing **715** fails. These benefits result in easier and quicker removal of the retainer **713** for maintenance or replacement when necessary. Also during operation, the thread relief **766** reduces stress in the retainer **713** at the transition from the dynamic threads **738** to the dynamic shoulder **739**.

[0386] In like manner to the tension bolts **714**, tension bolts **7127** provide an additional tensile load to the threaded joint formed by the static threads **724** of the dynamic body **707** and the dynamic threads **781** of the static section **705** above that produced by the torquing of the threads together. This also results in lower failure rates and easier assembly and disassembly of the fluid end section **778**.

[0387] Referring now to FIGS. **126-142**, another embodiment of a multi-piece fluid end **802** is shown. The multi-piece fluid end **802** comprises a fluid end body **803**, a plurality of plunger systems **804**, and a plurality of flow control systems **8146**. The fluid end body **803** comprises a static section **805** and a plurality of dynamic sections **806**. Each dynamic section **806** comprises a dynamic body **807**, a plunger system wear ring **808**, a plunger system wear ring seal **809**, a spacer ring **8129**, a flow control system wear ring **810**, and a flow control system wear ring seal **811** as shown in FIGS. **127-130**. Each plunger system **804** comprises a retainer seal **812**, a retainer **813**, a plurality of tension bolts **814**, packing **815**, a lantern ring seal **8130**, a packing nut **816**, a plunger seal **817**, and a plunger **818**, as shown in FIGS. **130-131**.

[0388] The dynamic body **807**, shown in FIGS. **132-136**, has a cylindrical shape. The dynamic body **807** comprises opposing front **819** and back **820** surfaces connected by an outer surface **821** and a flow bore **822**. The outer surface **821** and flow bore **822** are concentric and their cylindrical axis is the longitudinal axis of the dynamic body **807**.

[0389] Referring now to FIG. **134**, the outer surface **821** comprises multiple concentric sections. Beginning at the front surface **819** of the dynamic body **807** and continuing along the longitudinal axis to the back surface **820**, the outer surface **821** comprises a static seal section **823**, static threads **824**, locating shoulder **859**, and a back section **8131**. The back section **8131** comprises a plurality of spanner wrench holes **827**. The spanner wrench holes **827** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **807**. Each spanner wrench hole **827** originates from the back section **8131** of the outer surface **821** but does not intersect the flow bore **822**. In this embodiment the spanner wrench holes **827** are proximate the static threads **824**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the back section **8131** as long as access for the spanner wrench (not shown) is available.

[0390] Continuing with FIG. **134**, beginning at the front surface **819** of the dynamic body **807** and moving along the bore axis of the flow bore **822** to the back surface **820**, the flow bore **822** comprises a plurality of concentric sections. They are a flow control system wear ring section **8132**, a flow control system wear ring shoulder **8133**, a flow control system section **8134**, a plunger section **8135**, a plunger system wear ring shoulder **8136**, a plunger system wear ring section **8137**, a retainer shoulder **8138**, and a retainer thread section **8139**.

[0391] Referring now to FIG. **135**, the flow control system wear ring section **8132** comprises a tapered bore **8140**. The largest diameter of the tapered bore **8140** is at the front surface **819** and the smallest diameter is at the flow control system wear ring shoulder **8133**. The taper is complementary to the taper of the outer surface of the flow control system wear ring **810**, as shown in FIG. **127**. The flow control system wear ring section **8132** may further comprise a front chamfer **8141** to aid in the installation of the flow control system wear ring **810**. The flow control system

wear ring section **8132** may further comprise a transition radius **8142** at the base of the tapered bore **8140** to reduce stress in the transition between the tapered bore **8140** and the flow control system wear ring shoulder **8133**.

[0392] The tapered bore **8140** comprises a seal groove **8143** comprising two side walls **8144** connected by a base **8145**. Each side wall **8144** is perpendicular to the bore axis of the flow bore **822** and extends from the surface of the tapered bore **8140** radially away from the bore axis of the flow bore **822**. The base **8145** is flat, that is parallel to the bore axis of the flow bore **822**. The seal groove **8143** is located from the front surface **819** approximately one-half of the longitudinal distance between the front surface **819** and the flow control system wear ring shoulder **8133**.

[0393] The flow control system wear ring shoulder **8133** is formed by the reduction in diameter of the flow bore **822** between the flow control system wear ring section **8132** and the flow control system section **8134**. The flow control system wear ring shoulder **8133** is perpendicular to the bore axis of the flow bore **822**.

[0394] The flow control system section **8134** comprises a straight portion **8147**, that is the bore wall is parallel to the bore axis of the flow bore **822**, and a tapered portion **8148**. As can be seen in FIG. **127**, the flow control system section **8134** is configured to receive particular components of the flow control system **8146**. The flow control system section **8134** may further comprise a front chamfer **8206** to aid in the installation of the particular components of the flow control system **8146** and a transition radius **8203** from the straight portion **8147** to the tapered portion **8148**.

[0395] The plunger section **8135** is also straight and provides a volume for the fluid to enter on the suction stroke of the plunger system **804** and to exit when the plunger system **804** applies force, generating fluid pressure, on the pressure stroke, as shown in FIG. **127**. The plunger section **8135** may further comprise a transition radius **8204** from the flow control system section **8134**, shown in FIG. **135**, and another transition radius **8205** into the plunger system wear ring shoulder **8136**, shown in FIG. **136**.

[0396] Referring now to FIG. **136**, the plunger system wear ring shoulder **8136** is formed by the increase in diameter of the flow bore **822** between the plunger section **8135** and the plunger system wear ring section **8137**. The plunger system wear ring shoulder **8136** is perpendicular to the bore axis of the flow bore **822**.

[0397] The plunger system wear ring section **8137** comprises a tapered bore **8149**. The largest diameter of the tapered bore **8149** is at the retainer shoulder **8138** and the smallest diameter is at the plunger system wear ring shoulder **8136**. The taper is complementary to the taper of the outer surfaces of the plunger system wear ring **808** and spacer ring **8129**, as shown in FIG. **128**. The plunger system wear ring section **8137** may further comprise a back chamfer **8150** to aid in the installation of the plunger system wear ring **808** and spacer ring **8129**. The plunger system wear ring section **8137** may also comprise a transition radius **8151** at the base of the tapered bore **8149** to reduce stress in the transition between the tapered bore **8149** and the plunger system wear ring shoulder **8136**.

[0398] The tapered bore **8149** comprises a seal groove **8152** comprising two side walls **8153** connected by a base **8154**. Each side wall **8153** is perpendicular to the bore axis of the flow bore **822** and extends from the surface of the tapered bore **8149** radially away from the bore axis of the flow bore **822**. The base **8154** is flat, that is parallel to the bore axis of the flow bore **822**. The seal groove **8152** is located from the retainer shoulder **8138** approximately one-half of the longitudinal distance between the retainer shoulder **8138** and the plunger system wear ring shoulder **8136**.

[0399] The retainer shoulder **8138** is formed by the increase in diameter of the flow bore **822** between the plunger system wear ring section **8137** and the retainer thread section **8139**. The retainer shoulder **8138** is perpendicular to the bore axis of the flow bore **822**.

[0400] The retainer thread section **8139** comprises an internal thread **8200** and a thread relief **866**. The internal thread **8200** begins at the back surface **820** and ends at the thread relief **866**. The internal thread **8200** is configured to receive external thread **8156** of the retainer **813**. The thread

relief **866** extends from the retainer shoulder **8138** to the internal thread **8200**. The thread relief **866** shown is a typical thread relief but may be similar to thread reliefs **366**, **466**, or **766**.

[0401] The retainer **813**, shown in FIGS. **137-142**, has a generally cylindrical shape. The retainer **813** comprises a front surface **828** and back surface **829** connected by an outer surface **830** and plunger bore **831**. The outer surface **830** and plunger bore **831** are concentric and their cylindrical axes are the longitudinal axis of the retainer **813**.

[0402] The front surface **828** comprises a seal groove **843**. The seal groove **843** is circular and concentric with the plunger bore **831**. The seal groove **843** is configured to receive the retainer seal **812**.

[0403] The outer surface **830**, as shown in FIG. **139**, comprises a dynamic thread section **8201**, a dynamic shoulder **8155**, a straight section **832** and a tapered section **833**. The dynamic thread section **8201** comprises an external thread **8156**, and a thread relief **866**. The external thread **8156** begins at the front surface **828** and continues along the longitudinal axis until it intersects the thread relief **866**. The thread relief **866** extends from the external thread **8156** to the dynamic shoulder **8155**. The straight section **832** begins at the dynamic shoulder **8155** and continues along the longitudinal axis until it connects with the tapered section **833**. The tapered section **833** extends from the straight section **832** to the back surface **829**. The outer surface **830** may further comprise a transition radius **8157** between the straight section **832** and tapered section **833** and a transition radius **8158** between the tapered section **833** and the back surface **829**. The thread relief **866** shown is a typical thread relief but may be similar to thread reliefs **366**, **466**, or **766**.

[0404] The tapered section **833** comprises a lubrication port **835**. The lubrication port **835** is a radial through bore connecting the tapered section **833** of the outer surface **830** to the packing section **840** of the plunger bore **831**. The bore axis of the lubrication port **835** may be perpendicular to the tapered face of the tapered section **833**. The lubrication port **835** may comprise a threaded section (not shown) adjacent to the outer surface **830** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0405] The plunger bore **831** comprises multiple sections, as shown in FIG. **142**. Beginning at the front surface **828** and continuing along the longitudinal axis to the back surface **829**, the plunger bore **831** comprises the packing section **840**, a packing nut shoulder **841**, and packing nut threads **842**. The packing nut threads **842** comprise internal threads **8159** configured to receive the external threads of the packing nut **816** and a thread relief **866**. The thread relief **866** shown is a typical thread relief but may be similar to thread reliefs **366**, **466**, or **766**.

[0406] Referring now to FIGS. **140** and **142**, the retainer **813** further comprises a plurality of tension bolt holes **844**. Each tension bolt hole **844** is a partially threaded through hole originating on the back surface **829** and terminating at the front surface **828**. The threaded portion of the tension bolt hole **844** extends from the front surface **828** approximately half the tension bolt hole **844** length to the back surface **829** and is configured to receive a tension bolt **814**. The tension bolt holes **844** are distributed evenly on a bolt circle that is concentric with the retainer **813**.

[0407] Referring now to FIGS. **126-131**, the assembly of the multi-piece fluid end **802** begins with the assembly of the plurality of dynamic sections **806**. A flow control system wear ring seal **811**, a flow control system wear ring **810**, a plunger system wear ring seal **809**, plunger system wear ring **808**, and spacer ring **8129** are installed into the flow bore **822** of each dynamic body **807**.

[0408] Each dynamic section **806** is then attached to the static section **805** by threading the static threads **824** of the outer surface **821** of the dynamic body **807** into the dynamic threads **881** of the static section **805** until the locating shoulder **859** abuts the back surface **880** of the static section **805**, as shown in FIG. **129**. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **827** in the back section **8131** of the outer surface **821** of the dynamic body **807**. The applicable components of the flow control system **8146** may now be installed into the sections of the static sections **805** and flow bores **822** of the assembled dynamic bodies **807** configured to receive them.

[0409] Referring now to FIG. 130, a retainer seal **812** is installed in the seal groove **843** of the front surface **828** of a retainer **813**. Then, the retainer **813** is attached to a dynamic section **806** by threading the external threads **8156** of the dynamic thread section **8201** of the outer surface **830** of the retainer **813** into the internal threads **8200** of the retainer thread section **8139** of the flow bore **822** of the dynamic body **807**. Minimal torque is required to make up the engaged external **8156** and internal **8200** threads. As will be described later, the tension bolts **814** provide the necessary tensile load. The joint only needs to be made up tight enough to ensure the retainer seal **812** sufficiently engages the retainer shoulder **8138** of the flow bore **822** of the dynamic body **807**.

[0410] The plurality of tension bolts **814** are then inserted, on a one-to-one basis, into the tension bolt holes **844** from the back surface **829** of the retainer **813** and threaded into the threaded section of the tension bolt holes **844** until the front surface of each tension bolt **814** contacts the retainer shoulder **8138** of the flow bore **822** of the dynamic body **807**. Once contact is made by all tension bolts **814**, the tension bolts **814** may be torqued to specification in a piecewise manner.

[0411] Next, the packing **815** is inserted into the dynamic section **806** and the packing section **840** of the plunger bore **831** of the retainer **813**. Next, the plunger seal **817** is installed in the packing nut **816** and the packing nut **816** is threaded into the packing nut threads **842** of the plunger bore **831** of the retainer **813**. Next, the plunger **818** is installed in the packing nut **816**, packing **815**, and dynamic section **806**. Lastly, the packing nut **816** is torqued to specification.

[0412] In operation the tension bolts **814** place the threaded joint formed by the internal threads **8200** of the retainer thread section **8139** of the flow bore **822** of the dynamic body **807** and the external threads **8156** of the dynamic thread section **8201** of the outer surface **830** of the retainer **813** in tension providing an additional tensile load above that produced by the torquing of the threads together. This allows for minimal make-up torque by the operator simplifying assembly and disassembly. The additional tensile load also reduces the magnitude of cyclic loading usually produced in the joint by the large pressure fluctuations in the dynamic section **806**. This reduction in the magnitude of cyclic loading reduces thread to thread fretting and other issues inherent in the previously encountered full, or nearly full, cyclic loading of the threads. Also, the retainer seal **812** protects the retainer thread section **8139** and dynamic thread section **8201** from contamination when the packing **815** fails. These benefits result in easier and quicker removal of the retainer **813** for maintenance or replacement when necessary.

[0413] Referring now to FIGS. 143-175, another embodiment of a multi-piece fluid end **902** is shown. The multi-piece fluid end **902** comprises a fluid end body **903**, a plurality of plunger systems **904**, and a plurality of flow control systems **9146**. The fluid end body **903** comprises a static section **905** and a plurality of dynamic sections **906**. Each dynamic section **906** comprises a dynamic body **907**, a front plunger system wear ring **908**, a plunger system wear ring seal **909**, a back plunger system wear ring **9129**, a flow control system wear ring **910**, and a flow control system wear ring seal **911** as shown in FIGS. 144-148. Each plunger system **904** comprises a plurality of locating dowel pins **9160**, a retainer **913**, a plurality of studs **9161**, a plurality of nuts **9162**, packing **915**, a packing nut **916**, a plunger seal **917**, and a plunger **918**, as shown in FIGS. 145, 146, 148 and 149.

[0414] The dynamic body **907**, shown in FIGS. 149-152, has a cylindrical shape. The dynamic body **907** comprises opposing front **919** and back **920** surfaces connected by an outer surface **921** and a flow bore **922**. The outer surface **921** and flow bore **922** are concentric and their cylindrical axis is the longitudinal axis of the dynamic body **907**.

[0415] Referring now to FIG. 152, the outer surface **921** comprises multiple concentric sections. Beginning at the front surface **919** of the dynamic body **907** and continuing along the longitudinal axis to the back surface **920** the outer surface **921** comprises a static seal section **923**, static threads **924**, locating shoulder **959**, and a back section **9131**. The back section **9131** comprises a plurality of spanner wrench holes **927**. The spanner wrench holes **927** are radial blind bores with a bore axis that is perpendicular to the longitudinal axis of the dynamic body **907**. Each spanner wrench hole

927 originates from the back section **9131** of the outer surface **921** but does not intersect the flow bore **922**. In this embodiment the spanner wrench holes **927** are proximate the static threads **924**, aligned longitudinally, and evenly spaced circumferentially but may be spaced in any manner on the back section **9131** as long as access for the spanner wrench (not shown) is available.

[0416] Referring now to FIG. **153**, the flow bore **922** also comprises multiple sections and is configured to receive the plunger system wear ring seal **909**, front plunger system wear ring **908**, back plunger system wear ring **9129**, flow control system wear ring seal **911**, and flow control system wear ring **910**. The flow bore **922** comprises a flow control system wear ring section **9132**, a flow control system wear ring shoulder **9133**, a flow control system section **9134**, a plunger section **9135**, a plunger system wear ring shoulder **9136** and a plunger system wear ring section **9137**.

[0417] The flow control system wear ring section **9132** comprises a tapered bore **9140**. The largest diameter of the tapered bore **9140** is at the front surface **919** and the smallest diameter is at the flow control system wear ring shoulder **9133**. The taper is complementary to the taper of the outer surface of the flow control system wear ring **910**, as shown in FIG. **144**. The flow control system wear ring section **9132** may further comprise a front chamfer **9141** to aid in the installation of the flow control system wear ring **910**. The flow control system wear ring section **9132** may further comprise a transition radius **9142** at the base of the tapered bore **9140** to reduce stress in the transition between the tapered bore **9140** and the flow control system wear ring shoulder **9133**.

[0418] The tapered bore **9140** comprises a seal groove **9143** comprising two side walls **9144** connected by a base **9145**. Each side wall **9144** is perpendicular to the bore axis of the flow bore **922** and extends from the surface of the tapered bore **9140** radially away from the bore axis of the flow bore **922**. The base **9145** is flat, that is parallel to the bore axis of the flow bore **922**. The seal groove **9143** is located from the front surface **919** approximately one-half of the longitudinal distance between the front surface **919** and the flow control system wear ring shoulder **9133**.

[0419] The flow control system wear ring shoulder **9133** is formed by the reduction in diameter of the flow bore **922** between the flow control system wear ring section **9132** and the flow control system section **9134**. The flow control system wear ring shoulder **9133** is perpendicular to the bore axis of the flow bore **922**.

[0420] The flow control system section **9134** comprises a straight portion **9147**, that is the bore wall is parallel to the bore axis of the flow bore **922**, and a tapered portion **9148**. As can be seen in FIG. **144** the flow control system section **9134** is configured to receive particular components of the flow control system **9146**. The flow control system section **9134** may further comprise a front chamfer **9206** to aid in the installation of the particular components of the flow control system **9146** and a transition radius **9203** from the straight portion **9147** to the tapered portion **9148**.

[0421] The plunger section **9135** is also straight and provides a volume for the fluid to enter on the suction stroke of the plunger system **904** and to exit from as the plunger system **904** applies force, generating fluid pressure, on the pressure stroke, as shown in FIG. **144**. The plunger section **9135** may further comprise a transition radius **9204** from the flow control system section **9134** and another transition radius **9205** into the plunger system wear ring shoulder **9136**.

[0422] The plunger system wear ring shoulder **9136** is formed by the increase in diameter of the flow bore **922** between the plunger section **9135** and the plunger system wear ring section **9137**. The plunger system wear ring shoulder **9136** is perpendicular to the bore axis of the flow bore **922**.

[0423] The plunger system wear ring section **9137** comprises a tapered bore **9149**. The largest diameter of the tapered bore **9149** is at the back surface **920** and the smallest diameter is at the plunger system wear ring shoulder **9136**. The taper is complementary to the taper of the outer surface **9169** of the front plunger system wear ring **908** and the outer surface **9175** of the back plunger system wear ring **9129**, ensuring a precise fit, as shown in FIG. **145**. The plunger system wear ring section **9137** may further comprise a back chamfer **9150** to aid in the installation of the front plunger system wear ring **908** and back plunger system wear ring **9129**. The plunger system

wear ring section **9137** may further comprise a transition radius **9151** at the base of the tapered bore **9149** to reduce stress in the transition between the tapered bore **9149** and the plunger system wear ring shoulder **9136**. The transition radius **9151** may also be called the root **9151** of the tapered bore **9149**.

[0424] The tapered bore **9149** comprises a seal groove **9152** comprising two side walls **9153** connected by a base **9154**. Each side wall **9153** is perpendicular to the bore axis of the flow bore **922** and extends from the surface of the tapered bore **9149** radially away from the bore axis of the flow bore **922**. The base **9154** is flat, that is parallel to the bore axis of the flow bore **922**. The seal groove **9152** is located from the back surface **920** approximately one-half of the longitudinal distance between the back surface **920** and the plunger system wear ring shoulder **9136**.

[0425] Referring now to FIGS. **151** and **152**, the back surface **920** comprises a plurality of threaded blind holes **9163** configured to receive the studs **9161**. The plurality of threaded blind holes **9163** are arranged in a circular pattern, and equally spaced around the flow bore **922**. This embodiment has sixteen threaded blind holes **9163** but may have more or less, may not be arranged in a circular pattern, and may not be equally spaced around the flow bore **922**.

[0426] Referring now to FIGS. **151** and **153**, the back surface **920** may further comprise a plurality of blind holes **9164** configured to receive the locating dowel pins **9160**. In this embodiment there are two blind holes **9164** diametrically opposed at approximately the same diameter as the circular pattern of the threaded blind holes **9163** but spaced between them. There may be more or less blind holes **9164** spaced in any manner as long as they do not intersect the threaded blind holes **9163**.

[0427] The front plunger system wear ring **908**, shown in FIGS. **160-163**, has an annular shape, with a front **9167** and back surface **9168** connected by an outer **9169** and inner **9170** surface. The front **9167** and back **9168** surfaces are parallel to each other and perpendicular to the longitudinal axis of the front plunger system wear ring **908**. The inner surface **9170** is parallel to the longitudinal axis of the front plunger system wear ring **908**. The outer surface **9169** is tapered. The taper is complementary to the taper of the tapered bore **9149** of the plunger system wear ring section **9137** of the flow bore **922** of the dynamic body **907**, ensuring a precise fit, as shown in FIG. **145**. The front plunger system wear ring **908** further comprises a clearance chamfer **9171**. The clearance chamfer **9171** is located at the intersection of the front surface **9167** and outer surface **9169**. The front plunger system wear ring **908** may further comprise a plurality of small chamfers **9172** at the intersections of the remaining surfaces. These small chamfers **9172**, commonly referred to as 'break edges', facilitate safer handling and durability.

[0428] The back plunger system wear ring **9129**, shown in FIGS. **164-167**, has an annular shape, with a front **9173** and back surface **9174** connected by an outer **9175** and inner **9176** surface. The front **9173** and back **9174** surfaces are parallel to each other and perpendicular to the longitudinal axis of the back plunger system wear ring **9129**. The inner surface **9176** is parallel to the longitudinal axis of the back plunger system wear ring **9129**. The outer surface **9175** is tapered. The taper is complementary to the taper of the tapered bore **9149** of the plunger system wear ring section **9137** of the flow bore **922** of the dynamic body **907**, ensuring a precise fit, as shown in FIG. **145**. The back plunger system wear ring **9129** further comprises a plurality of clearance chamfers **9177**. One clearance chamfer **9177** is located at the intersection of the front surface **9173** and outer surface **9175**. A second clearance chamfer **9177** is located at the intersection of the back surface **9174** and inner surface **9176**. The back plunger system wear ring **9129** may further comprise a plurality of small chamfers **9172** at the intersections of the remaining surfaces. These small chamfers **9172**, commonly referred to as 'break edges', facilitate safer handling and durability.

[0429] The retainer **913**, shown in FIGS. **154-159**, has a generally cylindrical shape. The retainer **913** comprises a front surface **928** and back surface **929** connected by an outer surface **930** and plunger bore **931**. The outer surface **930** and plunger bore **931** are concentric and their cylindrical axes are the longitudinal axis of the retainer **913**. The retainer **913** further comprises a plurality of

stud through holes **9165**, a plurality of locating dowel pin through holes **9166**, and a lubrication port **935**.

[0430] The plunger bore **931** comprises multiple sections, as shown in FIG. **157**. Beginning at the front surface **928** and continuing along the longitudinal axis to the back surface **929** the plunger bore **931** comprises the packing section **940**, a packing nut shoulder **941**, and packing nut threads **942**. The packing nut threads **942** comprise internal threads **9159** configured to receive the external threads of the packing nut **916** and a thread relief **966**. The thread relief **966** shown is a typical thread relief but may be similar to thread reliefs **366**, **466**, or **766**.

[0431] Referring now to FIGS. **156** and **158**, the stud through holes **9165** have a bore axis that is parallel to the longitudinal axis of the retainer **913** and connect the front **928** and back **929** surfaces. There are the same number of stud through holes **9165**, with the same spacing, as the threaded blind holes **9163** in the back surface **920** of the dynamic body **907**.

[0432] Referring now to FIGS. **156** and **159**, the locating dowel pin through holes **9166** have a bore axis that is parallel to the longitudinal axis of the retainer **913** and connect the front **928** and back **929** surfaces. Each locating dowel pin through hole **9166** comprises a counterbore **9202** originating from the front surface **928** and extending approximately twenty-five percent of the total length of the locating dowel pin through hole **9166**.

[0433] Referring back to FIG. **157**, the lubrication port **935** is a radial through bore connecting the outer surface **930** to the packing section **940** of the plunger bore **931**. The bore axis of the lubrication port **935** is perpendicular to the longitudinal axis of the retainer **913**. The lubrication port **935** may comprise a threaded section **937** adjacent to the outer surface **930** configured to receive a lubrication fitting (not shown) or plug (not shown).

[0434] The packing **915** comprises a junk ring **9178**, a backup ring **9179**, a plurality of V-rings **9180**, a front lantern ring **9181**, and a back lantern ring **9182**, as shown in FIGS. **145** and **149**. The junk ring **9178**, backup ring **9179**, and V-rings **9180** are typical of those used in the industry.

[0435] The front lantern ring **9181**, shown in FIGS. **168-171**, has an annular shape and comprises a front surface **9183** and back surface **9184**, which are connected by an outer surface **9185** and inner surface **9186**. The front surface **9183** comprises a V-shaped cutout **9187**. The V-shaped cutout **9187** is complementary to the V-shaped extrusion of the V-ring **9180** with which the front lantern ring **9181** interfaces. The front **9183** and back **9184** surfaces, excepting the portion of the front surface **9183** that is the V-shaped cutout **9187**, are parallel to each other and perpendicular to the longitudinal axis of the front lantern ring **9181**. The outer surface **9185** and inner surface **9186** are parallel to the longitudinal axis of the front lantern ring **9181**. The front lantern ring **9181** may further comprise a plurality of small chamfers **9172** at the intersection of the back surface **9184** and outer surface **9185** and the intersection of the back surface **9184** and inner surface **9186**. These small chamfers **9172**, commonly referred to as 'break edges', facilitate safer handling and durability.

[0436] The back lantern ring **9182**, shown in FIGS. **172-175**, has an annular shape and comprises a front surface **9188** and back surface **9189**, which are connected by an outer surface **9190** and an inner surface **9191**. The front **9188** and back **9189** surfaces are parallel to each other and perpendicular to the longitudinal axis of the back lantern ring **9182**.

[0437] The outer surface **9190** comprises a front section **9192** and a back section **9193** connected by a transition section **9194**. As shown in FIG. **175**, the front section **9192** begins at the front surface **9188** and continues parallel to the longitudinal axis of the back lantern ring **9182** until intersecting the transition section **9194**. The back section **9193** begins at the back surface **9189** and continues parallel to the longitudinal axis of the back lantern ring **9182** until intersecting the transition section **9194**. The front section **9192** has a smaller diameter than the back section **9193**. The transition section **9194** connects the smaller diameter front section **9192** to the larger diameter back section **9193** and has a short length along the longitudinal axis of the back lantern ring **9182** causing the transition section **9194** to be angled with the longitudinal axis.

[0438] Continuing with FIG. 175, the inner surface **9191** also comprises a front section **9195** and a back section **9196** connected by a transition section **9197**. The front section **9195** begins at the front surface **9188** and continues parallel to the longitudinal axis of the back lantern ring **9182** until intersecting the transition section **9197**. The back section **9196** begins at the back surface **9189** and continues parallel to the longitudinal axis of the back lantern ring **9182** until intersecting the transition section **9197**. The front section **9195** has a larger diameter than the back section **9196**. The transition section **9197** connects the larger diameter front section **9195** to the smaller diameter back section **9196** and has a short length along the longitudinal axis of the back lantern ring **9182** causing the transition section **9197** to be angled with the longitudinal axis.

[0439] The back lantern ring **9182** further comprises a plurality of lubrication holes **9198**. Each lubrication hole **9198** is a through bore connecting the front section **9192** of the outer surface **9190** to the front section **9195** of the inner surface **9191**. Each lubrication hole **9198** has a bore axis that is perpendicular to the longitudinal axis of the back lantern ring **9182**. In this embodiment there are eight lubrication holes **9198** spaced evenly around the circumference of the back lantern ring **9182**. The lubrication holes **9198** are also aligned longitudinally and located approximately at the longitudinal center of the front sections **9192** and **9195** of the outer **9190** and inner **9191** surfaces. There may be more or less lubrication holes **9198**, spaced in any manner as long as they connect the front sections **9192** and **9195** of the outer **9190** and inner **9191** surfaces.

[0440] The back lantern ring **9182** may further comprise a plurality of small chamfers **9172** at the intersection of all the surfaces **9188**, **9189**, **9190**, and **9191**. These small chamfers **9172**, commonly referred to as ‘break edges’, facilitate safer handling and durability.

[0441] Referring now to FIGS. 143-149, the assembly of the multi-piece fluid end **902** begins with the assembly of the plurality of dynamic sections **906**. Each dynamic section **906** is assembled by first, installing the flow control system wear ring seal **911** into the seal groove **9143** of the tapered bore **9140** of the flow control system wear ring section **9132** of the flow bore **922**. Second, inserting the flow control system wear ring **910**, back surface **9199** first, into the tapered bore **9140** until the back surface **9199** of the flow control system wear ring **910** contacts the flow control system wear ring shoulder **9133**. This may be a press fit. Third, a front plunger system wear ring **908** is inserted, front surface **9167** first, into the tapered bore **9149** of the plunger system wear ring section **9137** of the flow bore **922** of the dynamic body **907** until the front surface **9167** of the front plunger system wear ring **908** contacts the plunger system wear ring shoulder **9136**. This may be a slip fit. Fourth, a plunger system wear ring seal **909** is installed in the seal groove **9152** of the tapered bore **9149**. Fifth, a back plunger system wear ring **9129** is inserted, front surface **9173** first, into the tapered bore **9149** until the back surface **9174** is flush with the back surface **920** of the dynamic body **907**. This may be a press fit. This assembly procedure leaves a predetermined distance between the front surface **9173** of the back plunger system wear ring **9129** and the plunger system wear ring shoulder **9136**. That distance is greater than the length of the front plunger system wear ring **908**.

[0442] Each dynamic section **906** is then attached to the static section **905** by threading the static threads **924** of the outer surface **921** of the dynamic body **907** into the dynamic threads **981** of the static section **905** until the locating shoulder **959** abuts the back surface **980** of the static section **905**, as shown in FIG. 146. A spanner wrench (not shown) may be used to apply the required torque using the spanner wrench holes **927** in the back section **9131** of the outer surface **921** of the dynamic body **907**. The applicable components of the flow control system **9146** may now be installed into the sections of the static section **905** and flow bores **922** of the assembled dynamic bodies **907** configured to receive them.

[0443] Next, the retainer **913** is attached to a dynamic section **906** by first, threading a stud **9161** into each of the threaded blind holes **9163** in the back surface **920** of the dynamic body **907** and torquing to specification. Second, the locating dowel pins **9160** are inserted into the blind holes **9164** in the back surface **920** of the dynamic body **907**, as shown in FIG. 149. Third, the retainer

913 is oriented with the front surface **928** facing the back surface **920** of the dynamic body **907** and the stud through holes **9165** and locating dowel pin through holes **9166** aligned with the studs **9161** and locating dowel pins **9160** now protruding from the back surface **920** of the dynamic body **907**. Fourth, the front surface **928** of the retainer **913** is moved toward the back surface **920** of the dynamic body **907**, while simultaneously inserting the studs **9161** and locating dowel pins **9160** now protruding from the back surface **920** of the dynamic body **907** into the stud through holes **9165** and locating dowel pin through holes **9166** of the retainer **913**, until the two surfaces **928** and **920** contact. Fifth, the nuts **9162** are threaded onto the studs **9161** now protruding from the back surface **929** of the retainer **913** and torqued to specification.

[0444] Next, the packing **915** is inserted into the dynamic section **906** and the packing section **940** of the plunger bore **931** of the retainer **913**. For this embodiment, the components of the packing **915** are inserted in the following order with the V-shaped cutouts **9187** ‘pointing’ toward the back surface **920** of the dynamic body **907** where applicable; junk ring **9178**, backup ring **9179**, two V-rings **9180**, front lantern ring **9181**, and back lantern ring **9182**.

[0445] Next, the plunger seal **917** is installed in the packing nut **916** and the packing nut **916** is threaded into the packing nut threads **942** of the plunger bore **931** of the retainer **913**. Next, the plunger **918** is installed in the packing nut **916**, packing **915**, and dynamic section **906**. Lastly, the packing nut **916** is torqued to specification.

[0446] Once the assembly is complete the front section **9192** of the outer surface **9190** of the back lantern ring **9182** may extend across the joint formed by the contact between the back surface **9174** of the back plunger system wear ring **9129** and the front surface **928** of the retainer **913**. The smaller diameter of the front section **9192** creates a radial clearance between the front section **9192** of the outer surface **9190** of the back lantern ring **9182** and the inner surface **9176** of the back plunger system wear ring **9129** and also the packing section **940** of the plunger bore **931** of the retainer **913** as can be seen in FIG. 145.

[0447] In operation the extremely high forces generated by a power end connected to this multi-piece fluid end **902** may cause an axial or radial misalignment of the plunger **918** within the packing **915** and packing nut **916**. Any misalignment creates a radial displacement of the plunger **918**.

[0448] In the prior art the radial displacement creates a bending load on both the one-piece lantern ring and one-piece plunger system wear ring. In the prior art this bending load is transferred to the root of the tapered bore containing the plunger system wear ring resulting in premature failures of the dynamic body.

[0449] In this improvement the lantern ring is divided into the front lantern ring **9181** and back lantern ring **9182** and the plunger system wear ring is divided into the front plunger system wear ring **908** and the back plunger system wear ring **9129**. Dividing these components reduces the radial displacement of the components, thereby reducing the bending stress at the root **9151** of the tapered bore **9149** of the plunger system wear ring section **9137** of the flow bore **922** of the dynamic body **907**.

[0450] The reduction in radial displacement is accomplished by the clearance between the front section **9192** of the outer surface **9190** of the back lantern ring **9182** and the inner surface **9176** of the back plunger system wear ring **9129** and also the packing section **940** of the plunger bore **931** of the retainer **913** as can be seen in FIG. 145. Any radial displacement of a plunger **918** will first be ‘absorbed’ by the clearance preventing contact between the back lantern ring **9182** and the back plunger system wear ring **9129**.

[0451] If the radial displacement is greater than the clearance provided by the back lantern ring **9182**, the front section **9192** of the outer surface **9190** of the back lantern ring **9182** will contact the inner surface **9176** of the back plunger system wear ring **9129**. This contact will create a bending stress throughout the back plunger system wear ring **9129**. The predetermined distance between the front surface **9173** of the back plunger system wear ring **9129** and the plunger system wear ring

shoulder **9136** provides another buffer preventing contact between the back plunger system wear ring **9129** and the front plunger system wear ring **908**. The bending stress is transmitted into the dynamic body **907** but not at the root **9151** of the tapered bore **9149**.

[0452] If the radial displacement of the plunger **918** is large enough to eliminate the clearance of the back lantern ring **9182** and the back plunger system wear ring **9129** such that contact is made with the front plunger system wear ring **908** then a third level of buffer is provided by the loose fit, that is non-press fit, of the front plunger system wear ring **908** into the tapered bore **9149**. This loose fit will require even more radial displacement of the plunger **918** before the outer surface **9169** of the front plunger system wear ring **908** contacts the tapered bore **9149** and transmits bending stress to the root **9151** of the tapered bore **9149**.

[0453] These three levels of clearance, or buffer, essentially decouple the bending stress from the root **9151** of the tapered bore **9149**. This decoupling greatly reduces the likelihood of failure of the dynamic body **907** at the root **9151** of the tapered bore **9149** increasing the life and reducing operating costs of the multi-piece fluid end **902**.

[0454] Referring back to FIGS. **143**, **147**, and **149**, the multi-piece fluid end **902** may further comprise a dual discharge conduit adapter **9207**, a single discharge conduit adapter **9208**, a discharge conduit plug **9209**, and fasteners **9210** to attach each to the static section **905** of the fluid end body **903**.

[0455] Referring to FIGS. **143**, **144** and **147**, the static section **905** may further comprise a left surface **9211**, right surface **9212**, and plurality of flow bores **979** to house the flow control systems **9146** on a one-to-one basis. Each flow bore **979** is concentric with a plunger system **904** and comprises a dynamic thread **981**. The static section **905** may further comprise an upper discharge conduit **9213** and a lower discharge conduit **9214**. Each discharge conduit **9213**, **9214** is a straight bore that is parallel to the transverse axis and intersects the left and right surfaces **9211**, **9212**. Each discharge conduit **9213**, **9214** also partially intersects every flow bore **979**. The upper discharge conduit **9213** intersects the top of each flow bore **979** while the lower discharge conduit **2914** intersects the bottom of each flow bore **979**.

[0456] As can be seen in FIG. **147**, the dual discharge conduit adapter **9207**, single discharge conduit adapter **9208**, and discharge conduit plug **9209** are attached to the static section **905** using fasteners **9210**. Depending on the orientation of the high-pressure pump **100** at the well site and the number of high-pressure pumps **100** in use, one or more of each type of discharge conduit adapter **9207**, **9208** or discharge conduit plug **9209** may be attached to the static section **905**.

[0457] In operation pressurized fluid flows from each flow bore **979** into the upper and lower discharge conduits **9213**, **9214** then out of the static section **905** through each discharge conduit adapter **9207**, **9208**. The volume of fluid flowing out of each discharge conduit adapter **9207**, **9208** and through each discharge conduit **9213**, **9214** is not consistent and depends on many factors such as which discharge conduit adapters **9207**, **9208** and/or discharge conduit plugs **9209** are used and the flow restrictions downstream of each conduit adapter **9207**, **9208**. Furthermore, the flow rate at any point within either discharge conduit **9213**, **9214** is also dependent on the number and type of discharge conduit adapters **9207**, **9208** and discharge conduit plugs **9209** used.

[0458] For example, FIG. **147** shows a dual discharge conduit adapter **9207** attached to the left surface **9211** of the static section **905** such that the flow from both the discharge conduits **9213**, **9214** are combined. Further, a single discharge conduit adapter **9208** is attached to the right surface **9212** of the static section **905** to receive only the flow from the upper discharge conduit **9213**.

Lastly, a discharge conduit plug **9209** blocks flow out of the lower discharge conduit **9214** on the right side of the static section **905**. This configuration results in no flow out of the right end of lower discharge conduit **9214** resulting in minimal or erosion of the lower discharge conduit **9214** at that point, however at the left end of the lower discharge conduit **9214** flow will be approximately five times that of the right end because all the fluid must exit at the left end of the lower discharge conduit **9214**. This results in higher erosion rates at the end with the greater flow

rate. The numerous possible discharge configurations sometimes result in reduced life of the static section **905** due to concentrated flow rates as described above.

[0459] Referring now to FIGS. **176-193**, another embodiment of a multi-piece fluid end **1002** is shown. The multi-piece fluid end **1002** comprises a fluid end body **1003**, a plurality of plunger systems **1004**, a plurality of flow control systems **10146**, a plurality of discharge conduit adapters **10208**, a plurality of discharge manifolds **10215**, a plurality of clamps **10216**, a plurality of clamp fasteners **10217**, a plurality of seal carriers **10218**, a plurality of clamp seals **10219**, and a plurality of discharge adapter seals **10237**.

[0460] The fluid end body **1003**, shown in FIGS. **176-182**, comprises a static section **1005**, a plurality of dynamic sections **1006**, and a plurality of radial static seals **10220**. The static section **1005** comprises a plurality of flow bores **1079**. The flow bores **1079** are evenly spaced transversely and centered vertically within the static section **1005**. Each flow bore **1079** is a through bore connecting the front and rear surfaces **1099**, **1080** of the static section **1005** having a bore axis that is parallel to the longitudinal axis. The flow bores **1079** are configured to receive a portion of the flow control systems **10146** on a one-to-one basis and to facilitate the attachment of the dynamic sections **1006** to the static section **1005** on a one-to-one basis. As shown in FIGS. **178**, **181**, and **182** each flow bore **1079** comprises a dynamic thread **1081** proximate the rear surface **1080**, a thread relief **10221**, an entry chamfer **10222**, a first straight section **10223**, a clearance bevel **10224**, a static seal groove **10225**, a second straight section **10226**, and a static shoulder **10227**. Each flow bore **1079** further comprises a discharge section **10228** between the discharge plug seal **10229** and the front fluid routing plug seal **10230** of the flow control system **10146** installed within the flow bore **1079**, as shown in FIG. **178**.

[0461] Referring again to FIGS. **178** and **179**, the static section **1005** further comprises a plurality of upper discharge conduits **10213** and a plurality of lower discharge conduits **10214**. Each upper discharge conduit **10213** is a through bore that extends from the top surface **10231** of the static section **1005** to the discharge section **10228** of a corresponding flow bore **1079**. Each upper discharge conduit **10213** has a vertical bore axis that intersects the bore axis of its respective flow bore **1079** on a one-to-one basis.

[0462] The lower discharge conduits **10214** are mirror images of the upper discharge conduits **10213**. Each lower discharge conduit **10214** extends from the bottom surface **10232** of the static section **1005** to the discharge section **10228** of its corresponding flow bore **1079**.

[0463] Each discharge conduit **10213**, **10214** comprises a flow section **10233** at its intersection with the flow bore **1079**, extending to a counterbore **10234** that includes a seal groove **10235**, and further extending to a threaded section **10236** that intersects the respective surface **10231**, **10232**.

[0464] Each dynamic section **1006** comprises a dynamic body **1007**, shown in FIGS. **178**, **181**, and **182**. The dynamic body **1007** comprises a front surface **1019** and outer surface **1021**. The outer surface **1021** comprises a static seal section **1023**, static threads **1024**, and locating shoulder **1059**.

[0465] Returning to FIG. **180**, each flow control system **10146** comprises a discharge plug **10238**, a discharge plug insert **10239**, a discharge plug seal **10229**, a discharge valve **10240**, a fluid routing plug **10241**, a front fluid routing plug seal **10230**, a suction valve **10242**, a suction valve guide **10243**, and a suction valve guide insert **10244**.

[0466] The discharge valve **10240**, shown in FIGS. **183-185**, comprises a stem **10245** and a guide bore **10246**. The stem **10245** and guide bore **10246** are concentric and opposed. As shown in FIG. **180**, the stem **10245** is configured to be received in the discharge plug insert **10239** and the guide bore **10246** is configured to receive the discharge valve guide **10247** of the fluid routing plug **10241**. In this embodiment the stem **10245** is a circular boss and the guide bore **10246** is circular but other cross-sectional shapes may be used.

[0467] The fluid routing plug **10241**, shown in FIGS. **186-190**, comprises a discharge valve guide **10247** and suction valve guide bore **10248**. The discharge valve guide **10247** and suction valve guide bore **10248** are concentric and opposed. As shown in FIG. **180**, the discharge valve guide

10247 is configured to be received by the guide bore **10246** of the discharge valve **10240**. The suction valve guide bore **10248** is configured to receive the guide legs **10249** of the suction valve **10242**. In this embodiment the discharge valve guide **10247** is a circular boss and the suction valve guide bore **10248** is circular but other cross-sectional shapes may be used.

[0468] The suction valve **10242**, shown in FIGS. **191-193**, comprises a plurality of guide legs **10249** and a stem **10250**. The guide legs **10249** are opposed to the stem **10250** and located entirely within, and spaced evenly around, the circumference of a circle that is concentric with the stem **10250**. The circle has a diameter slightly smaller than the diameter of the suction valve guide bore **10248**. As shown in FIG. **180**, the guide legs **10249** are configured to be received by the suction valve guide bore **10248** and the stem **10250** is configured to be received by the suction valve guide insert **10244**. In this embodiment there are four guide legs **10249** but there may be more or less. Also, in this embodiment the stem **10250** is a circular boss but other cross-sectional shapes may be used. Likewise, the placement of the guide legs **10249** may be such that they are located within any cross-section shape.

[0469] The discharge conduit adapter **10208**, shown in FIGS. **178** and **179**, comprises a threaded end **10251** and opposed flanged end **10252**. The threaded end **10251** is configured to mate with the threaded sections **10236** of the upper and lower discharge conduits **10213** and **10214**. The flanged end **10252** is configured for use in a clamp joint used to connect the discharge manifolds **10215** to the static section **1005**. The clamp joint is the type commonly used in the industry for high pressure connections.

[0470] The discharge manifold **10215**, shown in FIGS. **176-179**, comprises a manifold body **10253** a plurality of inlet ports **10254** and a plurality of outlet ports **10255**. Each inlet port **10254** comprises a flanged end **10256** configured for use in a clamp joint. The inlet ports **10254** are perpendicular to the manifold body **10253** and spaced so they will be coincident with the discharge conduit adapters **10208** once assembled. The outlet ports **10255** are configured to be connected to high pressure conduits (not shown) which may be attached to the other discharge manifold **10215**, or to other discharge manifolds of other fluid ends, or to the wellhead.

[0471] Referring now to FIGS. **181** and **182**, during assembly of the fluid end **1002**, the dynamic sections **1006** must be threaded into the dynamic threads **1081** of the static section **1005**. The weight of the dynamic sections **1006**, the clearance between the mating dynamic threads **1081** and static threads **1024**, and the uneven resistance of the radial static seal **10220** as the static seal section **1023** initially contacts the radial static seal **10220**, all contribute to a misalignment between the longitudinal axis of the dynamic body **1007** and the axis of the flow bore **1079**. This causes the static seal section **1023** of the outer surface **1021** of the dynamic body **1007** to contact the first straight section **10223** of the flow bore **1079**, scarring the static seal section **1023**. However, the clearance bevel **10224** does not contact the static seal section **1023**. Once the dynamic section **1006** is fully threaded into its operating position, the scarring is outside the contact area of the radial static seal **10220** allowing leak free operation.

[0472] Referring now to FIGS. **178** and **180**, the assembly of the fluid end **1002** also requires the insertion of the components of the flow control systems **10146** into the flow bores **1079**. Once the components are assembled, in each flow bore **1079**, the stem **10250** of the suction valve **10242** is inserted in the suction valve guide insert **10244**, the guide legs **10249** of the suction valve **10242** are inserted in the suction valve guide bore **10248** of the fluid routing plug **10241**, the discharge valve guide **10247** of the fluid routing plug **10241** is inserted in the guide bore **10246** of the discharge valve **10240**, and the stem **10245** of the discharge valve **10240** is inserted in the discharge plug insert **10239**.

[0473] Referring now to FIGS. **176-179**, the assembly of the discharge manifolds **10215** to the static section **1005** is as follows. The discharge adapter seals **10237** are inserted in the seal grooves **10235** of the upper and lower discharge conduits **10213** and **10214**. Next, the threaded end **10251** of the discharge conduit adapters **10208** are threaded into the threaded sections **10236** of the upper

and lower discharge conduits **10213** and **10214**. Third, the clamp seals **10219** are installed in the seal carriers **10218** and the seal carriers **10218** are inserted in the flanged ends **10252** of the discharge conduit adapters **10208**. Fourth, the flanged ends **10256** of the inlet ports **10254** are placed over the seal carriers **10218**. At this point in the assembly the two flanged ends **10252** and **10256** will be in contact. Fifth, the clamps **10216** are placed around the flanged ends **10252** and **10256** and the clamp fasteners **10217** are installed.

[0474] Referring now to FIGS. **178** and **180**, in operation, the discharge and suction valves **10240** and **10242** alternate between fully open and fully closed positions by moving parallel to the bore axis of the flow bore **1079**. At all times, regardless of position, at least a portion of the discharge valve guide **10247** remains inserted in the guide bore **10246** of the discharge valve **10240** and at least a portion of the stem **10245** of the discharge valve **10240** remains inserted in the discharge plug insert **10239**. Similarly, at all times and in any position, at least a portion of the guide legs **10249** of the suction valve **10242** remain inserted in the suction valve guide bore **10248** of the fluid routing plug **10241**, and at least a portion of the stem **10250** of the suction valve **10242** remains inserted in the suction valve guide insert **10244**.

[0475] In the prior art, valves only supported on one end, by stems within guide inserts, often experience an axial misalignment between the longitudinal axes of the valves and the longitudinal axes of the fluid routing plug and/or the guide inserts. This misalignment, or drooping, leads to uneven and premature wear of the stems, guide inserts, and sealing faces of the valves. The present improvement, which provides continuous engagement of guide elements at both the front and rear ends of the discharge and suction valves **10240** and **10242**, eliminates or significantly reduces axial misalignment, thereby increasing the operational life of the valves **10240** and **10242**.

[0476] Referring now to FIGS. **178** and **179**, during operation, the pressurized fluid exits each flow bore **1079** through the upper or lower discharge conduits **10213** and **10214**. The fluid then immediately exits the fluid end body **1003** and enters a discharge manifold **10215** through an inlet port **10254**. The resulting pressure within the manifold bodies **10253** is evenly distributed along their length, creating uniform back pressure on each discharge valve **10240** within the fluid end body **1003**. This even pressure distribution eliminates the uneven pressure on discharge valves seen in prior art designs, thereby extending the life of the discharge valves **10240**.

[0477] Referring now to FIGS. **194-202**, another embodiment of a multi-piece fluid end **1102** is shown. The multi-piece fluid end **1102** comprises a fluid end body **1103**, a plurality of plunger systems **1104**, a plurality of flow control systems **11146**, a plurality of discharge conduit adapters **11208**, a plurality of discharge manifolds **11215**, a plurality of hammer union seals **11219**, and a plurality of discharge adapter seals **11237**.

[0478] The fluid end body **1103**, shown in FIGS. **194-199**, comprises a static section **1105**, a plurality of dynamic sections **1106**, and a plurality of radial static seals **11220**. The static section **1105** comprises a plurality of flow bores **1179**. The flow bores **1179** are evenly spaced transversely and centered vertically within the static section **1105**. Each flow bore **1179** is a through bore connecting the front and rear surfaces **1199**, **1180** of the static section **1105** having a bore axis that is parallel to the longitudinal axis. The flow bores **1179** are configured to receive a portion of the flow control systems **11146** on a one-to-one basis and to facilitate the attachment of the dynamic sections **1106** to the static section **1105** on a one-to-one basis. As shown in FIGS. **196** and **199** each flow bore **1179** comprises a dynamic thread **1181** proximate the rear surface **1180**, a thread relief **11221**, an entry chamfer **11222**, a first straight section **11223**, a static seal groove **11225**, a second straight section **11226**, and a static shoulder **11227**. Each flow bore **1179** further comprises a discharge section **11228** between the discharge plug seal **11229** and the front fluid routing plug seal **11230** of the flow control system **11146** installed within the flow bore **1179**, as shown in FIGS. **196** and **198**. In alternative embodiments the static shoulder **11227** may comprise a groove (not shown) configured to receive a face seal (not shown). The face seal may be used in addition to, or instead of, the radial static seal **11220**.

[0479] Referring again to FIGS. **196** and **197**, the static section **1105** further comprises a plurality of upper discharge conduits **11213** and a plurality of lower discharge conduits **11214**. Each upper discharge conduit **11213** is a through bore that extends from the top surface **11231** of the static section **1105** to the discharge section **11228** of a corresponding flow bore **1179**. Each upper discharge conduit **11213** has a vertical bore axis that intersects the bore axis of its respective flow bore **1179** on a one-to-one basis.

[0480] The lower discharge conduits **11214** are mirror images of the upper discharge conduits **11213**. Each lower discharge conduit **11214** extends from the bottom surface **11232** of the static section **1105** to the discharge section **11228** of its corresponding flow bore **1179**.

[0481] Each discharge conduit **11213**, **11214** comprises a flow section **11233** at its intersection with the flow bore **1179**, extending to a counterbore **11234** that includes a seal groove **11235**, and further extending to a threaded section **11236** that intersects the respective surface **11231**, **11232**. In alternative embodiments the threaded section **11236** may be omitted, and a bolt pattern may be formed in the top or bottom surface **11231**, **11232** around the discharge conduits **11213**, **11214** to facilitate the attachment of discharge conduit adapters **11208** configured with a flange for attachment to the static section **1105**.

[0482] Each dynamic section **1106** comprises a dynamic body **1107**, shown in FIGS. **196** and **199**. The dynamic body **1107** comprises a front surface **1119** and outer surface **1121**. The outer surface **1121** comprises a static seal section **1123**, a guide section **11261**, static threads **1124**, and locating shoulder **1159**. The guide section **11261** has a larger diameter than the static seal section **1123** and extends from the static threads **1124** to the static seal section **1123**.

[0483] Returning to FIG. **198**, each flow control system **11146** comprises a discharge plug **11238**, a discharge plug insert **11239**, a discharge plug seal **11229**, a discharge valve **11240**, a fluid routing plug **11241**, a front fluid routing plug seal **11230**, a suction valve **11242**, a suction valve guide **11243**, and a suction valve guide insert **11244**.

[0484] The discharge valve **11240**, shown in FIGS. **200-202**, comprises a stem **11245**, a guide bore **11246**, a front surface **11257**, and a plurality of passages **11258**. The stem **11245** and guide bore **11246** are concentric and opposed. As shown in FIG. **198**, the stem **11245** is configured to be received in the discharge plug insert **11239** and the guide bore **11246** is configured to receive the discharge valve guide **11247** of the fluid routing plug **11241**. In this embodiment the stem **11245** is a circular boss and the guide bore **11246** is circular but other cross-sectional shapes may be used.

[0485] The passages **11258** interconnect the guide bore **11246** and the front surface **11257**. Specifically, the passages **11258** interconnect that portion of the front surface **11257** that does not contact the fluid routing plug **11241** when the discharge valve **11240** is in the closed position as shown in FIG. **198**. In this embodiment there are three linear passages **11258** with circular cross sections spaced evenly around the circumference of the discharge valve **11240**. The longitudinal axis of each passage **11258** forms an acute angle with the longitudinal axis of the discharge valve **11240**. In alternative embodiments there may be more or less than three passages **11258** and they may not be linear and may have a cross-section that is not circular.

[0486] The fluid routing plug **11241**, shown in FIG. **198**, comprises a discharge valve guide **11247**. The discharge valve guide **11247** is configured to be received by the guide bore **11246** of the discharge valve **11240**. In this embodiment the discharge valve guide **11247** is a circular boss but other cross-sectional shapes may be used.

[0487] The discharge conduit adapter **11208**, shown in FIGS. **194-197**, comprises a threaded end **11251** and opposed female hammer union end **11252**. The threaded end **11251** is configured to mate with the threaded sections **11236** of the upper and lower discharge conduits **11213** and **11214**. The female hammer union end **11252** is configured for use in a hammer union joint used to connect the discharge manifolds **11215** to the static section **1105**. The hammer union joint is commonly used in the industry for high pressure connections and may be called a hammer wing union or wing joint in some sectors of the industry. In alternative embodiments the threaded end **11251** may be

replaced by a bolt on flange (not shown) configured to be attached to the top or bottom surface **11231**, **11232** of the static section **1105**.

[0488] The discharge manifold **11215**, shown in FIGS. **194-197**, comprises a manifold body **11253** having a plurality of inlet ports **11254**, a plurality of outlet ports **11255**, a plurality of wing nuts **11259** and a plurality of split retainers **11260**. Each inlet port **11254** comprises a male hammer union end **11256** configured for use in a hammer union joint. The inlet ports **11254** are perpendicular to the manifold body **11253** and spaced so they will be coincident with the discharge conduit adapters **11208** once assembled. The outlet ports **11255** are configured to be connected to high pressure conduits (not shown) which may be attached to the other discharge manifold **11215**, or to other discharge manifolds of other fluid ends, or to the wellhead.

[0489] Referring now to FIG. **199**, during assembly of the multi-piece fluid end **1102**, the dynamic sections **1106** must be threaded into the dynamic threads **1181** of the static section **1105**. The weight of the dynamic sections **1106**, the clearance between the mating dynamic threads **1181** and static threads **1124**, and the uneven resistance of the radial static seal **11220** as the static seal section **1123** initially contacts the radial static seal **11220**, all contribute to a misalignment between the longitudinal axis of the dynamic body **1107** and the axis of the flow bore **1179**. This causes the guide section **11261** of the outer surface **1121** of the dynamic body **1107** to contact the first straight section **11223** of the flow bore **1179**, scarring the guide section **11261**. However, the smaller diameter static seal section **1123** does not contact any portion of the flow bore **1179** thus avoiding any scarring to the static seal section **1123** and allowing leak free operation.

[0490] Referring now to FIGS. **196** and **198**, the assembly of the multi-piece fluid end **1102** also requires the insertion of the components of the flow control systems **11146** into the flow bores **1179**. Once the components are assembled, in each flow bore **1179**, the discharge valve guide **11247** of the fluid routing plug **11241** is inserted in the guide bore **11246** of the discharge valve **11240**, and the stem **11245** of the discharge valve **11240** is inserted in the discharge plug insert **11239**.

[0491] Referring to FIGS. **194-197**, the assembly of each discharge manifold **11215** must be completed prior to its attachment to the static section **1105**. For each inlet port **11254** the following steps are performed: First, a wing nut **11259** is slid onto the male hammer union end **11256** of the inlet port **11254**. Next, the components of the split retainer **11260**, typically comprising three separate pieces, are positioned around inlet port **11254**, encircling it entirely. Once the split retainer **11260** is in place, the wing nut **11259** is slid back down toward the male hammer union end **11256**. The split retainer **11260** prevents the wing nut **11259** from sliding completely off the inlet port **11254** and the wing nut **11259** prevents the split retainer from falling off the inlet port **11254**. This also allows relative movement between the wing nut **11259** and inlet port **11254**. In alternate embodiments the split retainer **11260** may be attached to the inlet port **11254** with a fastener.

[0492] The assembly of the discharge manifolds **11215** to the static section **1105** is as follows. First, the discharge adapter seals **11237** are inserted in the seal grooves **11235** of the upper and lower discharge conduits **11213** and **11214**. Second, the threaded ends **11251** of the discharge conduit adapters **11208** are threaded into the threaded sections **11236** of the upper and lower discharge conduits **11213** and **11214**. Third, the hammer union seals **11219** are installed in the female hammer union ends **11252** of the discharge conduit adapters **11208**. Fourth, the male hammer union ends **11256** of the inlet ports **11254** are placed in corresponding female hammer union ends **11252** of the discharge conduit adapters **11208**. At this point in the assembly the two hammer union ends **11252** and **11256** will be in contact. Fifth, the internal threads of the wing nuts **11259** are threaded onto the external threads of the female hammer union end **11252** of the discharge conduit adapters **11208**. Sixth, a hammer, or other suitable impact tool, is used to tighten each wing nut **11259** securely, ensuring proper engagement and sealing.

[0493] Referring now to FIGS. **196** and **198**, in operation, any fluid or other material trapped within the guide bore **11246** exits the discharge valve **11240** through the passages **11258** as the

discharge valve **11240** is closed. The material is expelled into the annular space **11262** formed by the front surface **11257** of the discharge valve **11240** and the fluid routing plug **11241** when the discharge valve **11240** is closed. In alternative embodiments passages to facilitate the expulsion of trapped material may be included in the discharge valve guide **11247** in addition to, or instead of, the passages **11258** in the discharge valve **11240**.

[0494] Referring now to FIGS. **203-218**, another embodiment of a multi-piece fluid end **1202** is shown. The multi-piece fluid end **1202** comprises a fluid end body **1203**, a plurality of plunger systems **1204**, a plurality of flow control systems **12146**, a plurality of discharge conduit adapters **12208**, a plurality of discharge manifolds **12215**, a plurality of hammer union seals **12219**, a plurality of discharge adapter seals **12237**, and a plurality of fasteners **12263**.

[0495] The fluid end body **1203**, shown in FIGS. **203-206** and **208**, comprises a static section **1205**, a plurality of dynamic sections **1206**, and a plurality of axial static seals **12333**. The static section **1205** comprises a plurality of flow bores **1279**. The flow bores **1279** are evenly spaced transversely and centered vertically within the static section **1205**. Each flow bore **1279** is a through bore connecting the front and rear surfaces **1299**, **1280** of the static section **1205** having a bore axis that is parallel to the longitudinal axis. The flow bores **1279** are configured to receive a portion of the flow control systems **12146** on a one-to-one basis and to facilitate the attachment of the dynamic sections **1206** to the static section **1205** on a one-to-one basis. As shown in FIGS. **205** and **208** each flow bore **1279** comprises a dynamic thread **1281** proximate the rear surface **1280**, a thread relief **12221**, an entry chamfer **12222**, a straight section **12223**, and a static shoulder **12227**. The static shoulder **12227** comprises an axial static seal groove **12264**. Each flow bore **1279** further comprises a discharge section **12228** between the discharge plug seal **12229** and the front fluid routing plug seal **12230** of the flow control system **12146** installed within the flow bore **1279**, as shown in FIG. **207**. Each flow bore **1279** further comprises an internal front retainer thread **12317** proximate the front surface **1299**. The internal front retainer thread **12317** is configured to receive the front retainer **12315** of the flow control system **12146** and the installation tool **12309** as shown in FIG. **219**.

[0496] Referring again to FIGS. **205**, **206**, and **209** the static section **1205** further comprises a plurality of upper discharge conduits **12213** and a plurality of lower discharge conduits **12214**. Each upper discharge conduit **12213** is a through bore that extends from the top surface **12231** of the static section **1205** to the discharge section **12228** of a corresponding flow bore **1279**. Each upper discharge conduit **12213** has a vertical bore axis that intersects the bore axis of its respective flow bore **1279** on a one-to-one basis.

[0497] The lower discharge conduits **12214** are mirror images of the upper discharge conduits **12213**. Each lower discharge conduit **12214** extends from the bottom surface **12232** of the static section **1205** to the discharge section **12228** of its corresponding flow bore **1279**.

[0498] Each discharge conduit **12213**, **12214**, shown in FIG. **209**, comprises a flow section **12233** at its intersection with the flow bore **1279**, extending to a counterbore **12234** that intersects the respective top or bottom surface **12231** or **12232**. The counterbore **12234** includes a seal groove **12235**.

[0499] The top and bottom surfaces **12231** and **12232** of the static section **1205** comprise a plurality of blind threaded holes **12265** arranged around each discharge conduit **12213**, **12214**, as shown in FIG. **209**. The blind threaded holes **12265** facilitate the attachment of discharge conduit adapters **12208** to the static section **1205**.

[0500] Each dynamic section **1206** comprises a dynamic body **1207** and an insert **12307**, shown in FIGS. **205**, **207** and **208**. The dynamic body **1207** comprises a front surface **1219** and outer surface **1221**. The outer surface **1221** comprises a nose section **1223**, a nose chamfer **12266**, static threads **1224**, and locating shoulder **1259**.

[0501] Returning to FIG. **207**, each flow control system **12146** comprises a front retainer **12315**, a discharge plug **12238**, a discharge plug insert **12239**, a discharge plug seal **12229**, a discharge valve

spring **12328**, a discharge valve **12240**, a fluid routing plug **12241**, a front fluid routing plug seal **12230**, a rear fluid routing plug seal **12308**, a suction valve **12242**, a suction valve spring **12306**, a suction valve guide **12243**, and a suction valve guide insert **12244**.

[0502] The discharge valve **12240**, shown in FIGS. **210-212**, comprises a stem **12245**, a guide bore **12246**, a front surface **12257**, and a plurality of passages **12258**. The stem **12245** and guide bore **12246** are concentric and opposed. As shown in FIG. **207**, the stem **12245** is configured to be received by the discharge plug insert **12239** and the guide bore **12246** is configured to receive the discharge valve guide **12247** of the fluid routing plug **12241**.

[0503] In this embodiment the stem **12245** is a circular boss and the guide bore **12246** is circular but other cross-sectional shapes may be used. The front surface **12257** comprises the strike face of the valve insert **12267**, the strike face **12268**, a relief section **12269**, and a guide bore section **12270**, all of which are concentric with the stem **12245** and guide bore **12246**.

[0504] The strike face of the valve insert **12267** and strike face **12268** are known in the art. The relief section **12269** is a concave frustoconical section between the strike face **12268** and the guide bore section **12270**. The guide bore section **12270** is flat, perpendicular to the longitudinal axis, and provides a surface for the formation of the guide bore **12246**.

[0505] The passages **12258** interconnect the guide bore **12246** and the relief section **12269**. In this embodiment there are three linear passages **12258** with circular cross sections spaced evenly around the circumference of the discharge valve **12240**. The longitudinal axis of each passage **12258** forms an acute angle with the longitudinal axis of the discharge valve **12240**. In alternative embodiments there may be more or less than three passages **12258** and they may not be linear and may have a cross-section that is not circular.

[0506] The fluid routing plug **12241**, shown in FIGS. **213-218**, comprises a discharge surface **12271**, a plurality of discharge fluid passages **12272**, and a suction surface **12273**. The discharge surface **12271** comprises a discharge seat **12274**, a first transition radius **12275**, a counterbore **12276**, a second transition radius **12277**, a central base **12278**, and a discharge fluid passage circle **12292**.

[0507] The discharge seat **12274** is a frustoconical surface forming an acute angle **12279** with the longitudinal axis **12280** of the fluid routing plug **12241**, as shown in FIG. **218**.

[0508] The counterbore **12276** has a bore axis **12281** colinear with the longitudinal axis **12280** and extends from the first transition radius **12275** to the second transition radius **12277**. The counterbore **12276** also comprises a radius **12282**.

[0509] The central base **12278** is also a frustoconical surface forming an acute angle **12283** with the longitudinal axis **12280**, as shown in FIG. **218**. The central base **12278** extends from the second transition radius **12277** to the center of the discharge surface **12271**. The central base **12278** comprises a cylindrical discharge valve guide **12247**. The discharge valve guide **12247** is concentric with the longitudinal axis **12280** of the fluid routing plug **12241**. As shown in FIG. **205**, the discharge valve guide **12247** is configured to be received by the guide bore **12246** of the discharge valve **12240**.

[0510] The discharge valve guide **12247** is a circular boss that comprises a third transition radius **12284**, an outer surface **12285**, a front surface **12286**, a blind bore **12287**, and a plurality of passages **12288**. The blind bore **12287** is concentric with the discharge valve guide **12247**. The blind bore **12287** comprises a countersink **12289**, a counterbore **12290** and a threaded section **12291** configured to receive an externally threaded removal tool (not shown). In this embodiment the discharge valve guide **10247** is a circular boss but other cross-sectional shapes may be used.

[0511] The discharge fluid passage circle **12292** is concentric with the longitudinal axis **12280**. The radius **12293** of the discharge fluid passage circle **12292** is the distance from the longitudinal axis **12280** of the fluid routing plug **12241** to the point of intersection **12294** between the longitudinal axis **12295** of any discharge fluid passage **12272** and a discharge seat extension line **12296**, as shown in FIG. **218**. The discharge seat extension line **12296** extends from the discharge seat **12274**

at the same acute angle **12279** of the discharge seat **12274** in any cross-sectional view of the discharge seat **12274**, as shown in FIG. **218**. In this embodiment the radius **12282** of the counterbore **12276** is greater than 30% larger than the radius **12293** of the discharge fluid passage circle **12292**.

[0512] The passages **12288** interconnect the outer surface **12285** and the threaded section **12291** of the blind bore **12287**. In this embodiment there are three linear passages **12288** with circular cross sections spaced evenly around the circumference of the discharge valve guide **12247**. The longitudinal axis of each passage **12288** forms an acute angle with the longitudinal axis **12280** of the fluid routing plug **12241**. In alternative embodiments there may be more or less than three passages **12288**, they may not form an acute angle with the longitudinal axis **12280**, may not be linear, and may not have a circular cross-section.

[0513] Each discharge fluid passage **12272** comprises a longitudinal axis **12295** forming an acute angle **12332** with the longitudinal axis **12280** of the fluid routing plug **12241**, as shown in FIG. **218**. Each discharge fluid passage **12272** further comprises a discharge surface section **12297**, an intermediate section **12298**, and a suction surface section **12299**, as shown in FIG. **217**. The discharge surface section **12297** comprises a bevel **12300**. The suction surface section **12299** begins at the suction surface **12273** and ends at the intermediate section **12298**. The intermediate section **12298** begins at the end of the suction surface section **12299** and ends at the discharge surface section **12297**. The discharge surface section **12297** begins at the end of the intermediate section **12298** and ends at the discharge surface **12271**. The cross-sectional areas of the suction surface section **12299** and the intermediate section **12298** are equal. The cross-sectional area of the discharge surface section **12297** is equal to the cross-sectional area of the intermediate section **12298** at the point of transition from the intermediate section **12298** to the discharge surface section **12297** but, due to the bevel **12300**, increases until the end of the discharge surface section **12297** at the discharge surface **12271**. The cross-sectional area of the discharge surface section **12297** at the discharge surface **12271** is greater than 50% larger than the cross-sectional area of the intermediate section **12298** or the suction surface section **12299**.

[0514] Each discharge conduit adapter **12208**, shown in FIGS. **203-206** and **209** comprises a female hammer union end **12252**, a bolt-on flange **12301**, a circular boss **12251**, and a flow bore **12302**.

[0515] The bolt-on flange **12301** has the general form of a flat rectangular prism, with a height significantly smaller than its width and depth. The bolt-on flange **12301** comprises a top surface **12303**, an opposing and parallel bottom surface **12304**, and a plurality of through bores **12305** extending between the top surface **12303** and bottom surface **12304**. The through bores **12305** are configured to receive fasteners **12263** and are arranged with the same spacing as the blind threaded holes **12265** on the top and bottom surfaces **12231**, **12232** of the static section **1205**. Each through bore **12305** has a bore axis perpendicular to the top and bottom surfaces **12303**, **12304**.

[0516] The female hammer union end **12252** extends perpendicularly from the top surface **12303** of the bolt-on flange **12301** and is centered on the top surface **12303**. The female hammer union end **12252** is configured for use in a hammer union joint, which connects the discharge manifold **12215** to the static section **1205** via the discharge conduit adapters **12208**. This hammer union joint is commonly used in the industry for high pressure connections and may also be referred to as a hammer wing union or wing joint in certain sectors of the industry.

[0517] The circular boss **12251** extends perpendicularly from the bottom surface **12304** of the bolt-on flange **12301**. It is centered on the bottom surface **12304** and is concentric with the female hammer union end **12252**. Each circular boss **12251** is configured to mate with a counterbore **12234** of the upper or lower discharge conduits **12213** and **12214** in the static section **1205**.

[0518] The flow bore **12302** has a bore axis that is perpendicular to the top and bottom surfaces **12303**, **12304** of the bolt-on flange **12301**. The bore axis is concentric with both the female hammer union end **12252** and the circular boss **12251**.

[0519] The discharge manifold **12215**, shown in FIGS. **203-206** and **209**, comprises a manifold body **12253** having a plurality of inlet ports **12254**, a plurality of outlet ports **12255**, a plurality of wing nuts **12259** and a plurality of split retainers **12260**. Each inlet port **12254** comprises a male hammer union end **12256** configured for use in a hammer union joint. The inlet ports **12254** are perpendicular to the manifold body **12253** and spaced so they will be coincident with the discharge conduit adapters **12208** once assembled. The outlet ports **12255** are configured to be connected to high pressure conduits (not shown) which may be attached to the other discharge manifold **12215**, or to other discharge manifolds of other fluid ends, or to the wellhead.

[0520] Referring now to FIG. **208**, during assembly of the multi-piece fluid end **1202**, the dynamic sections **1206** must be threaded into the dynamic threads **1281** of the static section **1205**. The weight of the dynamic sections **1206** and the clearance between the mating dynamic threads **1281** and static threads **1224** can cause a misalignment between the longitudinal axis of the dynamic body **1207** and the axis of the flow bore **1279**. This misalignment may result in the nose section **1223** of the outer surface **1221** of the dynamic body **1207** contacting the straight section **12223** of the flow bore **1279**, potentially scarring the nose section **1223**. However, the axial static seal **12301** seals against the front surface **1219** of the dynamic body **1207**, rendering any scarring of the outer surface **1221** inconsequential.

[0521] Referring now to FIGS. **205** and **207**, the assembly of the multi-piece fluid end **1202** also requires the insertion of the components of the flow control systems **12146** into the flow bores **1279**. First, the suction valve guide insert **12244** is inserted in the suction valve guide **12243**. There may be an interference fit between the suction valve guide insert **12244** and the suction valve guide **12243** to prevent relative motion between the two components during operations. Second, the suction valve guide **12243**, with the assembled suction valve guide insert **12244**, is installed in the flow bore **1279**. Third, the suction valve spring **12306** is inserted in the flow bore **1279**. Fourth, the suction valve **12242** is installed in the flow bore **1279**, with the stem **12250** being inserted into the suction valve guide insert **12244**, thereby trapping the suction valve spring **12306** between the suction valve guide **12243** and the suction valve **12242**. Fifth, the front fluid routing plug seal **12230** and the rear fluid routing plug seal **12308** are installed on the fluid routing plug **12241**. Sixth, the fluid routing plug **12241** is inserted into the flow bore **1279**.

[0522] The fits between the rear fluid routing plug seal **12308** and the insert **12307**, and between the front fluid routing plug seal **12230** and flow bore **1279**, are so tight that a significant force must be applied to the discharge surface **12271** of the fluid routing plug **12241** to install it. The most accessible portion of the discharge surface **12271** is the front surface **12286** of the discharge valve guide **12247**. However, applying the required force to the front surface **12286** of the discharge valve guide **12247** will distort the discharge valve guide **12247** rendering it inoperable.

[0523] To address this problem an installation tool **12309** is used. Referring to FIGS. **219-223**, the installation tool **12309** comprises an anchor **12310** and a push rod **12311**. The anchor **12310** is generally shaped like a thin cylinder, with a rim **12312** connected to a hub **12313** by spokes **12314**. In this embodiment, there are three spokes **12314**, but the number of spokes **12314** may vary. The rim **12312** comprises a threaded outer surface **12316** configured to mate with the internal front retainer threads **12317** of the flow bore **1279**.

[0524] The hub **12313** is located concentrically within the rim **12312** of the anchor **12310** and is connected to the rim **12312** by the spokes **12314**. The hub **12313** comprises a concentric threaded through hole **12318** configured to receive the external threads **12319** of the stem **12320** of the push rod **12311**.

[0525] The push rod **12311** comprises the stem **12320**, the cup **12321** and the nut **12322**. The cup **12321** comprises a rear surface **12323**, a clearance bore **12324**, and a front surface **12325**. The rear surface **12323** is configured to contact the central base **12278** of the discharge surface **12271** of the fluid routing plug **12241**. Specifically, the rear surface **12323** features a frustoconical surface complementary to the frustoconical surface of the central base **12278**, as shown in FIG. **220**.

[0526] The clearance bore **12324** is a blind bore that is concentric with the stem **12320**. It is configured to receive the discharge valve guide **12247** without contacting the outer surface **12285** during the insertion of the fluid routing plug **12241** within the fluid bore **1279**.

[0527] The stem **12320** is a cylindrical shaft concentric with the cup **12321** and integrally formed on the front surface **12325** of the cup **12321**. The stem **12320** extends from the front surface **12325** and comprises an external thread **12319** along its entire length.

[0528] The nut **12322** is a standard hex nut with internal threads **12326** configured to thread on the external thread **12319** of the stem **12320**.

[0529] Prior to use in the assembly of the multi-piece fluid end **1202**, a fluid end section, or other fluid end, the installation tool **12309** must be assembled. Referring to FIGS. **221-223**, the external threads **12319** of the stem **12320** are first threaded into the threaded through hole **12318** of the hub **12313** until the end of the stem **12320** protrudes from the hub **12313** by more than half the length of the stem **12320**. Next, the internal threads **12326** of the nut **12322** are threaded onto the external threads **12319** of the protruding end of the stem **12320** until all the internal threads **12326** are fully engaged with the external threads **12319**. Once the nut **12322** is in place, it is welded to the stem **12320**. The welds **12327** can best be seen in FIG. **223**. As can be seen the anchor **12310** is trapped between the cup **12321** and the nut **12322** of the push rod **12311**. This completes the assembly of the installation tool **12309**.

[0530] Returning now to the assembly of the multi-piece fluid end **1202**, specifically to step six, the insertion of the fluid routing plug **12241** into the flow bore **1279**, the installation tool **12309** may be used. Referring now to FIG. **220**, once the fluid routing plug **12241** has been inserted as far as possible within the flow bore **1279** by hand, the installation tool **12309** may be employed. First, the push rod **12311** is turned until the rear surface **12323** of the cup **12321** abuts the hub **12313** of the anchor **12310**. Next, the threaded outer surface **12316** of the rim **12312** of the anchor **12310** is threaded into the internal front retainer threads **12317** of the flow bore **1279**. In this embodiment, the anchor **12310** is shown threaded to the full depth of the internal front retainer threads **12317** but may be threaded in until the only the full length of the threaded outer surface **12316** of the rim **12312** engages the internal front retainer threads **12317**. Third, using the nut **12322**, the push rod **12311** is turned clockwise (assuming right hand threads), extending the cup **12321** from the hub **12313** until the clearance bore **12324** goes over the discharge valve guide **12247** and the rear surface **12323** of the cup **12321** contacts the central base **12278** of the discharge surface **12271** of the fluid routing plug **12241**. Once the rear surface **12323** contacts the central base **12278**, torsional resistance increases to the point where a wrench or other torsional amplifier must be used to continue turning the push rod **12311** using the nut **12322**. As the nut **12322** is turned, the push rod **12311** continues to force the fluid routing plug **12241** into the flow bore **1279** until full insertion is achieved. Once the fluid routing plug **12241** is fully inserted, the push rod **12311** may be retracted by turning the nut **12322** counterclockwise and the anchor **12310** unscrewed from the internal front retainer threads **12317**, removing the installation tool **12309** for the continuation of the assembly process.

[0531] The remaining components of the flow control system **12146** may now be inserted in the flow bore **1279**. Continuing with step seven and referring to FIGS. **205** and **207**, the discharge valve **12240** is inserted in the flow bore **1279** with the front surface **12257** first, and the discharge valve guide **12247** of the fluid routing plug **12241** is inserted in the guide bore section **12270** of the front surface **12257**. Eighth, the discharge valve spring **12328** is inserted in the flow bore **1279**. Ninth, after the discharge plug seal **12229** and discharge plug insert **12239** have been installed in the discharge plug **12238**, the assembly is inserted in the flow bore **1279** such that the stem **12245** of the discharge valve **12240** is guided into the discharge plug insert **12239**. The discharge valve spring **12328** is now trapped between the discharge valve **12240** and the discharge plug **12238**. Finally, the front retainer **12315** is threaded into the internal front retainer threads **12317** and torqued to specification, thus completing the assembly of the flow control system **12146** within the

multi-piece fluid end **1202**.

[0532] Referring to FIGS. **203-206** and **209**, the assembly of each discharge manifold **12215** must be completed prior to its attachment to the static section **1205**. For each inlet port **12254** the following steps are performed: First, a wing nut **12259** is slid onto the male hammer union end **12256** of the inlet port **12254**. Next, the components of the split retainer **12260**, typically comprising three separate pieces, are positioned around inlet port **12254**, encircling it entirely. Once the split retainer **12260** is in place, the wing nut **12259** is slid back down toward the male hammer union end **12256**. The split retainer **12260** prevents the wing nut **12259** from sliding completely off the inlet port **12254**, and the wing nut **12259** prevents the split retainer **12260** from falling off the inlet port **12254**. This arrangement allows relative movement between the wing nut **12259** and inlet port **12254**. In alternate embodiments, the split retainer **12260** may be attached to the inlet port **12254** with a fastener.

[0533] The assembly of the discharge manifolds **12215** to the static section **1205** is as follows. First, the discharge adapter seals **12237** are inserted in the seal grooves **12235** of the upper and lower discharge conduits **12213** and **12214**. Second, the circular bosses **12251** of the discharge conduit adapters **12208** are inserted into the counterbores **12234** of the upper and lower discharge conduits **12213** and **12214** until the bottom surface **12304** of the bolt-on flange **12301** contacts the top surface **12231** of the static section **1205**. Third, the through bores **12305** of each bolt-on flange **12301** are aligned with the blind threaded holes **12265** of the top surface **12231** and bottom surface **12232** of the static section **1205**. Fourth, the fasteners **12263** are inserted through the through bores **12305** and torqued into the blind threaded holes **12265** thus securing the discharge conduit adapters **12208** to the static section **1205**. Fifth, the hammer union seals **12219** are installed in the female hammer union ends **12252** of the discharge conduit adapters **12208**. Sixth, the male hammer union ends **12256** of the inlet ports **12254** are placed in corresponding female hammer union ends **12252** of the discharge conduit adapters **12208**. At this point in the assembly the two hammer union ends **12252** and **12256** will be in contact. Seventh, the internal threads of the wing nuts **12259** are threaded onto the external threads of the female hammer union ends **12252** of the discharge conduit adapters **12208**. Eighth, a hammer, or other suitable impact tool, is used to tighten each wing nut **12259** securely, ensuring proper engagement and sealing. In this embodiment, each fastener **12263** comprises a stud **12329**, a leveling washer **12330**, and a flanged nut **12331**. In alternative embodiments, the fastener **12263** may omit the leveling washer **12330**, use a standard washer, comprise a flanged bolt, or use any other fasteners or fastening systems known in the art.

[0534] Referring now to FIGS. **205** and **207**, during operation, any fluid or other material trapped within the guide bore **12246** of the discharge valve **12240** or the blind bore **12287** of the discharge valve guide **12247** exits through the passages **12258** and **12288** as the discharge valve **12240** is closed. The material is expelled into the annular space **12262** formed by the front surface **12257** of the discharge valve **12240** and the fluid routing plug **12241** when the discharge valve **12240** is closed. In alternative embodiments only a single set of passages, either **12258** or **12288**, may be employed to facilitate the expulsion of trapped material.

[0535] Other improvements described herein provide further operational benefits by combining to reduce erosion of the front surface **12257** of the discharge valve **12240** and the discharge surface **12271** of the fluid routing plug **12241**. Specifically, as fluid exits the discharge fluid passage **12272**, the increased cross-sectional area of the discharge surface section **12297** reduces fluid velocity, thereby reducing erosion around the discharge fluid passage **12272** on the discharge surface **12271** of the fluid routing plug **12241**. The reduced velocity also reduces the impact force of the fluid on the front surface **12257** of the discharge valve **12240**. Additionally, the acute angle **12332** of the longitudinal axis **12295** of the discharge fluid passages **12272** ensures that fluid is not directly aimed at the discharge valve guide **12247**, thereby extending the service life of the fluid routing plug **12241** by reducing the erosion rate of discharge valve guide **12247**. Lastly, the relief section **12269** on the front surface **12257** of the discharge valve **12240** provides a smooth flow

transition as fluid flows past the front surface **12257** of the discharge valve **12240** into the discharge section **12228** area of the flow bore **1279**. These improvements collectively reduce erosion and extend the service life of both the discharge valve **12240** and the fluid routing plug **12241**.

[0536] Referring now to FIGS. **224-244**, another embodiment of a multi-piece fluid end **1302** is shown. The multi-piece fluid end **1302** comprises a fluid end body **1303**, a plurality of plunger systems **1304**, a plurality of flow control systems **13146**, a plurality of discharge conduit adapters **13208**, a plurality of discharge manifolds **13215**, a plurality of clamps **13216**, a plurality of clamp fasteners **13217**, a plurality of seal carriers **13218**, a plurality of clamp seals **13219**, a plurality of discharge adapter seals **13237**, and a plurality of fasteners **12263**.

[0537] The plunger system **1304** depicted in FIGS. **224-244** is largely similar to plunger system **904** shown in FIG. **149**, with a single-piece lantern ring. That is, the system **1304** comprises a plunger **1318**, bolted on retainer **1313** configured to thread to a packing nut **1316**, and various seals making up the packing **1315** (similar to packing **915**). These individual items are not repeated here, but an artisan can understand the systems **904** and **1304** to be largely the same.

[0538] The fluid end body **1303**, shown in FIGS. **224-228** and **230** comprises a static section **1305**, a plurality of dynamic sections **1306**, a plurality of radial static seals **13220**, and a plurality of axial static seals **13333**. The static section **1305** comprises a plurality of flow bores **1379**. The flow bores **1379** are evenly spaced transversely and centered vertically within the static section **1305**. Each flow bore **1379** is a through bore connecting the front and rear surfaces **1399**, **1380** of the static section **1305** having a bore axis that is parallel to the longitudinal axis. The flow bores **1379** are configured to receive a portion of the flow control systems **13146** on a one-to-one basis and to facilitate the attachment of the dynamic sections **1306** to the static section **1305** on a one-to-one basis. As shown in FIGS. **226** and **230** each flow bore **1379** comprises a dynamic thread **1381** proximate the rear surface **1380**, a thread relief **13221**, an entry chamfer **13222**, a first straight section **13223**, a radial static seal groove **13225**, a second straight section **13226**, and a static shoulder **13227**. The static shoulder **13227** comprises an axial static seal groove **13264**. Each flow bore **1379** further comprises a discharge section **13228** between the discharge plug seal **13229** and the front fluid routing plug seal **13230** of the flow control system **13146** installed within the flow bore **1379**, as shown in FIG. **229**. Each flow bore **1379** further comprises an internal front retainer thread **13317** proximate the front surface **1399**. The internal front retainer thread **13317** is configured to receive the front retainer **13315** as shown in FIG. **226**.

[0539] Referring again to FIGS. **226**, **227**, and **231** the static section **1305** further comprises a plurality of upper discharge conduits **13213** and a plurality of lower discharge conduits **13214**. Each upper discharge conduit **13213** is a through bore that extends from the top surface **13231** of the static section **1305** to the discharge section **13228** of a corresponding flow bore **1379**. Each upper discharge conduit **13213** has a vertical bore axis that intersects the bore axis of its respective flow bore **1379** on a one-to-one basis.

[0540] The lower discharge conduits **13214** are mirror images of the upper discharge conduits **13213**. Each lower discharge conduit **13214** extends from the bottom surface **13232** of the static section **1305** to the discharge section **13228** of its corresponding flow bore **1379**.

[0541] Each discharge conduit **13213**, **13214**, shown in FIG. **231**, comprises a flow section **13233** at its intersection with the flow bore **1379**, extending to a counterbore **13234** that intersects the respective top or bottom surface **13231** or **13232**. The counterbore **13234** includes a seal groove **13235**.

[0542] The top and bottom surfaces **13231** and **13232** of the static section **1305** comprise a plurality of blind threaded holes **13265** arranged around each discharge conduit **13213**, **13214**, as shown in FIG. **231**. The blind threaded holes **13265** facilitate the attachment of discharge conduit adapters **13208** to the static section **1305**.

[0543] Each dynamic section **1306** comprises a dynamic body **1307** and an insert **13307**, shown in

FIGS. 226, 229 and 230. The dynamic body 1307 comprises a front surface 1319 and outer surface 1321. The outer surface 1321 comprises a nose chamfer 13266, a radial static seal section 1323, a guide section 13261, static threads 1324, and locating shoulder 1359. The guide section 13261 has a larger diameter than the radial static seal section 1323 and extends from the static threads 1324 to the radial static seal section 1323.

[0544] Returning to FIGS. 226, 229, and 232 each flow control system 13146 comprises a front retainer 13315, a discharge plug 13238, a discharge plug insert 13239, a discharge plug seal 13229, a discharge valve spring 13328, a discharge valve 13240, a fluid routing plug 13241, a front fluid routing plug seal 13230, a rear fluid routing plug seal 13308, a suction valve 13242, a suction valve spring 13306, a suction valve guide wear ring 13334, a suction valve guide 13243, and a suction valve guide insert 13244.

[0545] The discharge plug 13238, shown in FIGS. 233-236, has a generally cylindrical shape and comprises a front surface 13335, a rear surface 13336, and an outer surface 13337. The front surface 13335 comprises a circular boss 13338 and a blind threaded hole 13339, both concentric with the outer surface 13337. The circular boss 13338 is configured fit within a counterbore 13340 on the rear surface 13341 of the front retainer 13315, as shown in FIG. 229. The blind threaded hole 13339 is configured to receive an assembly tool (not shown).

[0546] The outer surface 13337 comprises a large diameter section 13342 connected to a small diameter section 13343 by a tapered section 13344. The small diameter section 13343 comprises a seal groove 13345. As shown in FIG. 232, the large diameter section 13342, tapered section 13344, and small diameter section 13343 are configured to fit closely within the flow bore 1379, and the seal groove 13345 is configured to receive the discharge plug seal 13229.

[0547] The rear surface 13336 comprises a circular boss 13346, a blind hole 13347, a plurality of flow cutouts 13348, and a plurality of legs 13349. The circular boss 13346 and blind hole 13347 are concentric with the outer surface 13337. The blind hole 13347 is configured to receive the discharge plug insert 13239.

[0548] The discharge plug insert 13239 must be long enough to reduce rotation of the of the discharge valve stem 13245 about the transverse axis. To accommodate this, the circular boss 13346 extends from the rear surface 13336, increasing the depth of the blind hole 13347 so that the entire discharge plug insert 13239 engages the blind hole 13347, as shown in FIG. 229.

[0549] In this embodiment there are two flow cutouts 13348 and two legs 13349. Neither the cutouts 13348 nor the legs 13349 extend around the entire circumference of the discharge plug 13238. Each feature extends approximately ninety degrees around the circumference and alternates. Each flow cutout 13348 is parallel to the rear surface 13336 and removes material from the rear surface 13336, resulting in a shorter, or thinner, section of the discharge plug 13238 in that area, as shown in FIGS. 229 and 234.

[0550] Each leg 13349 is a cylindrical arc segment that extends circumferentially approximately ninety degrees and longitudinally from the rear surface 13336. The leg 13349 comprises a base section 13350, a nose section 13351, and inner surface 13352, an outer surface 13353, a rear surface 13354, and a wear surface 13355. The base section 13350 extends from the rear surface 13336 of the discharge plug 13238 to the nose section 13351, and the nose section 13351 extends from the base section 13350 to the rear surface 13354 of the leg 13349. The rear surface 13354 of the leg 13349 is perpendicular to the longitudinal axis of the discharge plug 13238 and is configured to engage the discharge surface 13271 of the fluid routing plug 13241 when assembled, as shown in FIG. 232. In this embodiment, the wear surface 13355 is an elastomeric material applied to the inner surface 13352 of the nose section 13351 of the leg 13349. The wear surface 13355 may be cast-in-place, pre-formed, or bonded and does not cover the rear surface 13354 of the leg 13349.

[0551] In alternate embodiments, the wear surface 13355 may be a pre-formed elastomeric piece that fits over the leg 13349, covering the inner surface 13352 and outer surface 13353 but not the

rear surface **13354** of the leg **13349**. In further alternate embodiments, the rear surface **13354** of the leg **13349** may have cutouts, or crenellations, formed to accommodate a pre-formed wear surface **13355** that wraps the inner surface **13352** and outer surface **13353**, and the cutout portions of the rear surface **13354** of the leg **13349** still leaving the portions of the rear surface **13354** of the leg **13349** that contact the fluid routing plug **13241** uncovered.

[0552] The discharge valve **13240**, shown in FIGS. **237-238**, comprises a stem **13245** opposite a front surface **13257**. The stem **13245** and front surface **13257** are concentric and opposed. As shown in FIG. **229**, the stem **13245** is configured to be received by the discharge plug insert **13239**. [0553] In this embodiment the stem **13245** is a circular boss but other cross-sectional shapes may be used. The front surface **13257** comprises the strike face of the valve insert **13267**, the strike face **13268**, a relief section **13269**, and a center section **13270**, all of which are concentric with the stem **13245**.

[0554] The strike face of the valve insert **13267** and strike face **13268** are known in the art. The relief section **13269** is a concave frustoconical section between the strike face **13268** and the center section **13270**. The center section **13270** is flat and perpendicular to the longitudinal axis of the discharge valve **13240**.

[0555] The fluid routing plug **13241**, shown in FIGS. **239-244**, comprises a discharge surface **13271**, a plurality of discharge fluid passages **13272**, a suction surface **13273**, and a plurality of suction passages **13358**. The discharge surface **13271** comprises a discharge seat **13274**, a first transition radius **13275**, a counterbore **13276**, a second transition radius **13277**, a central base **13278**, and a discharge fluid passage circle **13292**.

[0556] It should be understood that the mechanics of a fluid routing plug, such as fluid routing plug **13241**, are generally shown (as to function, not to specific features) in U.S. Pat. No. 11,300,111, issued to Thomas, et. al. Fluid is taken in at an intermediate surface from a suction manifold in response to the retreat of a plunger. The suction valve thus pulls away from the suction surface **13273** and fluid fills the dynamic internal flow bore. The plunger then pushes into the dynamic internal flow bore, forcing the suction valve to cover suction bores in the suction surface. Fluid flows instead through the discharge fluid passages **13272**, which are not covered by the suction valve.

[0557] Fluid flow through discharge fluid passages **13272** forces a discharge valve away from the discharge surface. High pressure fluid may then leave the fluid end **1302** through discharge manifolds **13215**.

[0558] The discharge seat **13274** is a frustoconical surface forming an acute angle **13279** with the longitudinal axis **13280** of the fluid routing plug **13241**, as shown in FIG. **244**.

[0559] The counterbore **13276** has a bore axis **13281** colinear with the longitudinal axis **13280** and extends from the first transition radius **13275** to the second transition radius **13277**. The counterbore **13276** also comprises a radius **13282**.

[0560] The central base **13278** is a planar surface that is perpendicular to the longitudinal axis **13280**, as shown in FIG. **243**. The central base **13278** extends from the second transition radius **13277** to the center of the discharge surface **13271**. The central base **13278** comprises a blind bore **13287**. The blind bore **13287** is concentric with the counterbore **13276**. The blind bore **13287** comprises a countersink **13289** and a threaded section **13291** configured to receive an externally threaded removal tool (not shown).

[0561] The discharge fluid passage circle **13292** is concentric with the longitudinal axis **13280**. The radius **13293** of the discharge fluid passage circle **13292** is the distance from the longitudinal axis **13280** of the fluid routing plug **13241** to the point of intersection **13294** between the longitudinal axis **13295** of any discharge fluid passage **13272** and a discharge seat extension line **13296**, as shown in FIG. **244**. The discharge seat extension line **13296** extends from the discharge seat **13274** at the same acute angle **13279** of the discharge seat **13274** in any cross-sectional view of the discharge seat **13274**. In this embodiment the radius **13282** of the counterbore **13276** is greater than

30% larger than the radius **13293** of the discharge fluid passage circle **13292**.

[0562] Each discharge fluid passage **13272** comprises a longitudinal axis **13295** that forms an acute angle **13332** with the longitudinal axis **13280** of the fluid routing plug **13241**, as shown in FIG.

244. Each discharge fluid passage **13272** further comprises a suction surface section **13299** and a bevel **13300**, as shown in FIG. **243**.

[0563] The suction surface section **13299** has a constant cross-sectional area and extends from the suction surface **13273** to the bevel **13300**. The bevel **13300** extends from the suction surface section **13299** to the discharge surface **13271**. The cross-sectional area of the bevel **13300** gradually increases from the suction surface section **13299** until the bevel **13300** intersects the discharge surface **13271**. The cross-sectional area of the bevel **13300** at the discharge surface **13271** is greater than 50% larger than the cross-sectional area of the suction surface section **13299**.

[0564] Each discharge conduit adapter **13208**, shown in FIGS. **224-227** and **231** comprises a bolt-on flange **13301**, a circular boss **13251**, an opposed flanged end **13252**, and a flow bore **13302**. The flanged end **13252** is configured for use in a clamp joint used to connect the discharge manifolds **13215** to the static section **1305**. The clamp joint is the type commonly used in the industry for high pressure connections.

[0565] The bolt-on flange **13301** has the general form of a flat rectangular prism, with a height significantly smaller than its width and depth. The bolt-on flange **13301** comprises a top surface **13303**, an opposing and parallel bottom surface **13304**, and a plurality of through bores **13305** extending between the top surface **13303** and bottom surface **13304**. The through bores **13305** are configured to receive fasteners **13263** and are arranged with the same spacing as the blind threaded holes **13265** on the top and bottom surfaces **13231**, **13232** of the static section **1305**. Each through bore **13305** has a bore axis perpendicular to the top and bottom surfaces **13303**, **13304**.

[0566] The circular boss **13251** extends perpendicularly from the bottom surface **13304** of the bolt-on flange **13301**. It is centered on the bottom surface **13304** and is concentric with the flanged end **13252**. Each circular boss **13251** is configured to mate with a counterbore **13234** of the upper or lower discharge conduits **13213** and **13214** in the static section **1305**.

[0567] The flow bore **13302** has a bore axis that is perpendicular to the top and bottom surfaces **13303**, **13304** of the bolt-on flange **13301**. The bore axis is concentric with both the flanged end **13252** and the circular boss **13251**.

[0568] The discharge manifold **13215**, shown in FIGS. **224-227**, and **231**, comprises a manifold body **13253**, a plurality of inlet ports **13254** and a plurality of outlet ports **13255**. Each inlet port **13254** comprises a flanged end **13256** configured for use in a clamp joint. The inlet ports **13254** are perpendicular to the manifold body **13253** and spaced so they will be coincident with the discharge conduit adapters **13208** once assembled. The outlet ports **13255** are configured to be connected to high pressure conduits (not shown) which may be attached to the other discharge manifold **13215**, or to other discharge manifolds of other fluid ends, or to the wellhead.

[0569] Referring now to FIGS. **229** and **230**, during assembly of the multi-piece fluid end **1302**, the dynamic sections **1306** must be threaded into the dynamic threads **1381** of the static section **1305**. The weight of the dynamic sections **1306**, the clearance between the mating dynamic threads **1381** and static threads **1324**, and the uneven resistance of the radial static seal **13220** as the radial static seal section **1323** initially contacts the radial static seal **13220**, all contribute to a misalignment between the longitudinal axis of the dynamic body **1307** and the axis of the flow bore **1379**. This causes the guide section **13261** of the outer surface **1321** of the dynamic body **1307** to contact the first straight section **13223** of the flow bore **1379**, scarring the guide section **13261**. However, the smaller diameter radial static seal section **1323** does not contact any portion of the flow bore **1379** thus avoiding any scarring to the radial static seal section **1323** and allowing leak free operation. Additionally, the axial static seal **13333** seals against the front surface **1319** of the dynamic body **1307** providing a safeguard in case the radial static seal **13220** does fail.

[0570] Referring now to FIGS. **226**, **228**, **229**, and **232**, the assembly of the multi-piece fluid end

1302 also requires the insertion of the components of the flow control systems **13146** into the flow bores **1379**. First, the suction valve guide insert **13244** is inserted in the suction valve guide **13243**. There may be an interference fit between the suction valve guide insert **13244** and the suction valve guide **13243** to prevent relative motion between the two components during operation. Second, the suction valve guide **13243**, with the assembled suction valve guide insert **13244**, is installed in the flow bore **1379**. Third, the suction valve guide wear ring **13334** is installed in the flow bore **1379** abutting the suction valve guide **13243**. Fourth, the suction valve spring **13306** is inserted in the flow bore **1379**. Fifth, the suction valve **13242** is installed in the flow bore **1379**, with the stem **13250** being inserted into the suction valve guide insert **13244**, thereby trapping the suction valve spring **13306** between the suction valve guide **13243** and the suction valve **13242**. Sixth, the front fluid routing plug seal **13230** and the rear fluid routing plug seal **13308** are installed on the fluid routing plug **13241**. Seventh, the fluid routing plug **13241** is inserted into the flow bore **1379**. [0571] The fluid routing plug **13241** may be inserted in the flow bore **1379** using the installation tool **12309**. However, since the fluid routing plug **13241** does not have the discharge valve guide **12247** feature of fluid routing plug **12241**, a cylindrical shaft (not shown) with a diameter larger than the blind bore **13287** and smaller than counterbore **13276** may be used apply force to the central base **13278** for insertion of the fluid routing plug **13241**. A removal tool (not shown) with a threaded end may be threaded into the threaded section **13291** of the blind bore **13287** and used to apply the force necessary to remove the fluid routing plug **13241** from the flow bore **1379** when necessary.

[0572] The remaining components of the flow control system **13146** may now be inserted in the flow bore **1379**. Continuing with step eight and referring to FIGS. **226-229**, and **232**, the discharge valve **13240** is inserted in the flow bore **1379** with the front surface **13257** first. Ninth, the discharge valve spring **13328** is inserted in the flow bore **1379**. Tenth, after the discharge plug seal **13229** and discharge plug insert **13239** have been installed in the discharge plug **13238**, the assembly is inserted in the flow bore **1379** such that the stem **13245** of the discharge valve **13240** is guided into the discharge plug insert **12339**. The discharge plug **13238** is inserted until the rear surfaces **13354** of the legs **13349** contact the fluid routing plug **13241**. The discharge valve spring **13328** is now trapped between the discharge valve **13240** and the discharge plug **13238**. Finally, the front retainer **13315** is threaded into the internal front retainer threads **13317** and torqued to specification, thus completing the assembly of the flow control system **13146** within the multi-piece fluid end **1302**.

[0573] Referring now to FIGS. **224-227**, and **231**, the assembly of the discharge manifolds **13215** to the static section **1305** is as follows. First, the discharge adapter seals **13237** are inserted in the seal grooves **13235** of the upper and lower discharge conduits **13213** and **13214**. Second, the circular bosses **13251** of the discharge conduit adapters **13208** are inserted into the counterbores **13234** of the upper and lower discharge conduits **13213** and **13214** until the bottom surface **13304** of the bolt-on flange **13301** contacts the top surface **13231** of the static section **1305**. Third, the through bores **13305** of each bolt-on flange **13301** are aligned with the blind threaded holes **13265** of the top surface **13231** and bottom surface **13232** of the static section **1305**. Fourth, the fasteners **13263** are inserted through the through bores **13305** and torqued into the blind threaded holes **13265** thus securing the discharge conduit adapters **13208** to the static section **1305**. Fifth, the clamp seals **13219** are installed in the seal carriers **13218** and the seal carriers **13218** are inserted in the flanged ends **13252** of the discharge conduit adapters **13208**. Sixth, the flanged ends **13256** of the inlet ports **13254** are placed over the seal carriers **13218**. At this point in the assembly the two flanged ends **13252** and **13256** will be in contact. Seventh, the clamps **13216** are placed around the flanged ends **13252** and **13256** and the clamp fasteners **13217** are installed. In this embodiment, each fastener **13263** comprises a stud **13329**, a leveling washer **13330**, and a flanged nut **13331**. In alternative embodiments, the fastener **13263** may omit the leveling washer **13330**, use a standard washer, comprise a flanged bolt, or use any other fasteners or fastening systems known in the art.

[0574] Referring now to FIGS. 228 and 232, for illustration purposes only, both the suction valve **13242** and discharge valve **13240** are shown in the open position. During operation, as the plunger **1318** retracts, the suction valve **13242** opens, allowing fluid to enter the flow bore **1379** by passing between the suction surface **13273** of the fluid routing plug **13241** and the suction valve **13242**. The fluid flow is represented by arrows **13356**. The fluid then impacts the suction valve guide wear ring **13334** as it enters the flow bore **1379**.

[0575] The suction valve guide wear ring **13334** comprises a harder material than the suction valve guide **13243** or the dynamic body **1307**, protecting both from erosion. The increased hardness of the suction valve guide wear ring **13334** results in longer maintenance intervals and allows for the replacement of only the suction valve guide wear ring **13334** if required, reducing the cost of operation.

[0576] In an alternate embodiment the suction valve guide wear ring **13334** may comprise a material that is not harder than the suction valve guide **13243**. In this alternate embodiment, the suction valve guide wear ring **13334** primarily serves as a sacrificial element, absorbing erosion rather than preventing it, thereby still providing protection for the suction valve guide **13243** and dynamic body **1307**.

[0577] As the plunger **1318** extends, the suction valve **13242** closes and the discharge valve **13240** opens. Fluid flows out of the bevel **13300** of the discharge fluid passage **13272** of the fluid routing plug **13241**, impacting the relief section **13269** of the front surface **13257** of the discharge valve **13240**, then flowing between the discharge surface **13271** of the fluid routing plug **13241** and the front surface **13257** of the discharge valve **13240**. The fluid then impacts the wear surface **13355** of the leg **13349** of the discharge plug **12238**. The fluid flow is represented by arrows **13357**.

[0578] As fluid exits the discharge fluid passage **13272**, the increased cross-sectional area of the bevel **13300** reduces fluid velocity, thereby reducing erosion around the bevel **13300** on the discharge surface **13271** of the fluid routing plug **13241**. The reduced velocity also decreases the impact force of the fluid on the front surface **13257** of the discharge valve **13240**. The relief section **13269** on the front surface **13257** of the discharge valve **13240** provides a smooth flow transition as fluid flows along the front surface **13257** of the discharge valve **13240**.

[0579] The wear surface **13355** comprises a material that reduces erosion and/or functions as a sacrificial element, thereby increasing the life of the leg **13349** and, consequently, the discharge plug **13238**. This results in extended maintenance intervals and reduced operational cost. If the wear surface **13355** is an embodiment configured for replacement, maintenance costs are further reduced by allowing only the wear surface **13355** to be replaced.

[0580] While the improvements disclosed herein are described in detail as being used with or applied to multi-piece fluid ends or fluid end sections, it is expected that one skilled in the art will recognize that these improvements can readily be adapted for use with either style of fluid end. Moreover, it is contemplated that the disclosed improvements are applicable to prior art fluid ends of any design or configuration.

[0581] The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion, and it is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

- 1.** A high-pressure pump, comprising: a fluid end body, comprising: a static section comprising at least one internal flow bore having internally-disposed threads; and a dynamic section comprising a dynamic internal flow bore, the dynamic section having external threads disposed about an external surface of the dynamic section; wherein the dynamic section is configured for threaded attachment to the static section such that the at least one internal flow bore and the dynamic internal flow bore are aligned.
- 2.** The high-pressure pump of claim 1 further comprising: a plunger system installed within the dynamic internal flow bore, comprising: a packing; a plunger, partially disposed within the packing; and a threaded packing nut disposed about the plunger and configured to retain the packing in the dynamic internal flow bore.
- 3.** The high-pressure pump of claim 2 further comprising: a retainer affixed to the dynamic section and configured to retain the threaded packing nut within the dynamic internal flow bore.
- 4.** The high pressure pump of claim 3 in which the retainer is affixed to the dynamic section by a plurality of studs.
- 5.** The high pressure pump of claim 3 in which the threaded packing nut is threaded to an internal surface of the retainer.
- 6.** The high pressure pump of claim 1 further comprising: an inlet manifold, in communication with the internal flow bore of the static section; an outlet manifold, in communication with the internal flow bore of the static section; and a fluid routing plug disposed within the internal flow bore, the fluid routing plug having a suction surface and a discharge surface and defining: a first fluid flow path from the inlet manifold, through the fluid routing plug, to the suction surface; and a second fluid flow path from the suction surface to the discharge surface.
- 7.** The high pressure pump of claim 6 further comprising: a suction valve disposed within the dynamic internal flow bore and configured to cover the first fluid flow path at the suction surface; and a discharge valve disposed within the static section and configured to cover the second fluid flow path at the discharge surface.
- 8.** The high pressure pump of claim 1 further comprising a discharge manifold, the discharge manifold attached to the static section and comprising: a first discharge conduit, the first discharge conduit extending from the static section and comprising: a first flange; a boss extending from the first flange; and a second flange, disposed on the discharge conduit at an end opposite the first flange, in which: the boss extends into a counterbore formed on the static section and; the first flange is attached to the static section with a plurality of bolts.
- 9.** The high pressure pump of claim 8 in which the discharge manifold further comprises: a manifold body having a plurality of inlet ports, the manifold body comprising a plurality of third flanges, each disposed about each of the plurality of inlet ports; and a clamp joint, joining one of the plurality of third flanges to the second flange of the discharge conduit.
- 10.** The high pressure pump of claim 1 in which the first discharge conduit is located above the static section, and further comprising: a second discharge conduit, the second discharge conduit extending from and being located below the static section and comprising: a first flange; a boss extending from the first flange; and a second flange, disposed on the discharge conduit at an end opposite the first flange, in which: the boss extends into a counterbore formed on the static section and; the first flange is attached to the static section with a plurality of bolts.
- 11.** The high pressure pump of claim 10 in which the discharge manifold further comprises: a top manifold body having a plurality of inlet ports, the top manifold body comprising a plurality of third flanges, each disposed about each of the plurality of inlet ports of the top manifold body; a bottom manifold body having a plurality of inlet ports, the bottom manifold body comprising a plurality of fourth flanges, each disposed about each of the plurality of inlet ports of the bottom manifold body; a first clamp joint, joining one of the plurality of third flanges to the second flange of the first discharge conduit; and a second clamp joint, joining one of the plurality of fourth

flanges to the second flange of the second discharge conduit.

12. The high pressure pump of claim 1 in which the static section comprises five internal flow bores.

13. The high pressure pump of claim 12 wherein five dynamic sections are threaded to the static section at corresponding internal flow bores.

14. The high pressure pump of claim 1 wherein the dynamic section comprises a front section extending into the static section further than the external threads of the dynamic section.

15. The high pressure pump of claim 14 further comprising an axial face seal disposed in the static section and contacting a front surface of the front section of the dynamic section.

16. The high pressure pump of claim 14 further comprising at least one radial seal disposed in the static section and contacting an outside surface of the front section of the dynamic section.

17. The high pressure pump of claim 16 further comprising an axial face seal disposed in the static section and contacting a front surface of the front section of the dynamic section.

18. The high pressure pump of claim 14 wherein the front section of the dynamic section comprises a chamfered nose.

19. The high pressure pump of claim 14 further comprising: a fluid routing plug disposed within the static section and configured to route fluid from a fluid inlet into the dynamic internal flow bore and, thereafter, across the fluid routing plug to a fluid outlet; and an insert disposed between the fluid routing plug and a shoulder of the front section of the dynamic section.

20. The high pressure pump of claim 19 in which the internal flow bore is closed by a threaded front retainer.

21. The high pressure pump of claim 20, further comprising: a suction valve configured to seal a first side of the fluid routing plug; a discharge valve configured to seal a second side of the fluid routing plug, in which the discharge valve comprises a stem; and a discharge plug seated in the threaded front retainer, the discharge plug having a blind hole configured to receive the stem of the discharge valve.

22. The high pressure pump of claim 21 further comprising a discharge plug insert disposed within the blind hole of the discharge plug, the discharge plug insert disposed about the stem.

23. The high-pressure pump of claim 22 further comprising: a plunger system installed within the dynamic internal flow bore, comprising: a packing; a plunger, partially disposed within the packing; and a threaded packing nut disposed about the plunger and configured to retain the packing in the dynamic internal flow bore.

24. The high-pressure pump of claim 23 further comprising a power end, wherein reciprocation of the plunger within the dynamic internal flow bore is driven by the power end.
